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The Global Fiscal Actions (GFA) Dataset

Hites Ahir, Miyoko Asai, Matthieu Bellon, Maria Chiara Cavalleri, Shuvam Das, Davide Furceri, Faizaan Kisat, Aiko Mineshima, Irene Rausell Moreno, Vishal Parmar, Nikhil Patel, and Adrian Peralta Alva

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ABSTRACT: We introduce the Global Fiscal Actions (GFA) Dataset, the first quarterly global narrative database of discretionary fiscal policy actions, covering 142 countries since 1990. GFA combines Economist Intelligence Unit country reports with large language models to systematically identify and classify tax and spending measures, together with their motivations, timing, implementation status, and supporting evidence. We validate the dataset against economists' assessments, existing narrative datasets, fiscal aggregates, and repeated AI extractions, demonstrating high accuracy and reproducibility. We illustrate its value through applications to fiscal cyclicality and country-specific fiscal multipliers, providing new evidence on the conduct and effectiveness of fiscal policy worldwide.

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Author's E-Mail Address:	HAhir@imf.org, MBellon@imf.org, MCavalleri@imf.org, DFurceri@imf.org, FKisat@imf.org, AMineshima@imf.org, IRausell@imf.org, VParmar@imf.org, NPatel@imf.org, APeralta-Alva@imf.org

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I. Introduction

Empirically assessing the stance and effectiveness of fiscal policy remains a central challenge for academics and policymakers. One key difficulty is identifying discretionary fiscal actions—that is, deliberate policy measures taken by the government to change spending, taxes, or transfers. Actual data on government spending and taxes reflect not only discretionary fiscal actions, but also automatic and endogenous responses to economic conditions—the so-called "automatic stabilizers." Although statistical methods can be used to purge actual fiscal data of the effects of economic conditions, these approaches provide only a crude and imperfect assessment of the fiscal stance and of discretionary policy actions. For example, the Cyclically Adjusted Primary Balance (CAPB)—that is, the primary balance adjusted for the business cycle—is a useful proxy but has important limitations: (i) it is sensitive to the specific filtering techniques used to estimate the output gap; (ii) it does not fully account for the various shocks driving economic activity, such as shocks to financial conditions and commodity prices; and (iii) it is ill-suited to assessing the fiscal stance and discretionary fiscal impulse during major crises or periods characterized by significant shifts in potential output.

A second key challenge is that discretionary fiscal policy actions may still be endogenous to the government's objectives. For example, discretionary increases in spending aimed at supporting economic activity during a crisis are endogenous to economic conditions and therefore cannot be used to assess the causal effect of spending on economic activity. Similarly, discretionary changes in the tax system motivated by distributional concerns are endogenous to inequality and therefore cannot be used to assess the redistributive effects of these policy actions.

The pioneering work of Romer and Romer (2010) addressed these challenges by applying narrative analysis to U.S. presidential speeches and Congressional reports to identify the size, timing, and principal motivation of all major postwar tax policy actions, and to distinguish legislated changes taken in response to prospective economic conditions from those motivated by more exogenous considerations. However, because this approach is highly labor-intensive, it has so far been applied only to a relatively small set of countries and largely on a one-off basis (see next section for details).

This paper leverages recent advances in generative artificial intelligence (AI)—specifically large language models (LLMs)—to construct the first global narrative dataset of discretionary fiscal policy actions: the Global Fiscal Actions Dataset (GFA). Specifically, the dataset records fiscal policy actions at a quarterly frequency for 142 countries since 1990.

In constructing the database, we closely follow the criteria identified by Romer and Romer (2019) for a "rigorous narrative analysis," while introducing a scalable and transparent implementation based on LLMs. For each identified discretionary fiscal action, we record a rich set of characteristics, including its direction (expansionary or contractionary), implementation status (e.g., announcement, approval, enactment), underlying motivation (such as debt sustainability, stabilization, long-term growth, redistribution, the provision of public goods, or political factors), and qualitative and quantitative measures of intensity (Table 1). Importantly, all extracted information is supported by verbatim textual evidence, ensuring traceability and facilitating validation.

The use of LLMs offers two main advantages over traditional manual approaches. First, it enables the large-scale and consistent processing of textual information across countries and over time, addressing a key

limitation of existing narrative datasets. Second, the structured prompt design ensures that information is extracted systematically and reproducibly, reducing the scope for subjective interpretation.

To address potential concerns regarding the accuracy, reliability, and consistency of our dataset, we provide several validation and reproducibility checks. First, we compare AI-assisted outputs with expert assessments by IMF economists and find that the level of agreement is comparable to that observed among economists themselves. Second, we benchmark our results against existing narrative datasets, including Romer and Romer (2019) and Adler et al. (2024), and find extremely high overlap in the identification of fiscal episodes. Third, we show that identified fiscal actions are systematically associated with subsequent movements in fiscal aggregates, supporting their empirical relevance. Fourth, we document the reproducibility of the AI-based procedure across repeated runs and prompt specifications. Taken together, these exercises provide strong evidence that the AI-based methodology produces reliable and economically meaningful measures of discretionary fiscal policy that can be used by academics and policymakers.

We use the dataset to establish three novel findings on the conduct and effectiveness of fiscal policy around the world. First, we document a set of stylized facts on the nature of fiscal actions. Fiscal actions are frequent, averaging about 2.8 measures per country per year. Across income groups, spending-based measures predominate, especially in low-income countries (LICs), where they account for the bulk of fiscal interventions. While fiscal sustainability is the most common motivation, expansionary measures are generally more prevalent, and individual actions tend to be small in size—particularly tax measures. Second, we document how fiscal policy is conducted over the cycle and how fiscal actions are correlated across countries. While spending measures have become less procyclical over time for a broad set of countries, especially emerging markets (EMs) and LICs—reflecting improvements in institutional quality—tax measures remain largely acyclical across countries, including in advanced economies (AEs). Moreover, fiscal measures are typically not synchronized across countries, reflecting the diverse objectives that governments pursue in the conduct of fiscal policy, except during periods of major global recessions, when fiscal stimuli are implemented across many countries to stabilize economic activity. Finally, we complement GFA with a novel quarterly dataset on spending and tax revenue covering a large set of advanced and developing economies, which we use to estimate country-specific spending and tax multipliers and examine how they vary across countries. The results point to two main findings: (i) tax multipliers are, on average, larger than spending multipliers; and (ii) there is significant heterogeneity in both types of multipliers, with differences substantially larger within income groups than across income groups.

The remainder of the paper is organized as follows. Section II discusses the paper's contribution to the literature. Section III describes the construction of the database, including the underlying data sources, the AI-assisted extraction methodology, and the design of the prompt. Section IV presents a set of stylized facts on discretionary fiscal actions across countries and over time. Section V validates the dataset by comparing the AI-generated classifications with economists' assessments, existing narrative datasets, fiscal aggregates, and repeated AI runs. Section VI illustrates selected empirical applications on fiscal cyclicality and country-specific fiscal multipliers. Section VII concludes.

II. Contribution to Literature

This paper contributes to several strands of the literature on fiscal policy and the construction of policy-action databases. Its first contribution is to the literature on fiscal narrative databases. We identify discretionary fiscal actions from EIU country reports for 142 countries at quarterly frequency from 1990 onward. Existing narrative

datasets are typically limited to a single country, such as Romer and Romer (2010) and Cloyne (2013); specific episodes, such as the financial-crisis episodes in Romer and Romer (2019); or fiscal consolidation episodes for relatively small groups of countries, as in Bhasin and Loungani (2026), Devries et al. (2011), Guajardo, Leigh, and Pescatori (2014), Carrière-Swallow, David, and Leigh (2021), Adler, Allen, Ganelli, and Leigh (2024), and Abdel-Latif, Bechchani, David, and Lemaire (2026). Recent AI-assisted narrative fiscal datasets also remain narrowly focused: Das, Furceri, Patel, and Peralta-Alva (2026) identify discretionary government spending actions for 64 countries; Alloza et al. (2025) identify spending and tax actions for a set of European Union countries; while Abdel-Latif, Bechchani, David, and Lemaire (2026) focus on fiscal consolidations in 14 Sub-Saharan African countries. Compared with this literature, our database provides significantly broader country and time coverage; is available at quarterly frequency, which is key for identification; and covers spending and tax actions as well as expansionary and contractionary actions within a single, globally consistent narrative source.

The second contribution is to the narrative fiscal identification literature by enriching the information available for identifying exogenous fiscal shocks**.** Existing narrative datasets rely on policy motivations to distinguish exogenous from endogenous fiscal actions, but they typically record only a limited set of characteristics for each observation. Romer and Romer (2019), for example, document the direction, size, and motivations of fiscal actions during financial crises, illustrating the value of richer narrative coding, but such hand-collected information is difficult to scale. GFA extends this approach by systematically recording motivations, underlying causes, implementation status, timing, direction, intensity, and supporting evidence for each fiscal action. This richer information allows researchers to construct alternative definitions of exogenous fiscal shocks tailored to the objective of their analysis. For example, measures recorded in existing consolidation databases may be exogenous to short-term economic fluctuations but endogenous to other factors, such as debt levels or inequality, and therefore may not be suitable for identifying the causal effect of fiscal policy on these outcomes. In contrast, our database can help researchers identify actions that are not motivated by debt-stabilization or redistributive goals. In addition, the additional information on timing and implementation status also facilitates the study of fiscal foresight and implementation lags, as emphasized by Ramey (2011) and Leeper, Walker, and Yang (2013).

The third contribution is to the emerging literature on AI-assisted economic measurement by extending its application to the construction of large-scale narrative fiscal datasets. Recent work has applied AI and LLMs to extract structured information from economic and policy documents, particularly in monetary policy and central-bank applications. Dunn, Meade, Sinha, and Kabir (2024), for example, assess whether generative AI models can classify topics discussed in FOMC minutes, while Wu, Xi, and Xie (2025) develop an LLM-based framework for economic expectations surveys. Broader central-bank applications are reviewed by Araujo, Doerr, Gambacorta, and Tissot (2024). Related ongoing work by Alloza et al. (2025) also applies LLM- and AI-based methods to proprietary government documents; however, the resulting dataset cannot be publicly disseminated because the underlying information is confidential. This paper contributes by applying a standardized LLM-based framework to a corpus of EIU country reports, which are publicly available through subscription, and by designing the extraction process to be transparent, reproducible, and systematically validated. The same prompts and coding rules are applied consistently across countries and over time, each identified fiscal action is linked to verbatim textual evidence, and the resulting classifications are validated against economists' assessments, existing narrative datasets, fiscal aggregates, and repeated AI runs.

The fourth contribution is to the fiscal multiplier literature by providing new country-specific quarterly estimates of spending and tax multipliers for a large set of countries. Seminal studies estimate fiscal multipliers using quarterly data, but they typically focus on individual advanced economies or small groups of countries. These

include structural VAR estimates for the United States (Blanchard and Perotti, 2002), five OECD economies (Perotti, 2005), narrative tax shocks for the United States (Romer and Romer, 2010; Mertens and Ravn, 2013), military spending news for the United States (Ramey and Zubairy, 2018), and narrative tax shocks for the United Kingdom (Cloyne, 2013). In addition, much of the cross-country literature estimates average multipliers for pooled samples or broad country groups rather than separate multipliers for individual countries, while some studies rely on annual data or focus on overall fiscal shocks rather than separately identified expenditure and tax actions. As a result, country-specific quarterly estimates remain particularly scarce for emerging market and low-income economies. We address this gap by providing quarterly narrative fiscal actions for a much broader set of advanced, emerging market, and low-income economies while distinguishing expenditure and tax measures within a unified framework. This makes it possible to estimate country-specific spending and tax multipliers at quarterly frequency, thereby substantially expanding the empirical basis for studying fiscal transmission and cross-country heterogeneity.

The fifth contribution is to the literature on fiscal cyclicality by using GFA to provide new evidence on the cyclical behavior of discretionary fiscal policy. While the cyclicality of the overall fiscal stance is the relevant metric for assessing whether fiscal policy amplifies or dampens business cycles, discretionary fiscal actions are an important component of that response and provide insight into the policy choices underlying fiscal outcomes. A large literature documents that fiscal policy has historically been more procyclical in developing and emerging economies than in advanced economies. Gavin and Perotti (1997) first documented this pattern for Latin America, while Kaminsky, Reinhart, and Végh (2004, 2005) showed that fiscal policy is procyclical across many developing countries. Talvi and Végh (2005) attribute this behavior to greater tax-base volatility, whereas Alesina, Campante, and Tabellini (2008) emphasize political-economy mechanisms. Frankel, Végh, and Vuletin (2013) show that stronger institutions facilitate “graduation” from fiscal procyclicality, while Végh and Vuletin (2015) demonstrate that tax policy is largely procyclical in developing countries but broadly acyclical in advanced economies. We contribute to this literature by providing a direct measure of discretionary fiscal actions, rather than inferring discretion from the behavior of government spending, tax revenues, or fiscal balances over the business cycle, which only imperfectly captures the degree of discretion in fiscal policy actions. This allows us to directly assess the cyclicality of discretionary fiscal policy and how it varies across countries. We also show that countries differ in the cyclicality of spending and tax policy, with stronger institutions mainly reducing spending procyclicality rather than shaping tax policy cyclicality.

Finally, we contribute to the literature on the international synchronization of macroeconomic policies by providing new cross-country evidence on the synchronization of discretionary fiscal actions. Existing studies have primarily examined the synchronization of business cycles (e.g., Kalemli-Ozcan, Papaioannou, and Peydró, 2013) and monetary policy (e.g., Forbes, Ha, and Kose, 2024), while evidence on the synchronization of discretionary fiscal actions remains limited because comparable high-frequency discretionary actions data have not been available. We fill this gap by providing a consistent quarterly panel of discretionary fiscal actions for a broad set of countries. Using a synchronization measure analogous to Kalemli-Ozcan, Papaioannou, and Peydró (2013), we show that discretionary fiscal actions are generally weakly synchronized across countries but become substantially more synchronized during major global shocks, particularly among advanced economies during the COVID-19 pandemic.

III. Construction of the Database

This section describes the construction of the AI-assisted Global Fiscal Actions Dataset (GFA). Following Romer and Romer’s (2023) recommendations for rigorous narrative analysis, we first describe the underlying

data sources and then outline the methodological framework used to extract information on fiscal actions. We subsequently document the narrative evidence and present new stylized facts on fiscal actions over time and across countries. Finally, we validate the database through several checks, including comparisons between AI-generated classifications and economists' assessments, as well as benchmarks against existing narrative fiscal action datasets.

Source of Information: EIU reports

Our primary source of information is the Economist Intelligence Unit (EIU), a leading provider of country-level intelligence that has published regular analytical reports covering up to 189 countries, with coverage for some countries extending back to the early 1950s. The EIU has been used in empirical research, including the narrative study by Romer and Romer (2019) and the World Uncertainty Index of Ahir, Bloom and Furceri (2022).

These reports provide a structured assessment of key economic, fiscal, and political developments in each country. Crucially, they offer detailed coverage of fiscal policy, including measures that have been implemented following earlier decisions, newly announced and enacted actions, as well as policy initiatives under consideration by key stakeholders. In addition, the reports often include qualitative—and in some cases quantitative—assessments of the fiscal and macroeconomic impacts of these actions, along with discussions of their underlying motivations.

As discussed in Ahir, Bloom and Furceri (2022), the preparation of these reports draws on an extensive network of specialists, including analysts based in the countries themselves and country experts at EIU headquarters. Headquarters-based analysts—typically with five to seven years of experience—are responsible for monitoring a small number of countries and maintain close familiarity through regular engagement and country visits. This structure enables the combination of detailed local knowledge with a comparative, cross-country perspective.

Importantly, EIU report production follows a standardized, multi-stage process. Field-based analysts prepare initial drafts, which are then reviewed and expanded by headquarters-based country experts to ensure consistency in structure and alignment with the EIU's regional and global assessments. The revised reports subsequently undergo detailed review by senior staff, followed by sub-editing to ensure clarity, internal consistency, factual accuracy, and verification. This rigorous and standardized process promotes comparability across countries, supporting the application of a unified methodological framework to extract consistent information.

Extracting Structured Information: Prompt Design

Our objective is to identify whether new fiscal actions occur in each quarter and, if so, to characterize them in a consistent and verifiable manner. We define fiscal actions as deliberate government measures that change tax policy or public spending (including transfers and other budgetary instruments). This definition excludes movements in fiscal balances driven by automatic stabilizers (e.g., unemployment benefits) or cyclical fluctuations in the tax base. Fiscal actions include, for example, announcements by government officials, cabinet approvals, parliamentary passage, enactment by the head of state, and the suspension or withdrawal of measures.

A central feature of the approach is the explicit temporal classification of fiscal actions, which allows contemporaneous actions to be identified in real time. For each quarterly report, actions are categorized as *past*, *current*, or *future* based on when the decision was taken/announced, rather than when the measure takes effect. Fiscal actions are further distinguished between announced and implemented. Accordingly, a current fiscal action is one for which the relevant policy decision occurs during the quarter covered by the report.

Fiscal actions and their characteristics are identified using an AI LLM (GPT-5.0 OpenAI) through a carefully designed prompt (see Annex A). The prompt provides detailed instructions, definitions, interpretive guidance, and examples to ensure consistent and replicable extraction across countries and over time.[1] The prompt begins with a preamble that sets out overarching instructions, including the definition of fiscal actions, the rules for temporal classification, and illustrative examples. The remainder of the prompt is organized as a sequence of questions corresponding to variables in the dataset.

The first set of questions (Q1–Q2) identifies whether the report mentions fiscal actions and extracts supporting evidence. If no fiscal action is identified, the model provides a rationale. If fiscal actions are present, the model extracts verbatim quotes from the report to ensure traceability and enable verification.

The second set of questions (Q3–Q4) focuses on the drivers of fiscal actions, covering both underlying causes and policy motivations. “Causes” are defined as the economic or institutional triggers that prompted policymakers to take fiscal actions—such as recessions, commodity price shocks, or financial crises that generate fiscal pressures or fiscal space—while “motivations” capture the policy intent behind the fiscal action itself. Identifying both causes and motivations improves the precision, consistency, and richness of the extracted information, and provides inputs that are particularly useful for assessing whether fiscal actions are plausibly exogenous. We follow an approach similar to Romer and Romer (2010) and define seven high-level motivation categories: (1) economic or financial stabilization, (2) debt sustainability and debt relief, (3) long-term growth, (4) redistribution, (5) provision of public goods and externalities, (6) politically related factors, and (7) others. The prompt also asks the AI LLM to explain its reasoning when assigning motivations.

The third set of questions (Q5–Q6) focuses on the status and timing of fiscal actions. The prompt asks the AI LLM to classify each action according to a standardized set of procedural stages—announced, administrative enactment, cabinet approval, parliament passage, signed, suspended, or withdrawn. It also requires extracting the effective date—the point at which the measure comes into force—and/or the relevant budget year, using the most precise timing available in the report (exact date, month/year, quarter/year, or year). In addition, the prompt records whether the fiscal action is part of a multi-year package (YES/NO/UNCLEAR) and, if so, captures its total duration in years along with the starting year.

The fourth set of questions (Q7–Q13) characterizes the direction, instruments, and intensity of fiscal actions. The prompt first asks the AI LLM to assess whether the identified action is expansionary, contractionary, neutral, or unclear relative to a counterfactual without the action, and to support this classification with direct quotes from the report, while also eliciting a qualitative assessment of the overall size. It then identifies the fiscal instruments involved by separately assessing the effects on tax revenues and government spending, including whether each component is expansionary or contractionary. To capture intensity more systematically, the prompt assigns numeric scores to the overall fiscal stance as well as to its tax and spending components,

[1] The database used in this paper is based on the prompt that was executed on April 6, 2026.

using a standardized scale from −10 to +10, where negative (positive) values indicate contractionary (expansionary) actions. The scores reflect the strength of the fiscal action based on available information, including the quantitative and qualitative magnitude of the measure and, where available, its aggregate fiscal impact. Because information varies across actions, some scores rely on limited information and should be interpreted with caution. Nevertheless, Section V reports that the intensity scores track movements in corresponding fiscal aggregates well. When available, the prompt also extracts quantitative estimates of the fiscal stance, along with their units and supporting quotations, ensuring a consistent and transparent characterization of both the direction and magnitude of fiscal actions across observations.

The final set of questions (Q14–Q16) collects additional important information needed for interpretation and validation. The prompt asks the AI LLM to list the specific tax and spending measures underlying each fiscal action using short phrases, and to distinguish between measures that are consistent with the overall fiscal stance (e.g., spending increases or tax cuts for expansionary actions; spending cuts or tax increases for contractionary actions) and those that are not aligned, including offsetting or ambiguous measures. It also allows the model to record brief additional notes of fiscal relevance (limited to 30 words). Finally, the prompt requires a short rationale explaining why alternative fiscal stances or motivations were not selected and why certain fiscal policy actions mentioned in the report were excluded—for example, because they refer to past or future measures, or actions that were withdrawn or not implemented.

The prompt is designed to guide the AI model's interpretation through detailed instructions, clarifying notes, and examples developed iteratively. In particular, a dedicated interpretive guidance section addresses common sources of misclassification identified in earlier versions. It clarifies key conceptual distinctions—such as the difference between causes and motivations—provides detailed instructions on timing, and emphasizes the need for internal consistency across responses (for example, alignment between the fiscal stance and the sign of estimated impacts). It also elaborates motivation categories with extensive examples, which significantly improves the reliability of the model's output.

Each EIU country report is processed individually using a fully automated pipeline. Reports are first converted into plain text and passed to the GPT-5.0 model via the Azure OpenAI application programming interface (API) in a ring-fenced environment. For each country-quarter report, the AI model is provided with the same prompt containing detailed instructions, definitions, interpretive guidance, and examples, together with the full text of the report as input. The prompt is applied uniformly and independently across all reports, with no country-specific customization. The model then returns a structured output in machine-readable JSON format, consisting of three separate objects corresponding to past, current, and future fiscal actions. The structured output of each report depends only on the prompt and text content of the report.

Database Documentation and Coverage

The accompanying database file (available at: https://data.imf.org/en/datasets/IMF:WP_GFA/IMF:GFA) reports the full narrative dataset. Exhibits 1–31 in Annex B provide detailed examples of how the AI classifies fiscal episodes, together with a description of the human assessment used to evaluate the AI output.

The current version of the database covers 142 countries from 1990 onward, spanning all major regions and income groups (Annex C). Its regional composition is broadly balanced: Africa accounts for 27 percent of countries, Europe for 24 percent, the Middle East for 17 percent, and Asia-Pacific and the Western Hemisphere

for about 16 percent each. By income group, EMs represent the largest share of countries (42 percent), followed by LICs (35 percent) and AEs (23 percent).

IV. Stylized Facts

Fiscal actions are frequent. EMs and AEs record similar numbers of fiscal actions per country—just below 100 over the full period, equivalent to about 2.8 actions per calendar year—while LICs record a moderately lower number, in the low 80s over the same period. A plausible explanation for this difference is that stronger institutional capacity and more developed fiscal frameworks in AEs and EMs allow for more regular policy adjustments, whereas capacity constraints and implementation challenges may limit the frequency of such actions in LICs (Figure 1).[2]

The rest of this section presents a novel set of stylized facts on the composition (spending vs. taxes), motivations, direction and intensity of fiscal actions, as well as their evolution over time and across income groups. It also provides new evidence on cross-country synchronization, highlighting how fiscal policy responses co-move across countries, particularly during periods of global shocks.

Stylized fact 1—Composition. *Spending-based measures dominate fiscal actions* (Figure 2). This dominance is particularly pronounced in LICs, where spending actions constitute the bulk of fiscal interventions. AEs exhibit a relatively larger share of combined tax-and-spending packages, while tax-only measures remain a minority across all groups. Over time a gradual increase in the share of combined tax-and-spending packages is observable—most notably in LICs, and to a lesser extent in EMs and AEs—particularly in more recent years. At the same time, tax-only measures not only remain limited in share but also appear to have declined gradually over time across income groups. These patterns suggest that while governments continue to rely primarily on spending instruments, there is a modest shift toward more integrated policy packages combining tax and spending measures, especially in lower-income settings. Overall these findings are consistent with previous empirical evidence suggesting that tax measures are often more politically difficult to implement than spending measures (Alesina et al., 2024).

Stylized fact 2—Motivation. *Fiscal sustainability is the prevalent motivation for fiscal actions* across all countries—especially in EMs and LICs—underscoring the central role of debt-related concerns in fiscal policymaking (Figure 3). Growth and development and redistribution are also widely represented across groups, with EMs accounting for a relatively large share of these actions in absolute terms. Stabilization and public goods provision display broadly similar patterns across income groups, while political economy considerations remain comparatively limited.

Differentiating across instruments, spending-based instruments dominate fiscal actions across most policy motives, accounting for roughly 50–70 percent of cases, while tax-only measures play a much smaller role (Figure 4). This pattern is particularly strong for objectives such as the provision of public goods and externalities (68.7 percent), redistribution (60.7 percent), and long-term growth (59.3 percent), where spending clearly outweighs other instruments. Combined tax-and-spending approaches represent a sizable but

[2] The database also distinguishes between announced measures and those that have already been implemented or formally enacted, with announcements accounting for a slightly larger share of fiscal actions than implemented measures.

secondary share, typically around 25–32 percent across most categories. In contrast, tax-only measures are generally limited—often below 15 percent—except for residual or "other" motives, where they become more prominent (46.8 percent). Overall, the evidence suggests a systematic preference for expenditure-based policies across core fiscal objectives, with more balanced or tax-driven approaches emerging mainly for stabilization and fiscal sustainability motives.

Stylized fact 3—Direction. *Fiscal actions are typically expansionary.* Expansionary measures consistently outnumber contractionary ones, especially since the early 2000s (Figure 5). This pattern is consistent with the existence of a "deficit bias"—that is, the political tendency of governments to run budget deficits and accumulate public debt (Caselli et al. 2022)—as well as recent empirical evidence from Cao, Dabla-Norris, and Di Gregorio (2024), which shows that political rhetoric in electoral democracies has shifted toward more expansionary policies and away from fiscal restraint. Episodes of fiscal expansion are evident during major downturns—most notably during the COVID-19 pandemic, when the share of countries undertaking expansionary actions rose to close to 80 percent.

Similar findings emerge when examining the evolution of net fiscal stance—defined as the average intensity score (which ranges from -10 to +10) for a particular country group over time (Figure 6). The fiscal stance has been consistently expansionary across income groups, especially during episodes of global recession, such as the Global Financial Crisis and the COVID-19 pandemic. At the same time, its evolution differs across country groups. EMs and LICs generally display more stable and moderately expansionary fiscal positions, particularly in the mid-2000s. By contrast, AEs exhibit greater volatility, including a sharp contraction around the European debt crisis followed by a rapid expansion that peaked during the COVID-19 pandemic.

Finally, the direction of fiscal actions varies by motivation (Figure 7). For expansionary measures, motivations are relatively diversified, with long-term growth and redistribution accounting for sizable shares across all instruments, alongside meaningful contributions from stabilization and the provision of public goods. In contrast, contractionary measures are, unsurprisingly, overwhelmingly driven by fiscal sustainability motives, which account for the large majority of actions—particularly for spending-based consolidations—while other motivations, such as growth and redistribution, play more limited roles.

Stylized fact 4—Intensity. *Fiscal actions are typically small, especially tax measures* (Figure 8 and 9). Comparing the average intensity of fiscal measures across income groups, distinguishing between tax and spending actions and between expansionary measures (from 0 to 10) and contractionary measures (from -10 to 0), spending actions generally exhibit larger average magnitudes than tax measures (Figure 8). This suggests that changes in public spending account for a disproportionate share of major fiscal policy interventions. Figure 9 further shows that large fiscal actions are rare, particularly on the tax side. Overall, the evidence suggests that fiscal policy adjustments are typically implemented through a series of relatively small measures rather than a few large interventions.

Stylized fact 5—Synchronization of fiscal measures. *Fiscal measures are generally not synchronized across countries, except during major recession-driven fiscal expansions.* To measure cross-country synchronization, we follow Kalemli-Ozcan, Papaioannou, and Peydro (2013), who use a similar approach to assess business cycle synchronization. Specifically, we quantify synchronization in fiscal actions between country *i* and country *j* at time *t* as:

$$synchronization\ score(t) = \frac{1}{N}\ \sum_{i,j} - |fa_{i,t} - fa_{j,t}| \qquad (1)$$

where $fa_{i,t}$ equals -1, 0, and 1 for contractionary, neutral, and expansionary fiscal actions, respectively. A higher synchronization score indicates a greater degree of synchronization in fiscal actions over time, with 0 denoting perfect synchronization.

Figure 10 reports the average synchronization of fiscal actions across income groups. It shows that fiscal actions are typically weakly correlated across countries, with only modest differences across income groups. However, a pronounced divergence emerges during the COVID-19 episode: fiscal actions in AEs become sharply more synchronized, driven by a common shift toward expansionary policies, while synchronization among EMs and LICs increases more modestly. This likely reflects differences in fiscal space, with AEs better positioned to implement large and coordinated fiscal expansions.

Overall, the limited cross-country correlation in fiscal actions contrasts sharply with the evidence of synchronization in monetary policy actions, especially across advanced economies (Forbes, Ha, and Kose, 2024). This difference likely reflects the fact that monetary policy actions are primarily implemented for stabilization purposes and therefore tend to respond more systematically to the global business cycle, whereas fiscal actions are driven by a broader range of objectives.

V. Validation

In this section, we validate the AI-assisted narrative fiscal database along several complementary dimensions, including its consistency with expert judgment, its alignment with established approaches in the fiscal policy literature, its empirical relationship with macro-fiscal aggregates, and its reproducibility. First, we compare the AI-generated classifications with independent assessments produced by economists using the same source reports. Second, we benchmark the database against existing narrative identification datasets. Third, we examine whether the resulting narrative indicators are systematically associated with subsequent movements in macro-fiscal aggregates. Finally, we assess the reproducibility of the AI classifications by repeatedly processing the same reports under identical prompts. Taken together, these exercises provide evidence on the consistency, interpretability, reproducibility, and empirical relevance of the AI-based classification framework.

Comparison with Economists' Output

The first validation exercise compares the AI-generated classifications against economists' assessments for a subset of EIU reports. We randomly selected a subset of 31 EIU reports to ensure broad coverage across regions and fiscal action motivations, while also allowing comparison with the literature (e.g., Romer and Romer 2019; Adler et al. 2024). Each report was independently reviewed by two economists, who answered a set of questions aligned with the AI prompt and provided short narrative assessments of current fiscal actions.

Overall, economists' assessments are broadly aligned with the AI output. The model reliably identifies relevant excerpts, correctly classifies the timing of fiscal actions, and captures key characteristics such as motivation and direction. Differences arise in a limited number of cases, typically reflecting ambiguity or lack of clarity in the underlying reports. Notably, these same cases often generate disagreement across economists, suggesting that discrepancies are driven more by inherent data limitations than by model performance.

To assess reliability and consistency more formally, we use three complementary approaches:

First, for numerical variables, we assess consistency using correlation coefficients. This applies to the intensity scores of fiscal actions, which correspond to the expansionary/contractionary impacts reported in Table 2 (Panel B). These scores capture the magnitude of fiscal effects on a standardized scale from −10 to +10, where negative (positive) values indicate contractionary (expansionary) impacts, and correlations measure how closely the AI's quantitative assessments align with those of economists.

Second, for categorical variables, we measure agreement as the share of identical classifications across two sets of results. This applies to variables such as the classification of expansionary/contractionary impacts (overall balance, tax, and spending) reported in Table 2 (Panel A), as well as implementation status (Panel D). Formally, for K observations and two sets of results n and m, agreement is defined as:

$$\frac{1}{\mathrm{K}}\sum_{k=1} I[X_k^n = X_k^m] \qquad (2)$$

Third, for multi-category variables, we assess overlap using a normalized intersection-over-union metric. This applies to the motivation categories reported in Table 2 (Panel C), where each observation X_k^n is a set of categories, restricted here to the seven high-level motivation groups. For example, an observation may include both "Redistribution" and "Politically motivated." The overlap measure captures the extent to which the sets of motivations identified across two results coincide and is defined as:

$$\frac{1}{\mathrm{K}}\sum_{k=1} \frac{number\ of\ common\ categories\ in\ X_k^n and\ X_k^m}{total\ number\ of\ distinct\ categories\ in\ \cup\{X_k^n, X_k^m\}} \qquad (3)$$

where the numerator denotes the number of common categories and the denominator the total number of distinct categories across the two sets.

Applying these approaches, we find consistently high levels of agreement across methods. For categorical measures (Panel A), agreement rates for expansionary/contractionary impacts—including overall balance, tax, and spending—exceed 90 percent in most cases, indicating that the AI closely replicates economists' classification of fiscal actions. Similarly, agreement on implementation status (Panel D) reaches about 82 percent, broadly in line with cross-economist comparisons.

For numerical measures (Panel B), correlations between AI-generated and economist-assessed expansionary/contractionary intensity scores range from 0.71 to 0.78, depending on whether all actions or only revenue- or spending-related measures are considered. These correlations are comparable to those observed across different economists, indicating a similar degree of consistency in assessing the magnitude of fiscal actions.

For multi-category variables (Panel C), the average overlap in motivation categories is about 75 percent, with at least one common category identified in most cases. Differences typically reflect the AI assigning a slightly larger set of motivations, rather than systematic misclassification.

Taken together, these results indicate that the AI model performs at a level comparable to human assessors across key dimensions—direction, magnitude, implementation, and motivation—with remaining differences largely attributable to ambiguity in source material rather than systematic biases in the models.

Comparison with Existing Fiscal Narrative Datasets

The existing literature using the narrative approach to identify fiscal actions can be organized into two broad groups: (i) crisis- and event-based multi-country datasets; and (ii) multi-country, fiscal consolidation-specific datasets. While these approaches share a common objective—identifying discretionary fiscal actions that are exogenous—they differ substantially in country and time coverage, frequency, coverage of fiscal direction (expansions and contractions) and fiscal instruments (tax and spending), and how policy motivation is incorporated into the identification strategy.

First group—Crisis- and Event-Based Multi-Country Datasets: Romer and Romer (2019) examine fiscal policy responses following financial crises in 30 OECD economies. Using EIU country reports, they classify fiscal measures by direction, size, and motivation, focusing on crisis episodes rather than constructing a continuous series of exogenous fiscal shocks. Although this approach provides rich qualitative insights into fiscal decision-making during crises, it is event-based, confined to crisis periods, and does not systematically quantify fiscal actions outside those episodes.

Second group—Multi-Country Fiscal Consolidation Datasets: This group applies the narrative approach to action-based fiscal consolidation datasets. Guajardo, Leigh, and Pescatori (2014) identify tax increases and spending cuts motivated by deficit reduction across 17 OECD economies over 1978–2009; Carrière-Swallow, David, and Leigh (2020) develop a comparable dataset for 14 Latin American and Caribbean countries over 1989–2016; Adler et al. (2024) extend coverage for OECD and Latin American and Caribbean economies through 2020; and Abdel-Latif et al. (2026) construct a similar dataset for 14 Sub-Saharan African countries over 1990–2024, using IMF staff reports and AI-assisted coding. In these studies, motivation plays a restrictive role: only measures aimed at deficit reduction or fiscal sustainability are included, while short-run stabilization measures are excluded. Exogeneity is therefore defined by deficit-reduction intent rather than independence from current macroeconomic conditions. This design provides asymmetric coverage, focusing on contractionary episodes and omitting expansionary fiscal actions.[3]

Table 3 summarizes the key design features of these approaches. As the table makes clear, despite substantial progress in narrative fiscal identification, no existing dataset simultaneously achieves broad country coverage, high frequency, symmetric treatment of expansionary and contractionary actions, and comprehensive inclusion of both tax and spending measures. Our dataset extends the literature along these dimensions by providing quarterly coverage for 142 countries and systematically coding both tax and spending measures in both directions. To assess the credibility of this broader framework, we next compare our dataset with existing narrative datasets.

We conduct two complementary validation exercises. First, for crisis-based studies, we evaluate agreement in direction, size, and motivation during documented crisis episodes. Second, for multi-country consolidation

[3] A related but methodologically distinct study employs automated textual analysis to construct measures of fiscal policy–related sentiment from parliamentary debates and policy documents. Latifi et al. (2024) develop sentiment measures that distinguish between exogenous and endogenous fiscal policy language. While such approaches reduce reliance on manual coding and can be implemented at high frequency, they capture the tone and content of fiscal discussions rather than coding discrete fiscal actions with direction and magnitude at the episode level. As a result, sentiment-based measures are complementary to, but not directly comparable with, action-based narrative identification datasets.

datasets, we assess whether our quarterly data detects annual consolidation episodes identified in prior work and whether episodes identified as large consolidations in prior work are also classified as large in our dataset. We also examine country-years in which our dataset identifies large consolidation episodes that are not recorded in the benchmark consolidation datasets. Together, these exercises benchmark our dataset against established approaches while accounting for differences in design and scope.

Validation Against Crisis-Based Narrative Identification. For crisis-based narrative identification, we benchmark our results against the fiscal expansion and contraction episodes reported in Table 3 of Romer and Romer (2019). Because their dataset focuses on fiscal actions taken during documented financial crises rather than constructing a continuous time-series of shocks, validation is conducted at the episode level. In Figure 11, we compare our classifications along three dimensions: (i) direction, combining expansionary and contractionary episodes; (ii) size, focusing on large fiscal actions; and (iii) motivation, focusing on episodes classified as exogenous in their study. Consistent with the next validation exercise, we compute detection rates defined as the share of episodes identified in Romer and Romer (2019) that are also detected in our dataset along each dimension. The detection rate is 95 percent (35 out of 37 episodes) for direction, 89 percent (17 out of 19 episodes) for large fiscal actions, and 90 percent (43 out of 48 episodes) for exogenous motivations. The close alignment indicates that our dataset closely mirrors the directional stance, magnitude classification, and motivational coding of fiscal actions documented in Romer and Romer (2019), despite differences in report selection and classification design (see Annex D.1 and Tables D.1.1 and D.1.2 for details).

Validation Against Multi-Country Consolidation Datasets. To validate our dataset against multi-country consolidation studies, we benchmark our quarterly dataset against annual consolidation episodes reported in prior work. For each fiscal consolidation dataset, we compute a detection rate defined as the share of consolidation episodes identified in that study that are also detected in our data (see Annex D.2 for the country–year comparisons underlying these detection rates). A consolidation year is considered detected if at least one quarter within that year records a contractionary fiscal action motivated by deficit reduction or fiscal discipline. Figure 12 reports detection rates for direction (fiscal consolidation for all episodes). We find high detection across datasets: 95 percent for Guajardo, Leigh, and Pescatori (2014), 99 percent for Carrière-Swallow, David, and Leigh (2020), and 94 percent for Abdel-Latif et al. (2026).

We next assess agreement in the identification of non-large as well as large fiscal consolidation episodes. We define small fiscal consolidation as less than or equal to 2 percent of GDP, and a large consolidation as exceeding 2 percent of GDP, consistent with magnitude thresholds commonly used in the fiscal consolidation literature (Abdel-Latif et al., 2026), and compute, for each study, the share of externally identified large consolidation episodes that are also classified as large in our data. As shown in Figure 13 (Panel A), detection rates for non-large fiscal consolidation episodes remain high for Guajardo, Leigh, and Pescatori (2014) (95 percent), Carrière-Swallow, David, and Leigh (2020) (99 percent), and Abdel-Latif et al. (2026) (94 percent). Meanwhile, Figure 13 (Panel B) shows that detection rates for large fiscal consolidation episodes also remain high for Guajardo, Leigh, and Pescatori (2014), Carrière-Swallow, David, and Leigh (2020), and Abdel-Latif et al. (2026)—all at 100 percent. Overall, these results indicate that our dataset closely matches existing narrative consolidation datasets in the direction of identified episodes as well as in the magnitude.

As an additional reverse comparison, we also examine country-years in which our dataset identifies large consolidation episodes that are not recorded in the benchmark consolidation datasets. For this exercise, we define a large consolidation year in our dataset as a year with at least two quarters satisfying the following criteria: the timing scope is Current, the fiscal action is contractionary, the motivation includes deficit reduction

or fiscal discipline, and the size of the fiscal action is classified as large. The resulting country-year lists are reported in Table D.2.4 and should be interpreted as additional coverage relative to the benchmark datasets, rather than as discrepancies, given differences in source material, frequency, and coding design.

Comparison with Fiscal Data

We complement the LLM-generated dataset with several additional variables to study the relationship between fiscal actions and country-specific and macroeconomic characteristics. We collect quarterly data on key fiscal variables such as government revenues, expenditures, deficits, and debt from Haver and the EIU. In addition, we retrieve quarterly tax revenue data from the IMF's Quarterly Government Finance Statistics database, and include nominal and real GDP data from the IMF's National Economic Accounts database. For variables where countries do not report seasonally adjusted values, we perform seasonal adjustment using the US Census Bureau's X-13 ARIMA-SEATS program (Sax and Eddelbuettel 2018). To improve data coverage and check the consistency of the quarterly data, we also include annual data for these fiscal and macroeconomic variables from the October 2025 vintage of the IMF's World Economic Outlook (WEO) database. We also collect income group classifications from the WEO and consider the following three income groups in subsequent sections: AEs, EMs, and LICs. The resulting dataset has substantial coverage across countries and is, to the best of our knowledge, the first quarterly dataset of major fiscal aggregates for most countries in the world. Annex E provides more details on the list of variables and data availability across countries.

To benchmark our narrative fiscal indicators against more standard macro-fiscal outcomes, we estimate panel local projections using quarterly country-level data. For this analysis, we focus only on the assessment of the AI regarding the current quarter and disregard the past and future actions.

To map the qualitative narrative assessments into a quantitative measure, the AI-generated classifications of fiscal actions are converted into a discrete indicator. Specifically, each identified action is coded as $+1$ if the effect is classified as "positive," -1 if "negative," and 0 if there is no discernible effect or the action is classified as unclear. This transformation yields an indicator variable capturing the direction of the fiscal action. Separate indicators are constructed for expenditure and tax actions, corresponding to the respective narrative classifications. These variables are then used as the $\text{Action}_{i,t}^{Y}$ series in the empirical analysis. Specifically, for each horizon h, we regress the log difference of the outcome variable on the contemporaneous fiscal action, controlling for lags of the action, lagged macro controls, and country fixed effects. In the quarterly specifications we also include quarter dummies, and standard errors are clustered at the country level.

Formally, for each horizon $h = 0, \ldots, 4$, we estimate

$$\Delta_h \log(Y_{i,t}) = \beta_h \, \text{Action}_{i,t}^{Y} + \sum_{k=0}^{4} \gamma_{h,k} \, \Delta \log(RGDP_{i,t-k}) + \sum_{k=0}^{4} \delta_{h,k} \, \text{Action}_{i,t-k}^{Y} + \alpha_i + \lambda_q + \varepsilon_{i,t+h}, \qquad (4)$$

where

$$\Delta_h \log(Y_{i,t}) = \log(Y_{i,t+h}) - \log(Y_{i,t-1}) \qquad (5)$$

$Y_{i,t}$ denotes the fiscal outcome of interest—either government expenditure or government revenue, $RGDP_{i,t}$ refers to real GDP, $\text{Action}_{i,t}^{Y}$ corresponds to the associated narrative fiscal action (expenditure or tax, respectively), α_i are country fixed effects, and λ_q are quarter dummies. Standard errors are clustered at the country level.

Figure 14 presents the cumulative responses of government spending and revenue to the identified fiscal actions. The left panel shows the cumulative response of government expenditure following an expenditure action, while the right panel shows the cumulative response of government revenue following a tax action. For expenditure actions, government spending increases on impact and continues to increase over the one-year horizon. For tax actions, the cumulative tax-revenue response is initially close to zero, rises through approximately horizon 2, and subsequently declines. Overall, the results indicate that the narrative indicators are associated with subsequent movements in the corresponding fiscal aggregates, with a more immediate and sustained response for spending than for revenues.[4]

Reproducibility

To assess reproducibility—that is, the extent to which the AI model generates the same classification when the same report and prompt are processed repeatedly—we ran the prompt on the set of 31 EIU country reports used in the earlier validation exercises. The extracted text of each report was processed 100 times using OpenAI GPT-5.0 API, yielding a total of 3,100 report–run observations. The prompt concentrates on six core dimensions of the fiscal action: (i) detection; (ii) the fiscal instrument(s) (tax and/or spending), (iii) policy direction, (iv) qualitative size; (v) its quantitative intensity score, and (vi) motivation(s).

We measure reproducibility for all dimensions—except motivation—using the modal share, calculated for each report as the fraction of runs in which the most frequently assigned value appears. A value of one indicates perfect reproducibility, meaning that the same output is returned in every run. Lower values indicate greater variation across runs. For example, consider a "master" run in the main database that identifies a fiscal action in a given quarter. If 94 of the 99 replicated report-runs also identify a fiscal action in that quarter, the modal share would be 0.95. We summarize the distribution of modal shares across the 31 reports by reporting the average modal share and other key moments such as the median, the minimum and the 25th and 75th percentile.

We assess motivation separately because the model may assign multiple motivations to a single fiscal action, ordered by relevance. Focusing on the seven high-level motivation categories, we evaluate reproducibility using three complementary measures. First, we calculate the modal share of the primary motivation assigned in each run. Second, we assess whether pairs of runs share at least one motivation category. Third, we compute the mean pairwise Jaccard similarity—defined as the ratio of overlapping motivation categories to the total number of distinct categories assigned across the two runs—across all unique run pairs for each report.

We first assess the reproducibility results for two key variables in the database: whether a fiscal action is detected and whether the qualitative size of the action is classified as large or not, across the 31 reports. These two variables underpin several applications in the paper. To compute the modal share, both variables are converted into binary indicators equal to one if a fiscal action is identified and if its qualitative size is classified as large, respectively, and zero otherwise. The results suggest high replicability. Across the 31 EIU reports, the average modal share is 0.95 for fiscal action detection and 0.90 for the identification of large fiscal actions (Figure 15), indicating that these key classifications are remarkably stable across repeated AI extractions.

[4] These results are robust to the use of the intensity variables instead of the indicator variables.

The remaining dimensions also indicate a high degree of reproducibility. Instrument type exhibits modal shares concentrated near one, indicating near-perfect consistency across runs. Policy direction is also highly reproducible, with an average modal share of 0.90 across reports. Regarding motivation, the broadest criterion—whether two AI extractions share at least one motivation category—exhibits the agreement share of 0.98 (Table 4). Agreement is slightly lower when assessed using exact agreement on the primary motivation or the Jaccard score, which compares the complete set of assigned motivations. This pattern reflects the fact that many fiscal actions plausibly pursue multiple policy objectives. As a result, repeated runs may identify the same broad motivations while differing in the ordering of motivations or the inclusion of secondary motivations, rather than indicating instability in the model's classifications (Figure 16).

We further explore the model's reproducibility among a subset of the 31 reports identified as having a "large" fiscal action in either of the economists' independent reviews or if the absolute value of the intensity score estimated by AI is greater than or equal to seven. Reproducibility improves across nearly all variables in this subset (Table 5). Fiscal action detection and motivation (measured by any overlap) exhibit perfect agreement, and instrument type exhibits an average modal share of 0.98 across the reports. Qualitative size classification and policy direction both exceed 0.90 (Figure 18). Exact agreement on the primary motivation also remains high, and the Jaccard similarity index increases relative to the full sample (Figure 19).

We assess the reproducibility of intensity by examining the dispersion of intensity scores, recognizing that the model selects from a range of possible integer values ([-10, +10]). The broad set of possible values makes assessing reproducibility by modal share subject to overrepresentation of disagreement, even if marginal. We examine dispersion by calculating interquartile range, which captures the spread of the middle 50 percent of scores, and the mean absolute deviation (MAD), which captures the deviation of individual report-run intensity scores from the report-level modal intensity score. Across the 30 reports for which an intensity score is assigned, both measures indicate limited variation. Notably, the MAD dispersion score of 0.62 implies that the typical deviation from the modal assessment is less than one point, despite the model selecting from a 21-point scale (Figure 17). The IQR dispersion improves from 0.97 to 0.79 when we consider the subset of reports with "large" fiscal action (Figure 20). This suggests that repeated runs generally produce highly similar intensity assessments even when exact agreement is not achieved.

Overall, the results suggest that the AI-based classification framework is highly reproducible, especially for the dimensions most central to the construction of the narrative fiscal actions database—such as fiscal action detection, qualitative size, instrument type, and policy direction. Reproducibility is even stronger for large fiscal actions.

VI. Applications

Cyclicality of Fiscal Actions

A large body of literature has examined the conduct of discretionary fiscal policy over the business cycle and how fiscal cyclicality—in both spending and taxation—varies across countries (e.g., Végh and Vuletin 2015) and institutional settings (e.g., Frankel, Végh, and Vuletin 2013). While the cyclicality of the overall fiscal stance is the relevant metric for assessing whether fiscal policy amplifies or dampens business cycles, understanding the cyclical behavior of discretionary fiscal actions remains important, as they reflect policy choices and institutional constraints that may shape overall fiscal outcomes. We contribute to this literature by providing a

direct measure of discretionary fiscal actions, rather than inferring discretion from the behavior of government spending, tax revenues, or fiscal balances over the business cycle, which only imperfectly captures the degree of discretion in fiscal policy actions.

The analysis first estimates fiscal cyclicality separately for each country and fiscal instrument—overall fiscal stance, spending actions, and tax actions—and then examines how these measures vary both across countries and over time, and relate to institutional quality. Specifically, for each country i, fiscal instrument $j \in$ {net,spending,tax}, and year t, the degree of fiscal cyclicality is estimated as:

$$FA_{i,t}^{j} = \alpha_i^{j} + \beta_i^{j} X_{i,t} + \varepsilon_{i,t}^{j}, \qquad (6)$$

where $FA_{i,t}^{j}$ the annual intensity of fiscal actions, constructed as the sum of the four quarterly fiscal action scores, each ranging from −10 (strongly contractionary) to +10 (strongly expansionary), and therefore measured on a standardized annual scale from −40 to +40. $X_{i,t}$is real GDP growth. The coefficient β_i^{j}captures the cyclicality of fiscal policy in country i. A positive coefficient indicates procyclical policy, whereby stronger economic conditions are associated with more expansionary fiscal action intensity; conversely, a negative coefficient indicates countercyclical policy, whereby fiscal actions help stabilize output.

Figure 21 reports the coefficients estimated from Equation (6) over the full 1990–2024 period. Several findings emerge. First, fiscal cyclicality varies substantially across countries, with coefficients ranging from strongly procyclical values of about 1.5 to strongly countercyclical values of around −1.7. Second, consistent with much of the previous literature, fiscal policy is typically more countercyclical in AEs than in EMs and LICs: many EMs and LICs display positive cyclicality coefficients—indicating more expansionary fiscal actions during periods of stronger growth and more contractionary actions during downturns—whereas AEs are more concentrated around acyclical or mildly countercyclical behavior. At the same time, a non-negligible number of EMs and LICs exhibit coefficients close to zero or negative values, suggesting that some countries in these groups conduct countercyclical fiscal policies. Third, in contrast to some earlier findings, countries that exhibit countercyclical spending do not necessarily pursue countercyclical tax policy, as shown by the non-negligible number of advanced economies for which tax policy is procyclical. Moreover, spending actions exhibit considerably wider variation in cyclicality coefficients than tax actions, suggesting that cross-country differences in fiscal cyclicality are driven primarily by the expenditure side.

Next, we examine how cyclicality has evolved over time by comparing the distribution of country-specific coefficients across two subperiods: 1990–2014 and 2015–2024. This split is motivated by major changes in the global fiscal policy environment following the Global Financial Crisis and the European sovereign debt crisis, which intensified debates on fiscal austerity, fiscal space, and state-dependent multipliers. The boxplots in Figure 22 reveal several notable shifts. Most importantly, they point to a broad, albeit incomplete, move toward less procyclical fiscal policy. In the later period, the distribution of coefficients shifts downward, with a larger share of countries displaying coefficients close to zero or negative values. This pattern is consistent with the notion of “graduation” from fiscal procyclicality documented in earlier work, although the transition remains uneven and less complete than previously documented. The improvement is also not uniform across country groups. AEs tend to display coefficients clustered around zero or mildly countercyclical values in both periods, reflecting relatively stable policy frameworks. By contrast, EMs and LICs continue to exhibit wider dispersion, with some countries moving toward countercyclicality while others remain strongly procyclical. Finally,

dispersion in cyclicality within country groups has declined. The interquartile range narrows in the later period for several country groups, suggesting more consistent policy behavior.

Finally, we examine how institutional quality shapes the degree of fiscal countercyclicality across countries. We begin by plotting in Figure 23 the country-specific cyclicality coefficients reported in Figure 21 against a measure of institutional quality—the principal component of bureaucracy quality, corruption, financial risk, and economic risk ratings from ICRG. The results corroborate earlier findings that stronger institutional quality is generally associated with lower fiscal procyclicality. At the same time, while this relationship is evident for the overall fiscal stance and spending actions, it appears substantially weaker for tax actions, for which cyclicality coefficients remain clustered around zero across different levels of institutional quality.

Next, to better isolate and quantify the role of institutional quality, we estimate the following panel specification with country and year fixed effects:

$$FA_{i,t}^{j} = \alpha_i + \lambda_t + \beta_1 X_{i,t} + \beta_2 IQ_{i,t-1} + \beta_3\left(X_{i,t} \times IQ_{i,t-1}\right) + \varepsilon_{i,t}, \quad (7)$$

where $FA_{i,t}^{j}$denotes the intensity of fiscal actions for instrument $j \in$ {net,tax,spending}, $Growth_{i,t}$is real GDP growth, and $IQ_{i,t-1}$is the lagged indicator of institutional quality. The coefficient $\beta_3$captures how cyclicality varies systematically with institutional quality. Figure 24 presents the correlation between GDP growth and the intensity of fiscal actions at different levels of institutional quality, as derived from the panel interaction specification. The estimation results confirm that, for both the overall fiscal stance and spending actions, the correlation with GDP growth declines as institutional quality improves. In countries with weaker institutions, fiscal policy is strongly procyclical, with stronger growth associated with more expansionary fiscal action intensity. By contrast, in countries with stronger institutions, this relationship weakens and approaches acyclicality.

Taken together, the results presented in this section reinforce and extend the existing literature in three important ways. First, by using narrative data, they provide direct evidence that fiscal procyclicality reflects deliberate policy actions rather than mechanical responses of fiscal aggregates to the business cycle. This helps address a key limitation of earlier approaches based on the cyclical components of revenues and expenditures, which are inherently imperfect measures of discretionary fiscal policy. Second, in contrast to some earlier findings, countries that conduct countercyclical spending policy do not necessarily pursue countercyclical tax policy. Third, while stronger institutional quality is associated with weaker procyclicality in spending, it is not systematically related to the cyclicality of tax policy actions.

Country-specific Fiscal Multipliers

This section estimates country-specific spending and tax multipliers for a large set of AEs, EMs, and LICs, and examines how these multipliers vary within and across income groups.

Empirical Framework

To estimate country-specific multipliers, we focus on fiscal actions that are not motivated by stabilization objectives, such as financial rescues or responses to the economic cycle and commodity-price fluctuations, which may be endogenous to current or expected conditions.

Multipliers are estimated using a country-by-country local projections (LP) framework (Jordà 2005), implemented with B-spline smoothing (Barnichon and Brownlees 2019). For each country i and horizon $h = 0, 1, ..., H$ (with H = 8 quarters). Specifically, we estimate separate local projections for GDP and the fiscal variable of interest—that is, spending or taxes—as follows:

$$y_{i,t+h} = \alpha_{i,h} + \beta_{i,h} \cdot z_{i,t} + \boldsymbol{\Gamma}_h \boldsymbol{X}_{i,t-1} + \varepsilon_{i,t+h} \qquad (8)$$
$$f_{i,t+h} = \alpha_{i,h} + \beta_{i,h} \cdot z_{i,t} + \boldsymbol{\Gamma}_h \boldsymbol{X}_{i,t-1} + \varepsilon_{i,t+h}$$

where y denotes trend-normalized real GDP[5]; f is the fiscal variable of interest - government expenditure or revenue – normalized by the same trend GDP series; $z_{i,t} \in \{-1,0,+1\}$ is the narrative shock series derived from the GFA database; $\mathbf{X}_{i,t-1}$ is a vector of $p = 6$ lags of normalized GDP, normalized fiscal variable, and the shock series itself, and $\alpha_{i,h}$ is a country-horizon-specific intercept.

For each country *i*, the system of LP equations is estimated separately using penalized OLS with cubic B-spline smoothing of the shock-response path and time-clustered robust standard errors (clustered within country). Inference is frequentist and based on draws from the estimated asymptotic (normal) distribution of the smoothed LP coefficients. Draws are retained subject to admissibility conditions requiring positive cumulative fiscal and output responses and bounded multipliers at selected horizons. The cumulative multiplier at horizon *h* is:

$$\mathcal{M}(h) = \frac{\sum_{h=0}^{H} \widehat{\mathrm{IRF}}_y(h)}{\sum_{h=0}^{H} \widehat{\mathrm{IRF}}_f(h)} \qquad (9)$$

where $\widehat{\mathrm{IRF}}_{\mathrm{y}}(\mathrm{h})$ and $\widehat{\mathrm{IRF}}_f(h)$ are the impulse responses of the normalized output and fiscal variables respectively at horizon h.

This empirical framework is applied separately to spending and revenue fiscal multipliers. For spending shocks, the fiscal variable of interest is real government spending; for tax policy shocks, the model uses general government revenues as the propagating channel. Estimates are based on quarterly data from the early 1990s through the end of 2023.

To compute average multipliers for the full sample and for each income group, we aggregate country-level estimates using inverse-variance weighting (IVW), with weights based on country-specific standard errors obtained from the accepted coefficient-draw simulations. Specifically, the average full-sample multiplier is given by:

$$M^{IVW}(h) = \sum_i w_i(h) M_i(h) / \sum_i w_i(h) \qquad (10)$$

[5] Normalization was performed by dividing it by country-specific trend real GDP (Ramey and Zubairy, 2018). The trend is estimated using a HP filter with a smoothing parameter λ=1,600.

where w_i is the country-specific weight computed in inverse proportion to the variance of the estimated coefficient. This approach assigns greater weight to countries where multipliers are more precisely estimated and lower weight to more imprecise estimates.

Spending and Tax multipliers

Figure 25 presents the pooled IVW impulse responses of output and spending to a spending shock for the full sample. A positive spending shock raises output, which peaks after about four quarters and remains positive thereafter (Panel A), while spending increases persistently and is precisely estimated (Panel B). The corresponding spending multiplier—computed as the ratio of the two impulse responses—is 0.57 on impact and 0.42 after one year (Panel C), within the range typically reported in the literature (Ramey 2019).

These average multipliers, however, mask substantial heterogeneity across countries, as shown in Panel D. The figure shows that the average and median multipliers within each income group are higher than the IVW average, suggesting that smaller multipliers tend to be estimated more precisely. In addition, although average and median multipliers are higher in AEs and LICs than in EMs, there are no systematic differences across income groups. More importantly, variation in estimated multipliers within each income group is larger than the differences across groups, suggesting that previous findings on multipliers for specific income groups may be less informative than commonly assumed.

Tax multiplier estimates are mostly limited to advanced economies and emerging markets because quarterly tax revenue data are scarce for low-income countries.[6] Figure 26 presents the impulse responses of output and tax revenue, along with the corresponding estimated multipliers. The results show that an expansionary tax shock is followed by a hump-shaped output response (Panel A) and persistent declines in tax revenue (Panel B). At the one-year horizon, the pooled IVW revenue multiplier is 0.83 (Panel C), significantly higher than the average spending multiplier of 0.42. As with spending multipliers, tax multipliers vary markedly both across and within income groups, with wide interquartile ranges and outliers.

Finally, we assess whether the estimated multipliers are sensitive to the size of fiscal shocks. Specifically, we re-estimate the multipliers using only fiscal actions classified by the AI as qualitatively large. Figure 27 compares these estimates with the baseline results. For spending shocks, the estimated multipliers on impact are larger than in the baseline and decline more rapidly to reach a similar order of magnitude at the end of the estimation horizon, suggesting that the baseline results are not driven by the inclusion of small fiscal actions. For tax shocks, the impact multiplier is considerably smaller than in the baseline at impact, while the one-year multiplier is broadly similar. Overall, the comparison suggests that the baseline findings are robust, although the dynamic responses differ somewhat for larger fiscal actions.[7]

[6] In Annex F, we repeat the analysis using a broader country sample based on overall government revenue, which includes tax revenue, grants, royalties, dividends, natural resource rents, and other resource-related receipts. The results are qualitatively similar: output exhibits a hump-shaped response, revenue declines persistently, and the cumulative multiplier is positive, peaking around the one-year horizon before gradually fading.

[7] We also test whether the estimated multipliers are sensitive to the number of lags included in the local projection regressions. Figure 28 compares the baseline specification, which includes six lags, with alternatives using four and eight lags. For government spending shocks, dynamics and multipliers remain broadly stable across specifications. For revenue shocks, the lag length affects the magnitude of the estimates more noticeably, with the baseline generally producing larger multipliers than with the eight-lag specification and smaller multipliers than in the four-lag specification. However, the responses and qualitative results remain similar.

VII. Conclusion

This paper introduces the AI-assisted Global Fiscal Actions Dataset (GFA), a novel quarterly database of discretionary fiscal policy actions covering 142 countries from 1990 onward. By combining a rich narrative source—the Economist Intelligence Unit country reports—with a transparent and reproducible Large Language Model (LLM)-based extraction framework, the dataset provides the first globally comparable, high-frequency record of discretionary tax and spending measures across advanced economies, emerging markets, and low-income countries. The paper documents the construction of the database, validates the methodology against human expert classifications and existing narrative datasets, and demonstrates its empirical relevance through applications to fiscal cyclicality and fiscal multiplier analysis. Overall, the results suggest that LLM-assisted approaches can generate reliable and scalable policy databases while preserving the richness of traditional narrative methods.

The analysis uncovers several important stylized facts about discretionary fiscal policymaking around the world. Fiscal actions are frequent but heterogeneous across countries and income groups, with spending measures generally more prevalent than tax measures. Expansionary policies occur more often than contractionary ones, while concerns related to fiscal sustainability, growth, and redistribution emerge as key policy motivations. The database also highlights substantial cross-country differences in the timing, composition, and intensity of fiscal actions, as well as spending and tax multipliers. These findings illustrate the value of moving beyond aggregate fiscal outcomes and focusing directly on the policy actions that shape them.

Beyond its immediate contribution to the fiscal policy literature, GFA can support a wide range of future empirical and policy-oriented analyses. The combination of broad country coverage, quarterly frequency, detailed policy classification, and information on motivations and implementation status creates opportunities that were previously unavailable to researchers and policy institutions. In particular, the dataset can be used to study how discretionary fiscal actions respond to economic conditions, institutional quality, political cycles, fiscal rules, and debt dynamics across different groups of countries; improve the estimation of fiscal multipliers by distinguishing between tax and spending measures, assessing state-dependent effects, and examining cross-country, sectoral, and macroeconomic heterogeneity for a large set of countries; analyze fiscal policy responses to major shocks, including financial crises, pandemics, climate-related disasters, geopolitical events, commodity-price fluctuations, and other external disturbances; assess whether measures motivated by growth, redistribution, fiscal sustainability, or stabilization objectives have different economic and fiscal outcomes; and support real-time policy surveillance and cross-country benchmarking by helping policy analysts track discretionary actions across jurisdictions, identify emerging policy trends, compare responses among peer countries, and strengthen evidence-based fiscal advice. Going forward, we plan to update the dataset regularly and make it available to researchers and policymakers.

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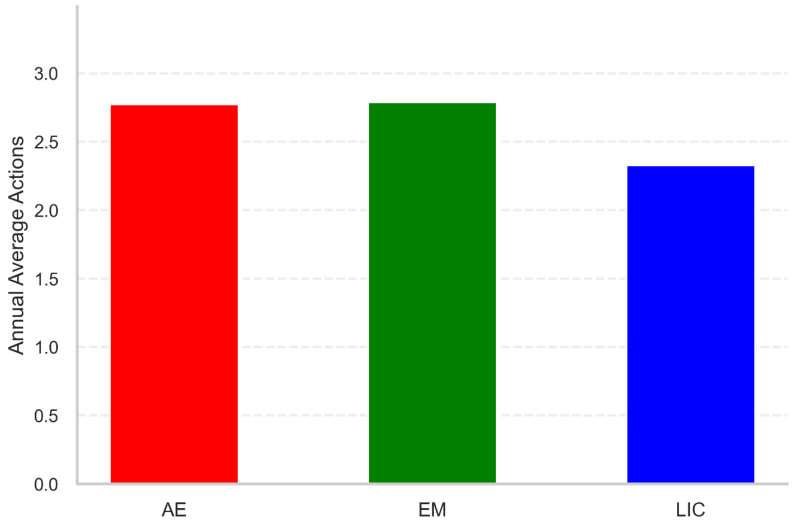
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Figures

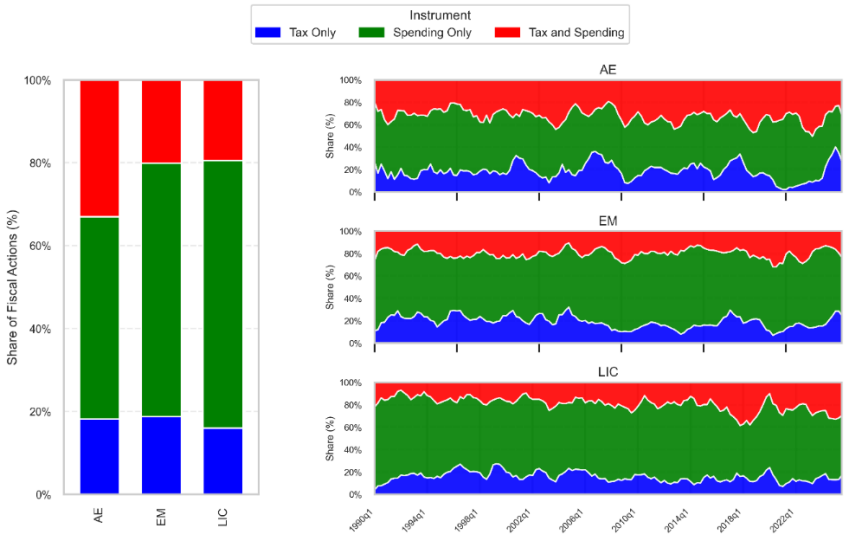
Figure 1. Average Annual Fiscal Actions per Country



Notes: The figure reports the average number of current fiscal actions per country over the sample period (1990 onward), separately for advanced economies (AE), emerging markets (EM), and low-income countries (LIC). The sample includes only observations identified as fiscal actions by the AI classification.

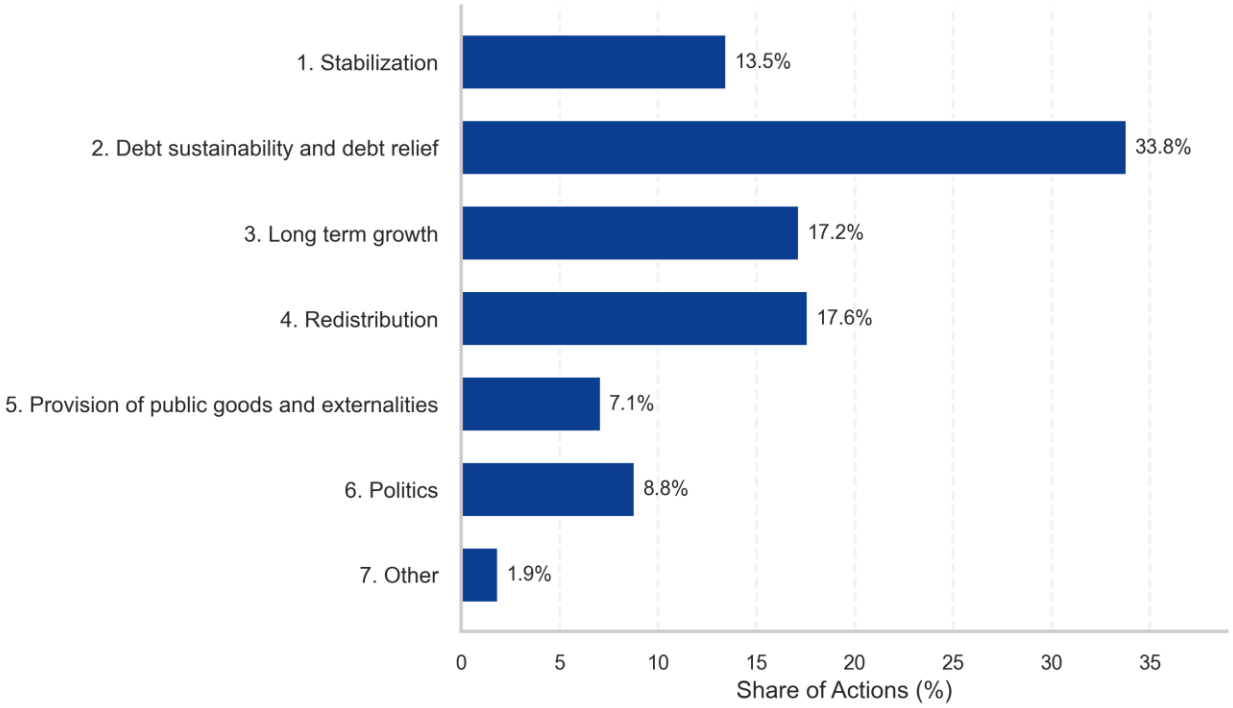
Figure 2. Fiscal Actions by Instrument by Income Group

Panel A. Full Sample Composition Panel B. Time Evolution



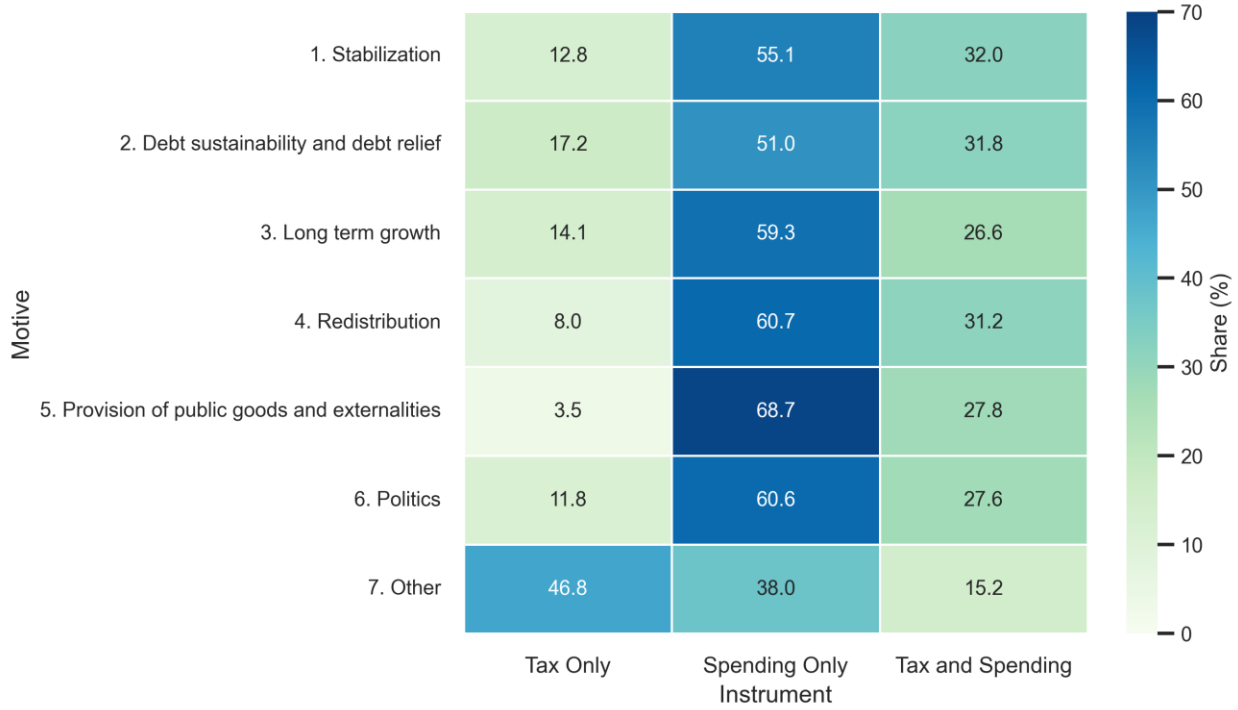
Notes: The left panel reports the distribution of fiscal actions across instrument types (tax-only, spending-only, and combined tax-and-spending measures) within each income group. Shares are expressed as percentages of all instrument-classified fiscal actions in each income group. The right panel reports the evolution of these shares over time using four-quarter moving averages. The sample includes current fiscal actions from 1990 onward.

Figure 3. Fiscal Actions by Motivation



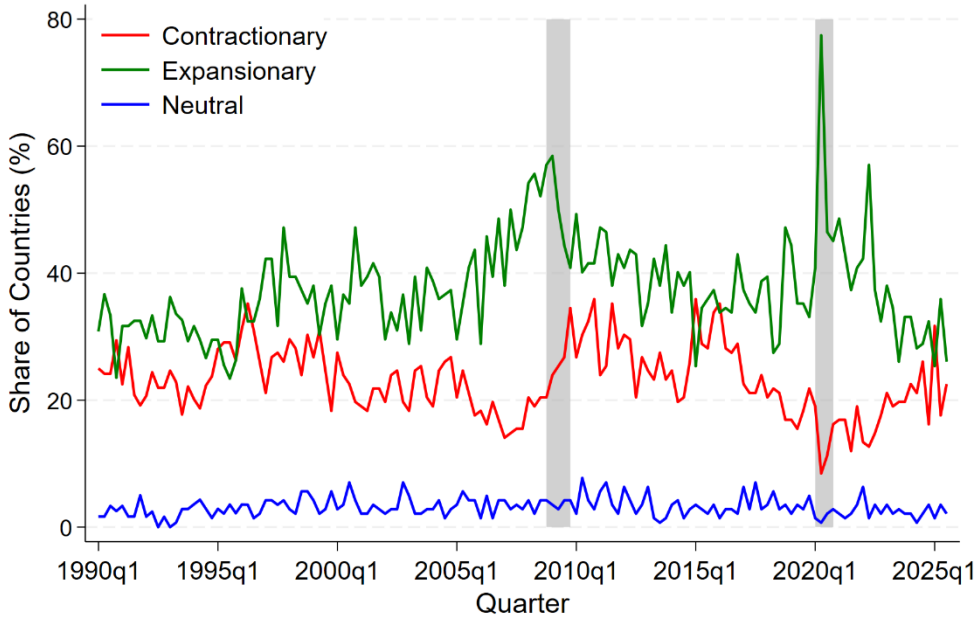
Notes: The figure reports the distribution of motivations across all current fiscal actions. Since a fiscal action may have multiple motivations, each identified motivation is counted separately. Percentages therefore sum the frequency of motivations, not fiscal actions.

Figure 4. Fiscal Actions by Motivation and Fiscal Instruments



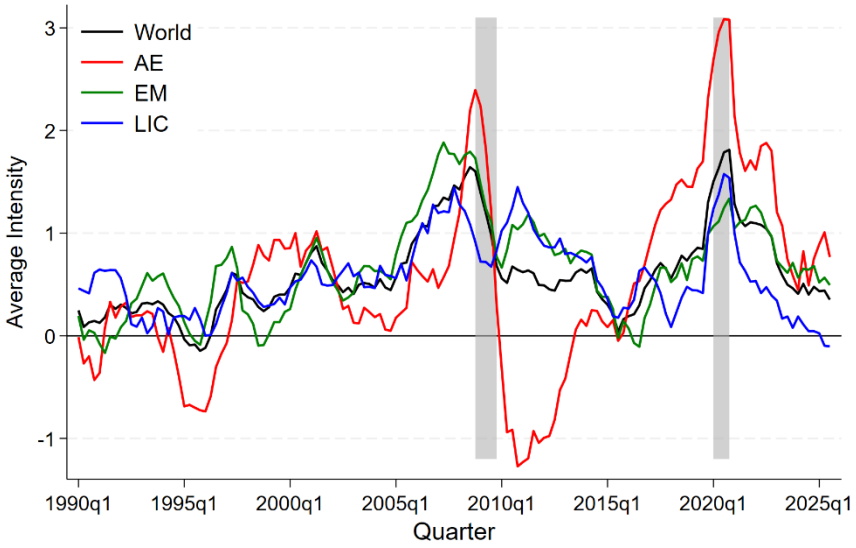
Notes: The heatmap reports the distribution of fiscal instruments within each motivation category. Each row sums to 100 percent and shows the share of actions implemented through tax-only measures, spending-only measures, or combined tax-and-spending measures among all fiscal actions associated with that motivation. Motivations are based on all motivations identified for each fiscal action.

Figure 5. Direction of Fiscal Actions Over Time Across Countries



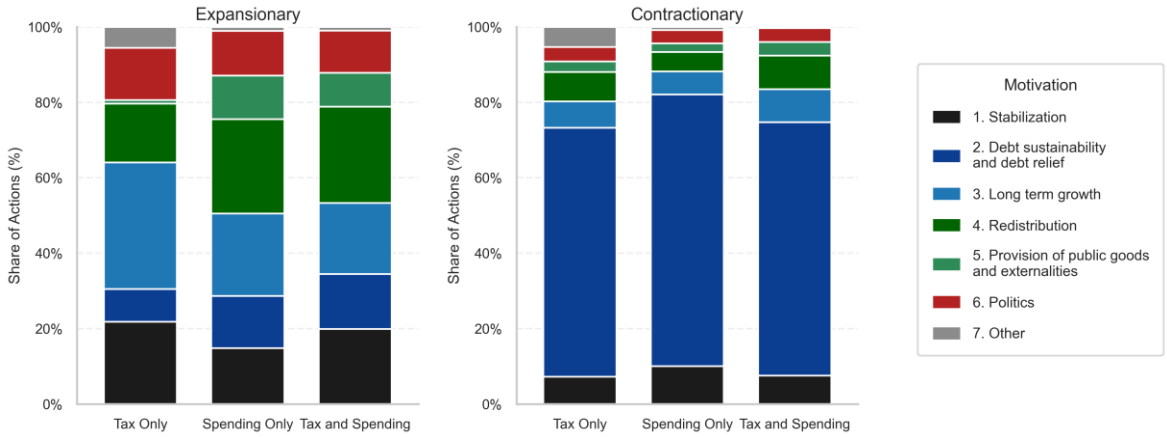
Notes: The figure plots the average fraction of countries undertaking contractionary, expansionary, and neutral fiscal actions over time. The denominator includes observations where no fiscal actions are undertaken. Shaded areas correspond to the period of the Global Financial Crisis (GFC, 2008Q4-2009Q4) and COVID-19 (2020Q1-2020Q4).

Figure 6. Direction of Fiscal Actions Over Time Across Countries



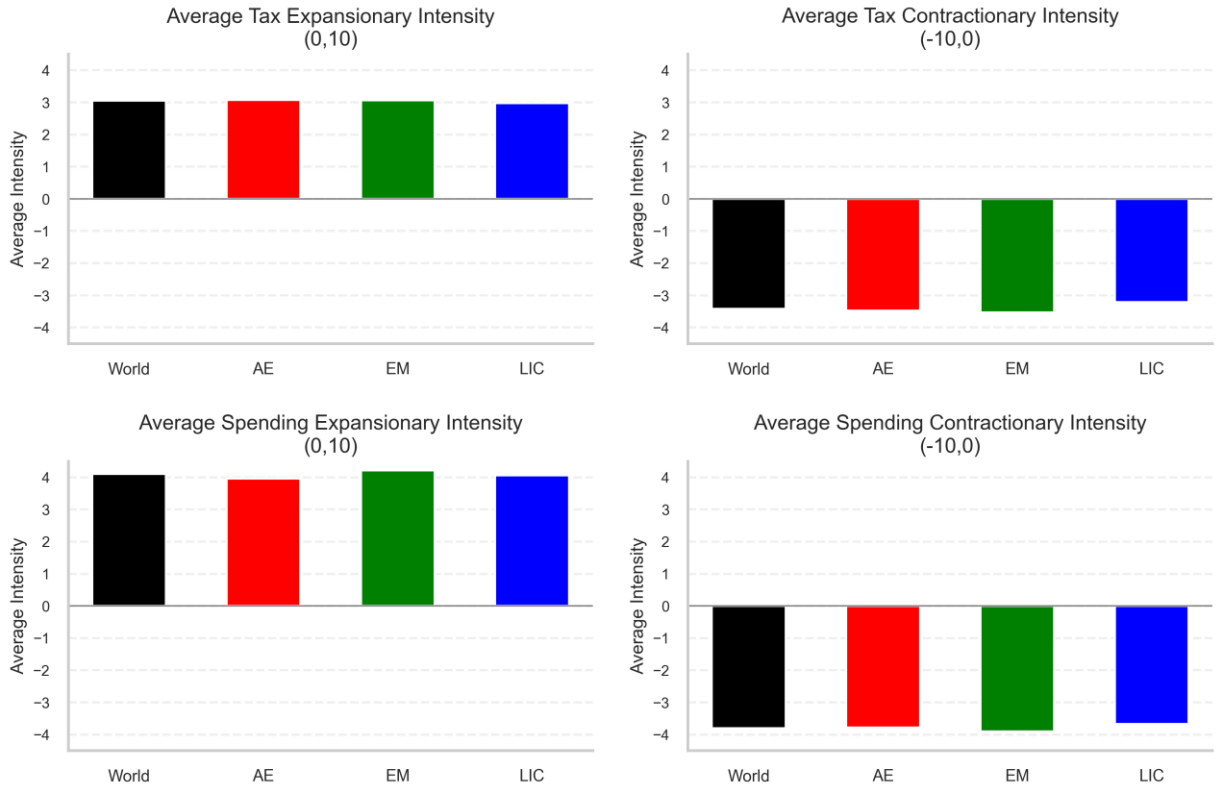
Notes: The figure plots the average qualitative intensity of fiscal measures over time by country income groups. The intensity variable ranges from -10 (for strongly contractionary actions) to +10 (for strongly expansionary actions). Results are reported for the full sample ("World") and separately for advanced economies (AE), emerging markets (EM), and low-income countries (LIC). Quarterly averages are plotted as a moving average across five quarters (two previous, one current, and two future quarters). Shaded areas correspond to the period of the Global Financial Crisis (GFC, 2008Q4-2009Q4) and COVID-19 (2020Q1-2020Q4).

Figure 7. Motivations across Fiscal Instruments by Direction



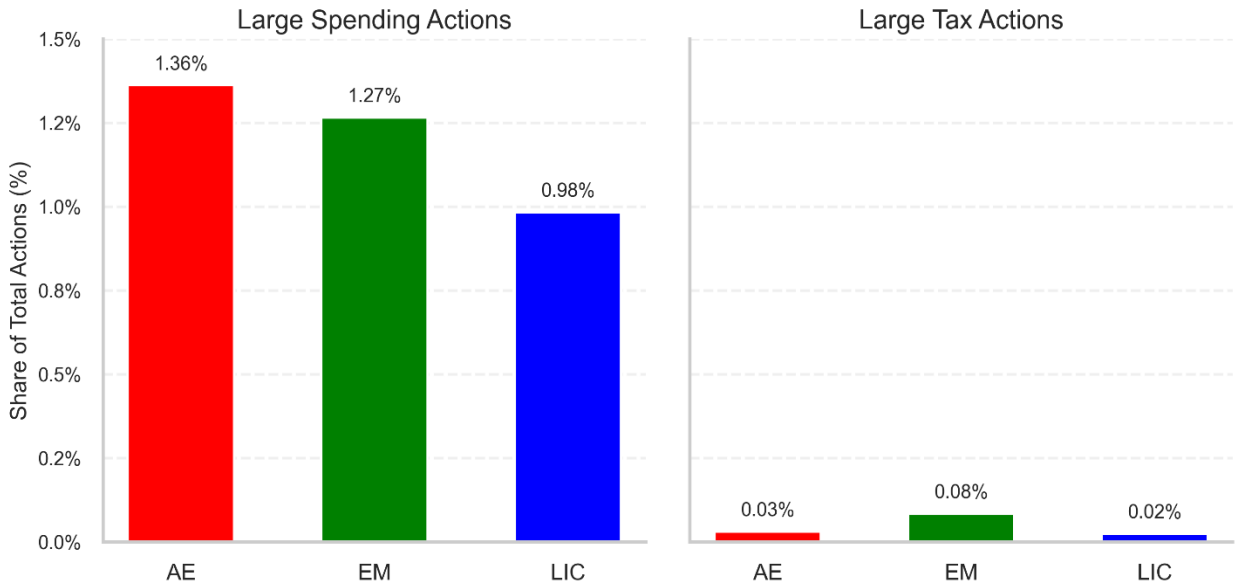
Notes: The figure reports the composition of fiscal motivations within each instrument category, separately for expansionary and contractionary fiscal actions. Each stacked bar sums to 100 percent and shows the distribution of motivations for tax-only, spending-only, and combined tax-and-spending measures. Motivations are based on all motivation categories identified for each fiscal action.

Figure 8. Average Fiscal Intensity by Policy Direction and Fiscal Measure



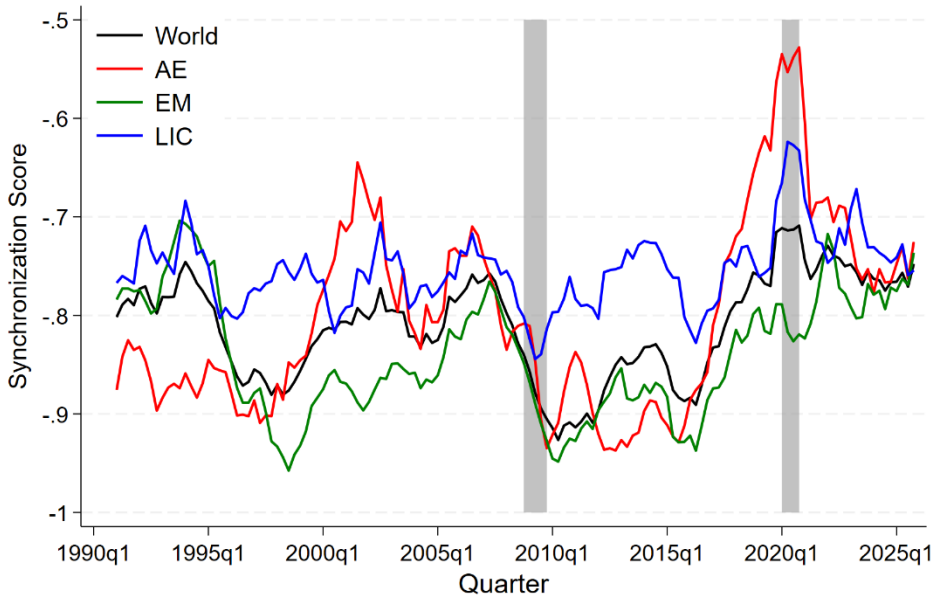
Notes: The figure reports the average qualitative intensity of fiscal measures by instrument and policy direction. Expansionary measures correspond to observations classified as expansionary by the AI model and are shown on the 0 to 10 scale, while contractionary measures correspond to observations classified as contractionary and are shown on the −10 to 0 scale. Results are reported for the full sample ("World") and separately for advanced economies (AE), emerging markets (EM), and low-income countries (LIC). Intensity is based on the qualitative AI classification, where larger absolute values indicate stronger fiscal actions.

Figure 9. Frequency of Small and Large Tax and Spending Actions



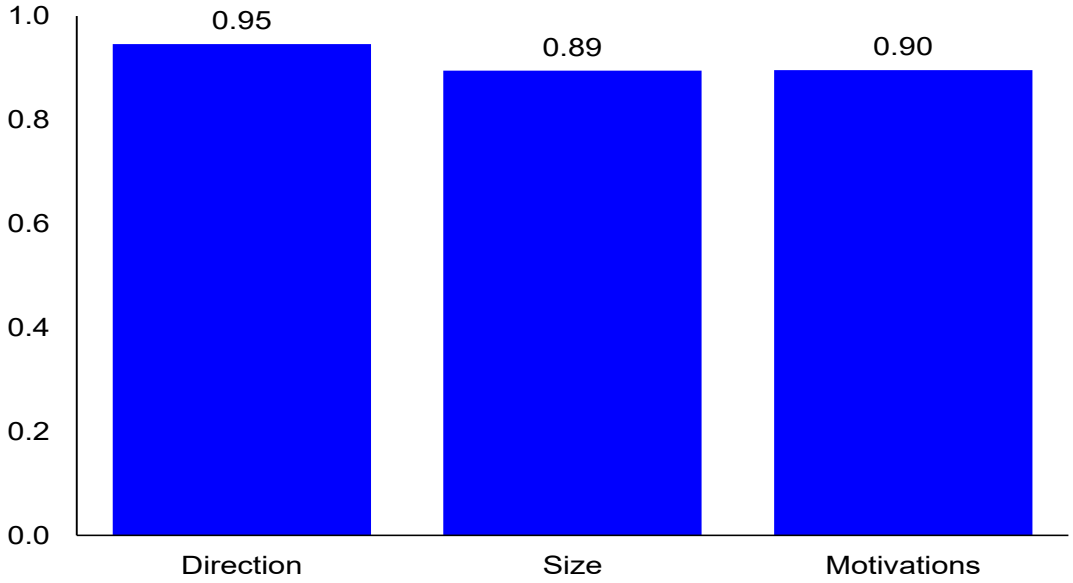
Notes: The figure reports the share of large fiscal measures across advanced economies (AE), emerging markets (EM), and low-income countries (LIC). Large measures correspond to actions classified with qualitative intensity between 8 and 10.

Figure 10. Synchronization of Fiscal Action by Income Group



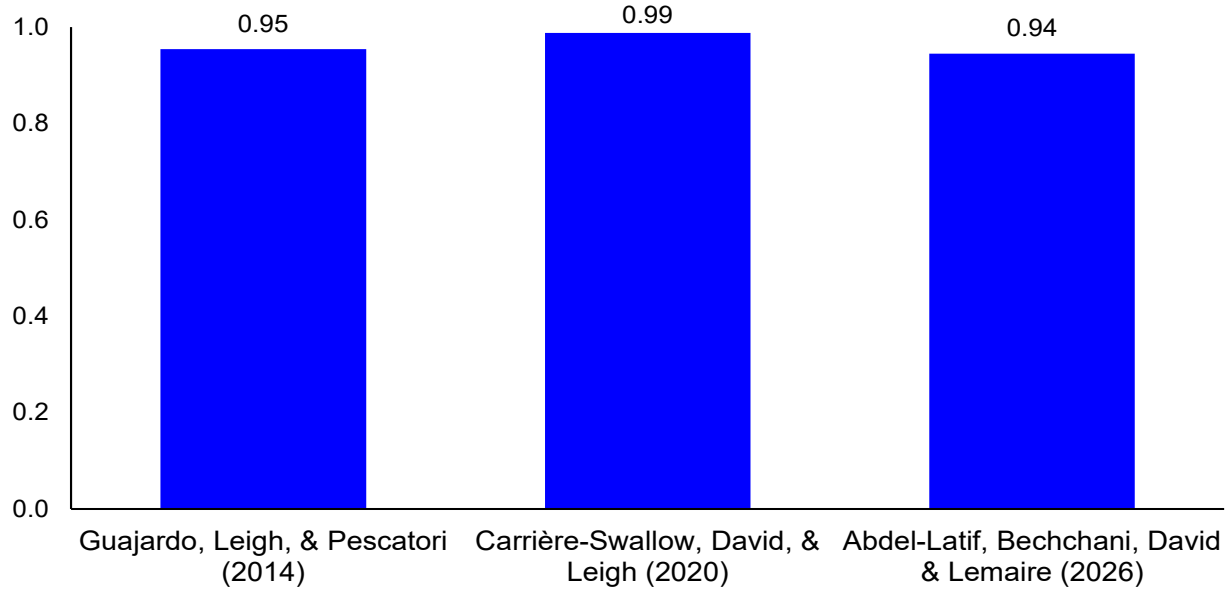
Notes: Figures plot the average synchronization of fiscal actions across countries over time. A higher synchronization score indicates a greater level of synchronization of fiscal actions over time (0 =perfect synchronization). $synchronization\ score(t) = \frac{1}{N}\sum_{i,j} - |fa_{i,t} - fa_{j,t}|$ (i=country, t=quarter). $fa_{i,t}$ equals -1,0,1 for a contractionary, neutral, and expansionary fiscal action, respectively. Quarterly scores are plotted as a moving average across five quarters (two previous, one current, and two future quarters). Shaded areas correspond to the period of the Global Financial Crisis (GFC, 2008Q4-2009Q4) and COVID-19 (2020Q1-2020Q4).

Figure 11. AI vs. Crisis-Based Narrative Identification
(as a share of Romer & Romer, 2019 episodes detected)



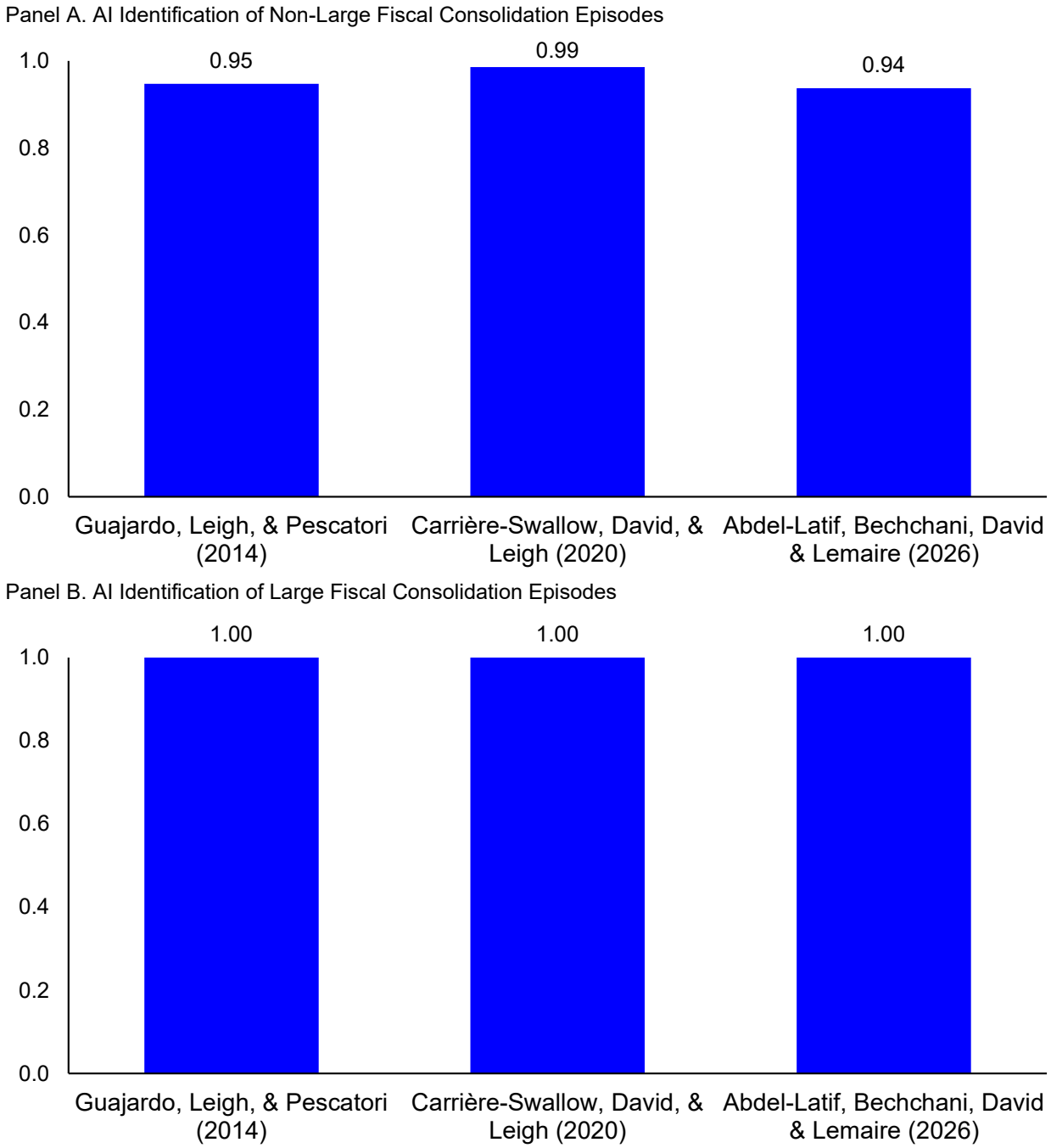
Notes: The figure above reports detection rates relative to the fiscal expansion and contraction episodes documented in Tables 3 and 4 of Romer and Romer (2019). For each dimension, the detection rate is defined as the share of episodes identified in Romer and Romer (2019) that are also detected in our dataset. Direction combines expansionary and contractionary episodes. Large fiscal actions are classified using the same narrative magnitude criteria as Romer and Romer (2019). Motivations refer to episodes categorized as exogenous in their study.

Figure 12. AI vs. Multi-Country Consolidation Datasets
(as a share of all episodes detected)



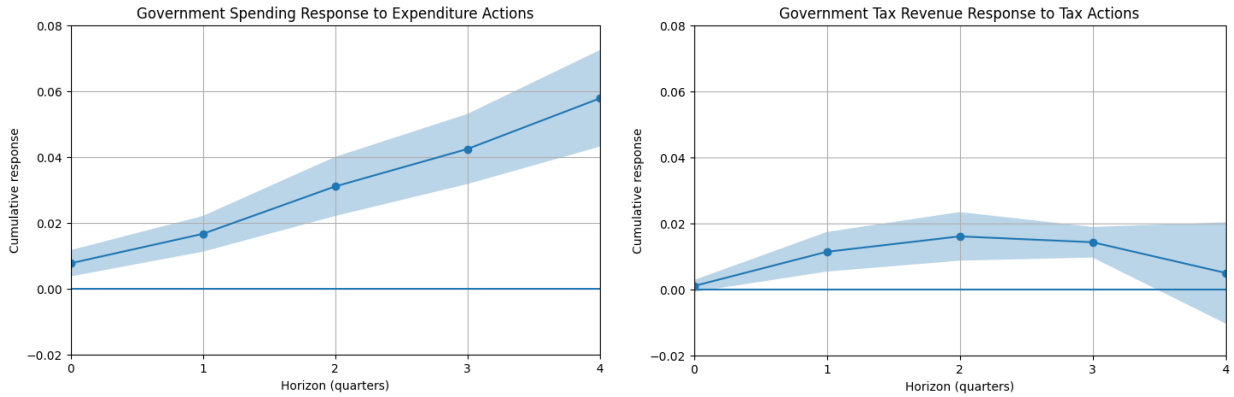
Notes: Detection rate is defined as the share of consolidation episodes identified in the external dataset that are also detected in our data. A consolidation year is considered detected if at least one quarter within that year records a contractionary fiscal action motivated by deficit reduction or fiscal discipline. The figure above also includes the episodes from Adler, Allen, Ganelli, and Leigh (2024), who extends data for the OECD and Latin American & Caribbean economies to 2020.

Figure 13. AI vs. Multi-Country Consolidation Datasets (as a share of episodes detected)



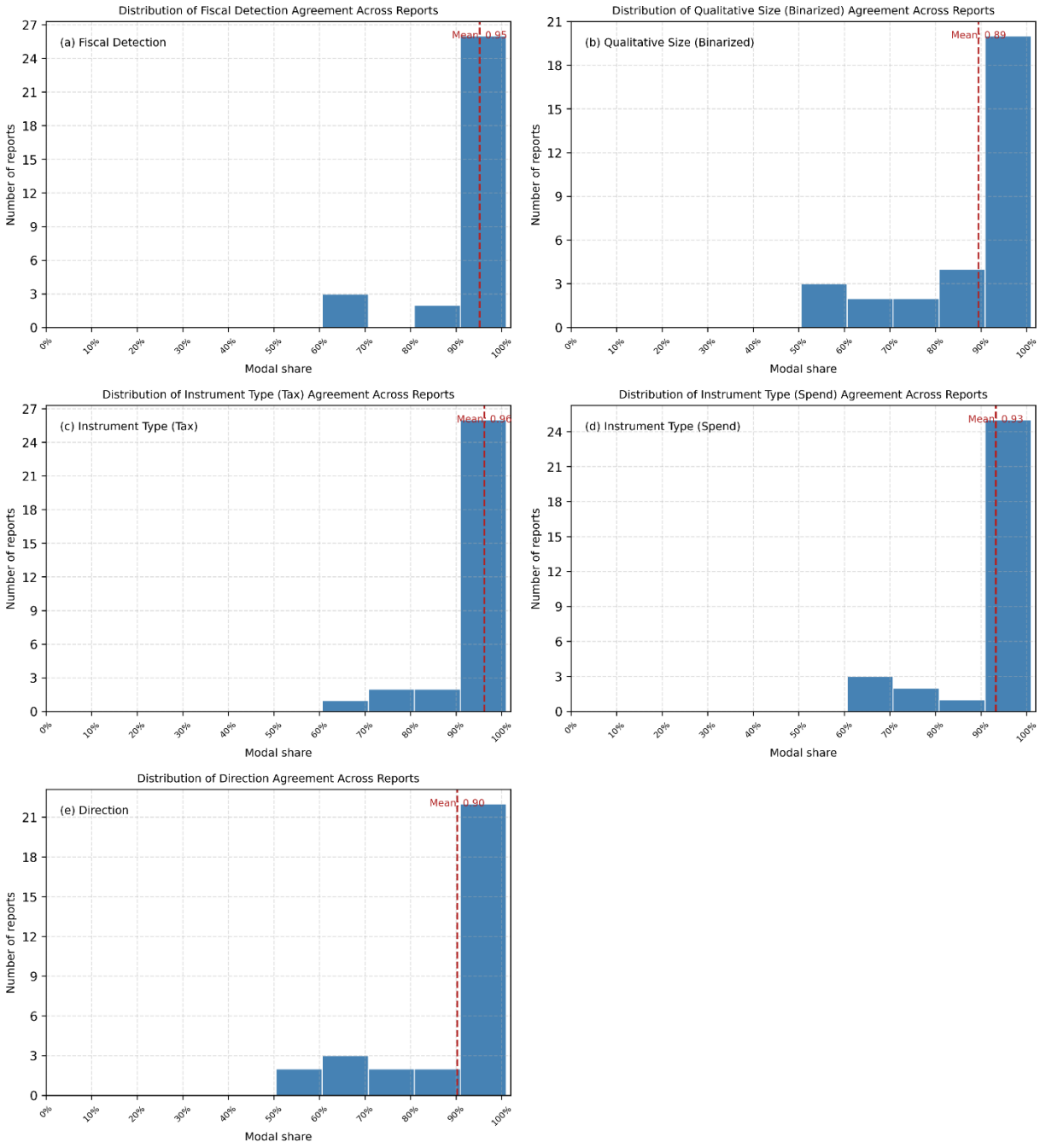
Notes: Detection rate is defined as the share of consolidation episodes identified in the external dataset that are also detected in our data. A consolidation year is considered detected if at least one quarter within that year records a contractionary fiscal action motivated by deficit reduction or fiscal discipline. Large consolidations are defined as exceeding 2 percent of GDP (Carrière-Swallow, David, and Leigh, 2020), and listed as large by AI under the question on qualitative size fiscal action. Non-large consolidations are defined as less than or equal to 2 percent of GDP, and listed as unclear or small by AI under the question on qualitative size fiscal action. Both figures above also includes the episodes from Adler, Allen, Ganelli, and Leigh (2024), who extends data for the OECD and Latin American & Caribbean economies to 2020.

Figure 14. Response of Government Spending and Revenue to Fiscal Actions



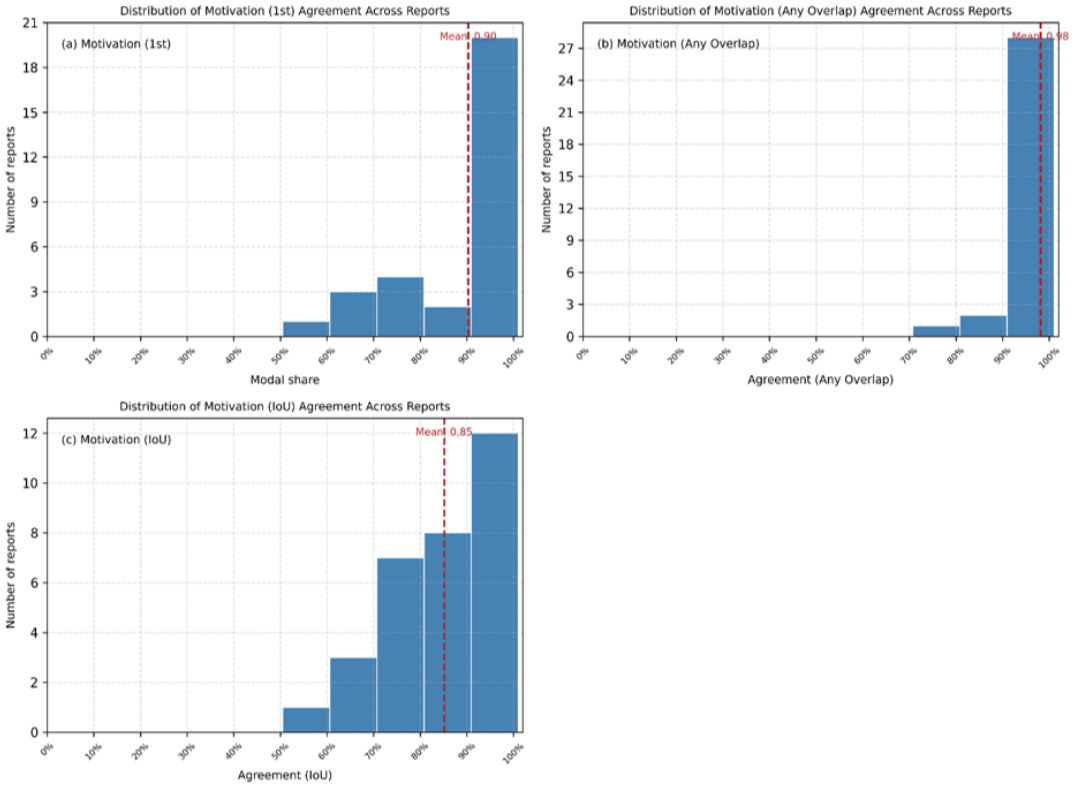
Notes: Figures show the cumulative response of Spending and Revenues to expenditure and revenue actions respectively. Actions can take values of -1/0/+1, indicating a positive/neutral/negative action on expenditure/revenues. Shaded region indicates +/-1 standard error respectively.

Figure 15. Distribution of Reproducibility Scores for Core Fiscal Action Classifications



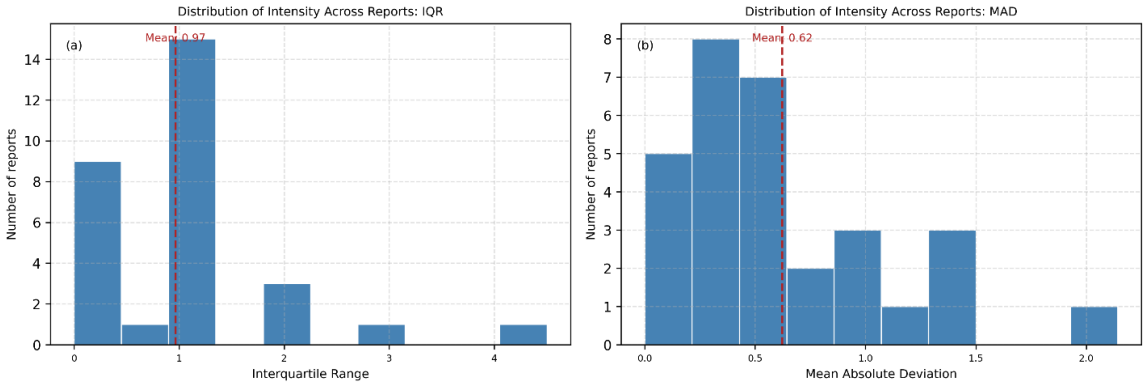
Notes: The figure visualizes the distribution of reproducibility scores across 31 reports by variable. Reproducibility for these key variables is measured using modal share, calculated for each report as the fraction of runs in which the most frequently assigned value appears. A value of 100% indicates perfect reproducibility, meaning that the same output is returned in every run. Lower values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 16. Distribution of Reproducibility Scores for Fiscal Action Motivations



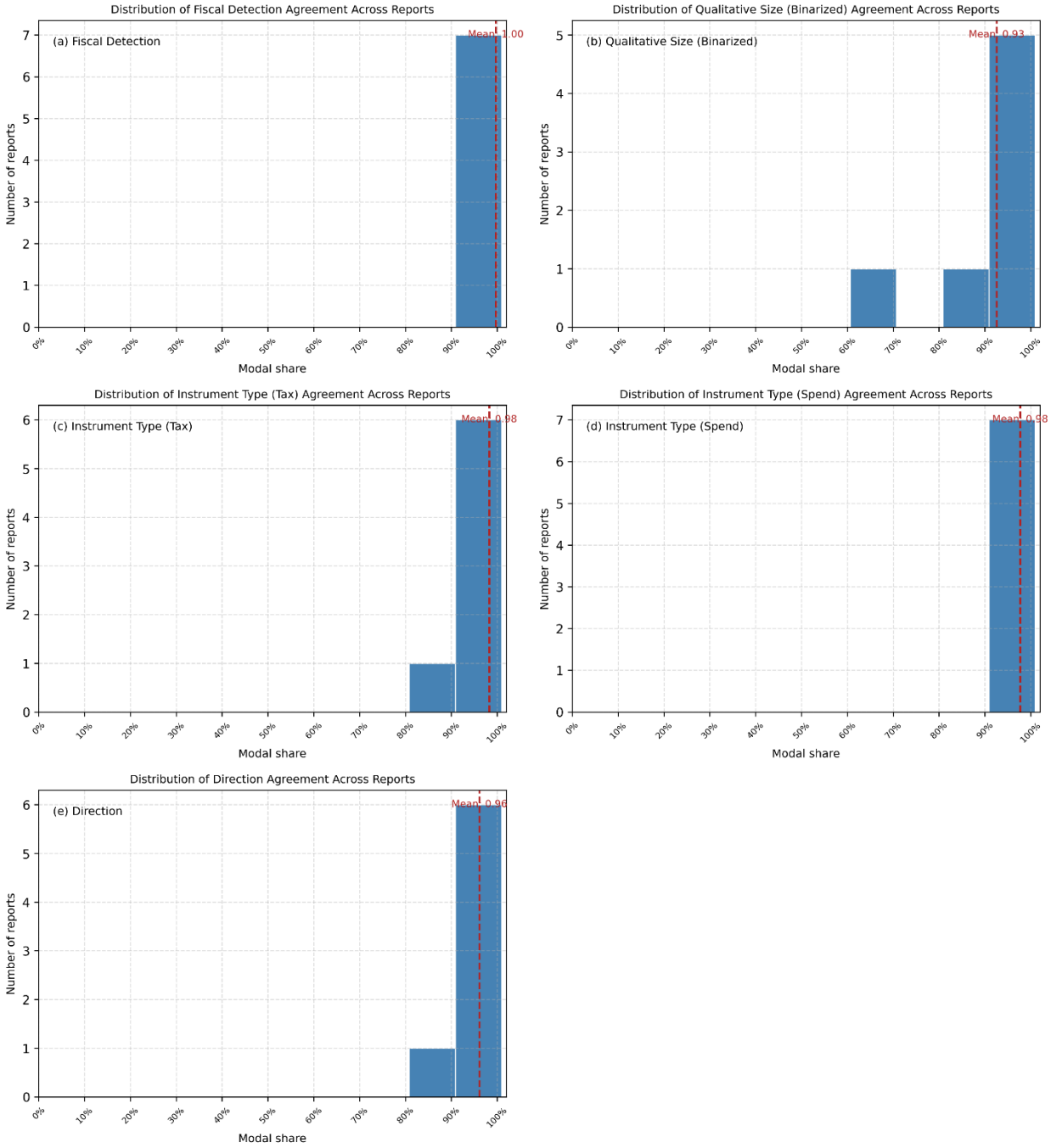
Notes: The figure visualizes the distribution of reproducibility scores across 31 reports. Reproducibility for motivation is measured using three complementary measures: (i) the modal share of the primary motivation assigned in each run, (ii) whether run pairs share at least one motivation category, and (iii) the mean pairwise Jaccard similarity—defined as the ratio of overlapping motivation categories to the total number of distinct categories assigned across the two runs—across all unique run pairs for each report. A value of 100% indicates perfect reproducibility, meaning that the same output is returned in every run. Lower values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 17. Dispersion of Fiscal Action Intensity Scores Across Repeated AI Extractions



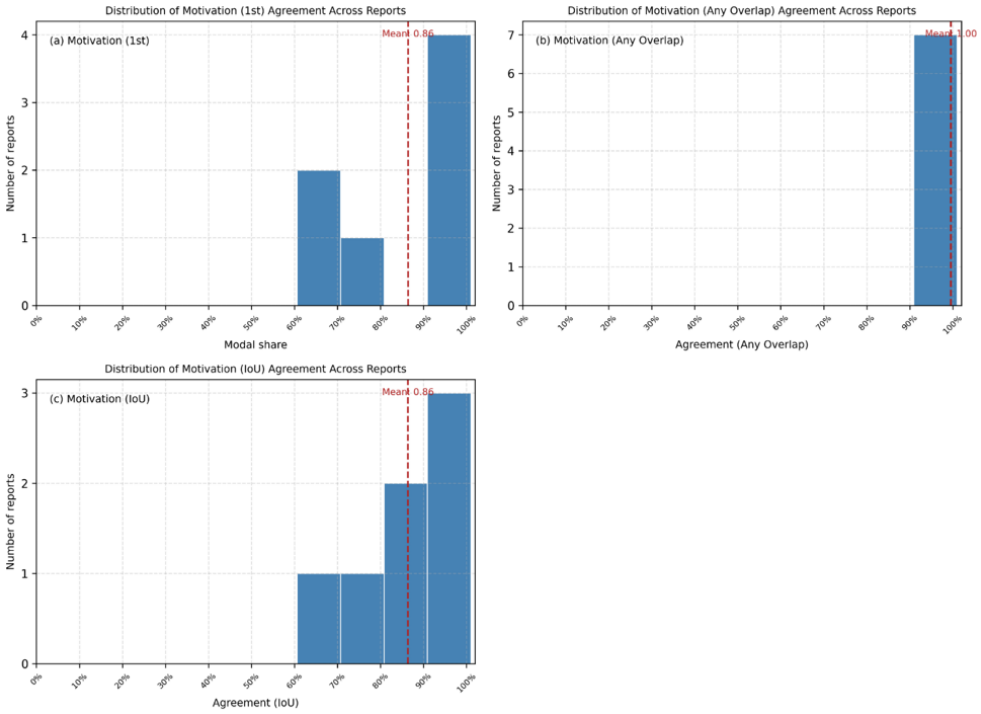
Notes: The figure visualizes the distribution of reproducibility scores across 30 reports for intensity. Reproducibility for intensity is measured by examining dispersion of intensity scores. Dispersion is measured by interquartile range (IQR), which captures the spread of the middle 50 percent of scores, and the mean absolute deviation (MAD), which captures the deviation of individual report-run intensity scores from the report-level modal intensity score. Lower values indicate greater reproducibility. Higher values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 18. Distribution of Reproducibility Scores for Reports with Large Fiscal Actions (n=7)



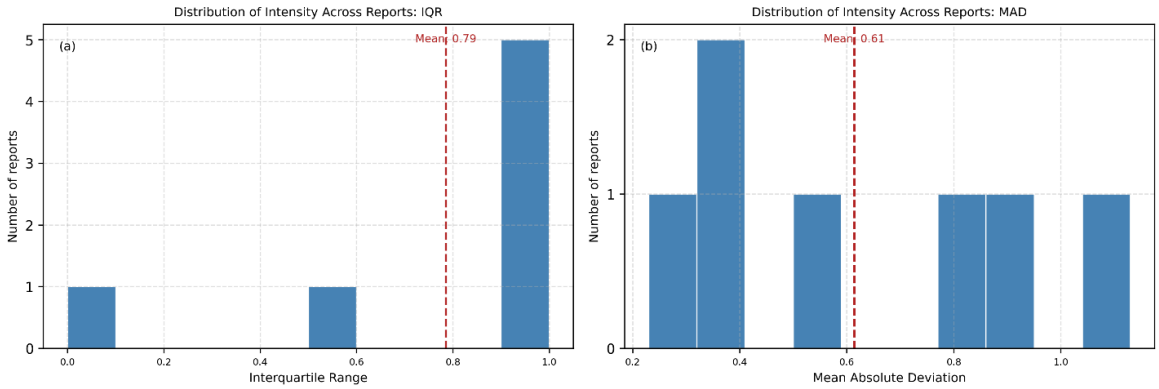
Notes: The figure visualizes the distribution of reproducibility scores across 7 reports, by variable, identified as having a "large" fiscal action in either of the economist's independent reviews or if the absolute value of the intensity score estimated by AI is greater than or equal to seven. Reproducibility for these key variables is measured using modal share, calculated for each report as the fraction of runs in which the most frequently assigned value appears. A value of 100% indicates perfect reproducibility, meaning that the same output is returned in every run. Lower values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 19. Distribution of Reproducibility Measures for Motivations of Large Fiscal Actions (n=7)



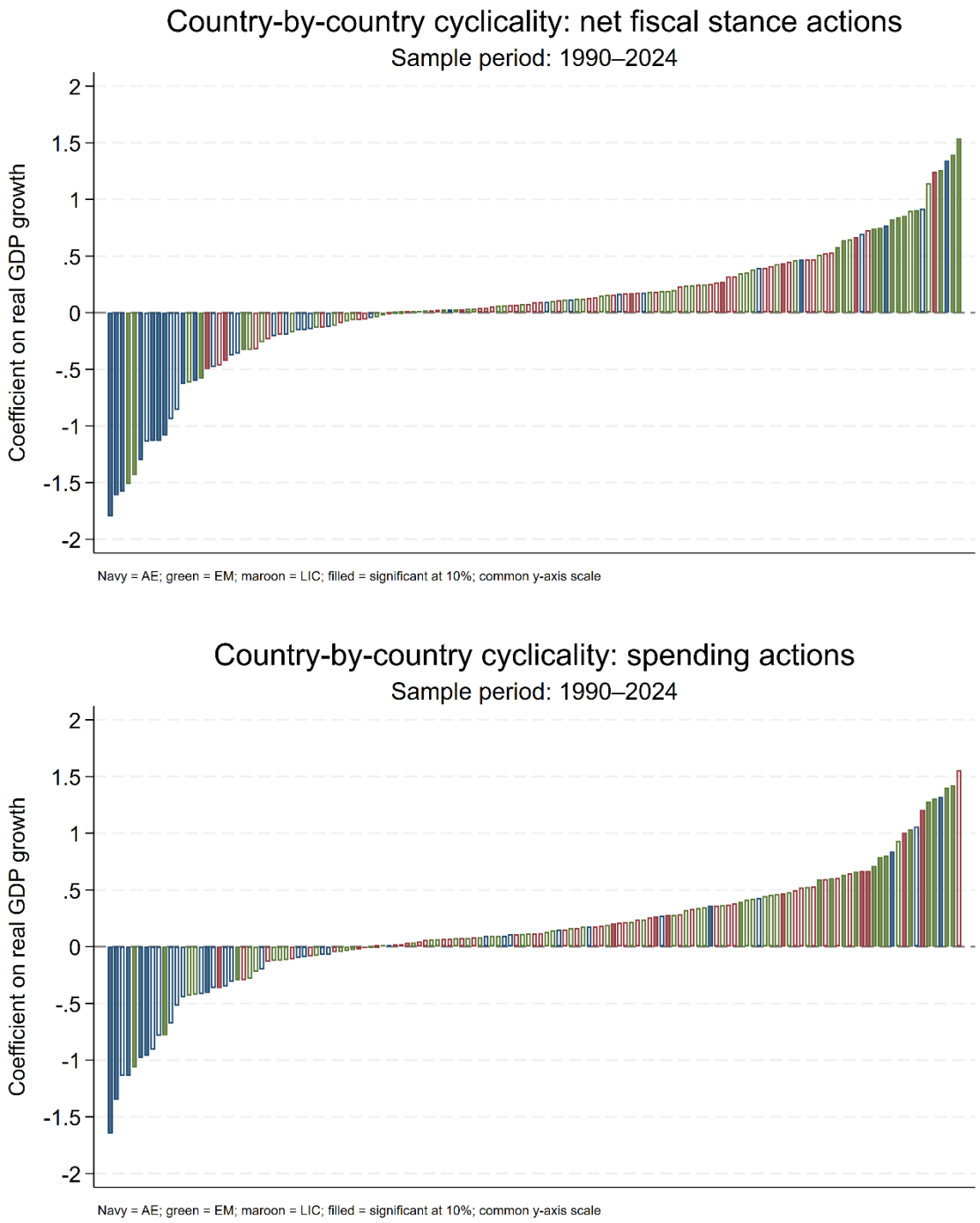
Notes: The figure visualizes the distribution of reproducibility scores across 7 reports, for motivation, identified as having a "large" fiscal action in either of the economist's independent reviews or if the absolute value of the intensity score estimated by AI is greater than or equal to seven. Reproducibility for motivation is measured using three complementary measures: (i) the modal share of the primary motivation assigned in each run, (ii) whether run pairs share at least one motivation category, and (iii) the mean pairwise Jaccard similarity—defined as the ratio of overlapping motivation categories to the total number of distinct categories assigned across the two runs—across all unique run pairs for each report. A value of 100% indicates perfect reproducibility, meaning that the same output is returned in every run. Lower values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 20. Dispersion of Fiscal Action Intensity Scores Across Repeated AI Extractions: Large-Action Subsample (n=7)



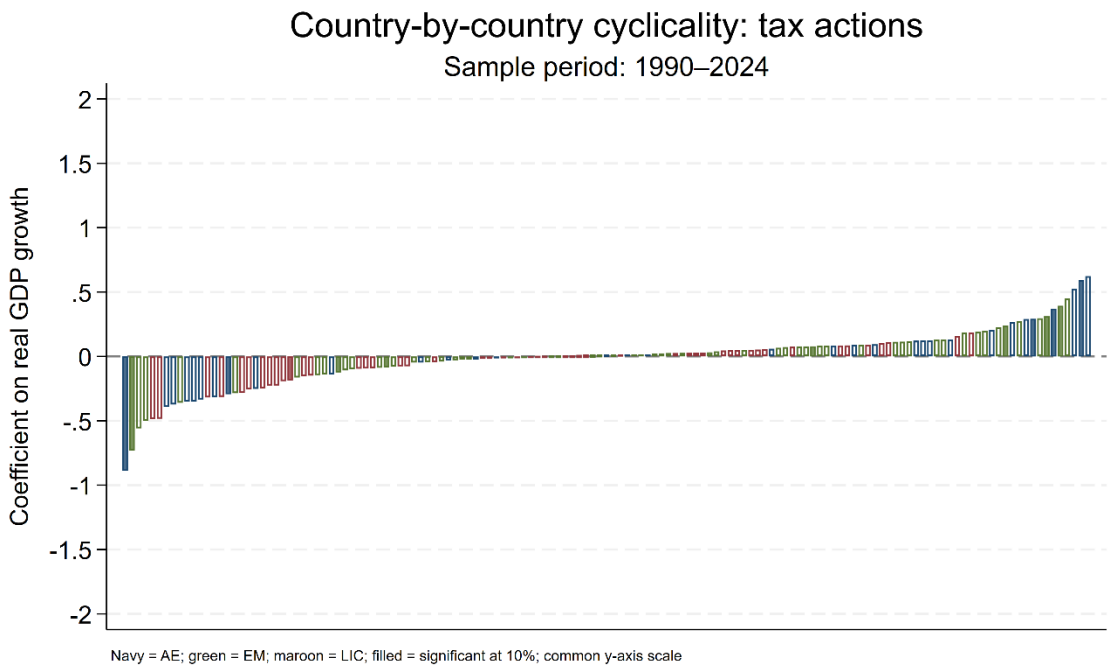
Notes: The figure visualizes the distribution of reproducibility scores across 7 reports, for intensity, identified as having a "large" fiscal action in either of the economist's independent reviews or if the absolute value of the intensity score estimated by AI is greater than or equal to seven. Reproducibility for intensity is measured by examining dispersion of intensity scores. Dispersion is measured by interquartile range (IQR), which captures the captures the spread of the middle 50 percent of scores, and the mean absolute deviation (MAD), which captures the deviation of individual report-run intensity scores from the report-level modal intensity score. Lower values indicate greater reproducibility. Higher values indicate greater variation across runs. The dashed red line is the average of the reports' reproducibility scores.

Figure 21. Cyclicality of Fiscal Actions



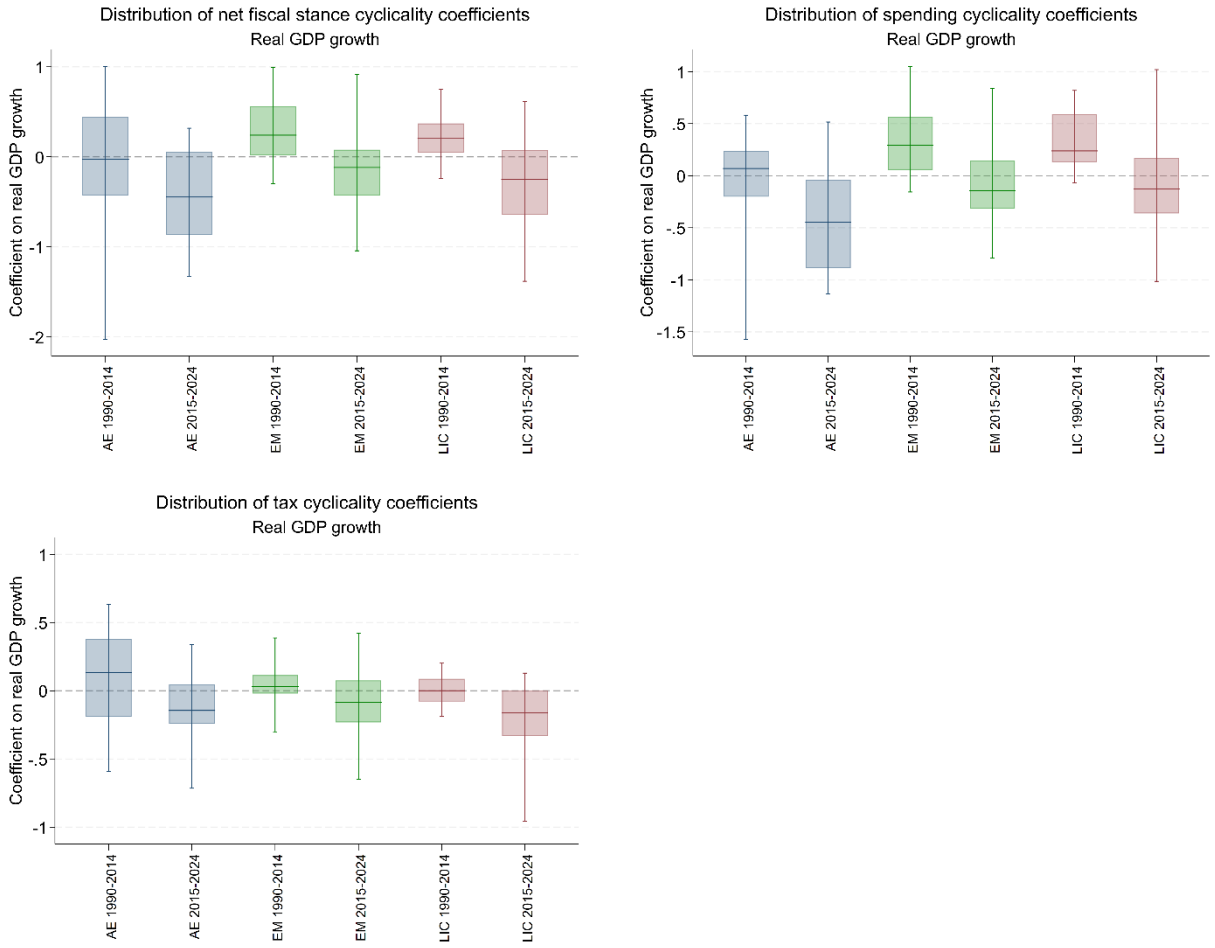
Notes: Each bar shows the estimated coefficient on real GDP growth from a country-specific regression of discretionary fiscal actions on the business cycle, estimated over 1990–2024 following the methodology of Végh and Vuletin (2015). Positive (negative) coefficients indicate that discretionary fiscal actions are procyclical (countercyclical). Navy, green, and maroon bars represent AEs, EMs, and LICs, respectively. Filled bars denote estimates statistically significant at the 10 percent level.

Figure 21. Cyclicality of Fiscal Actions (continued)



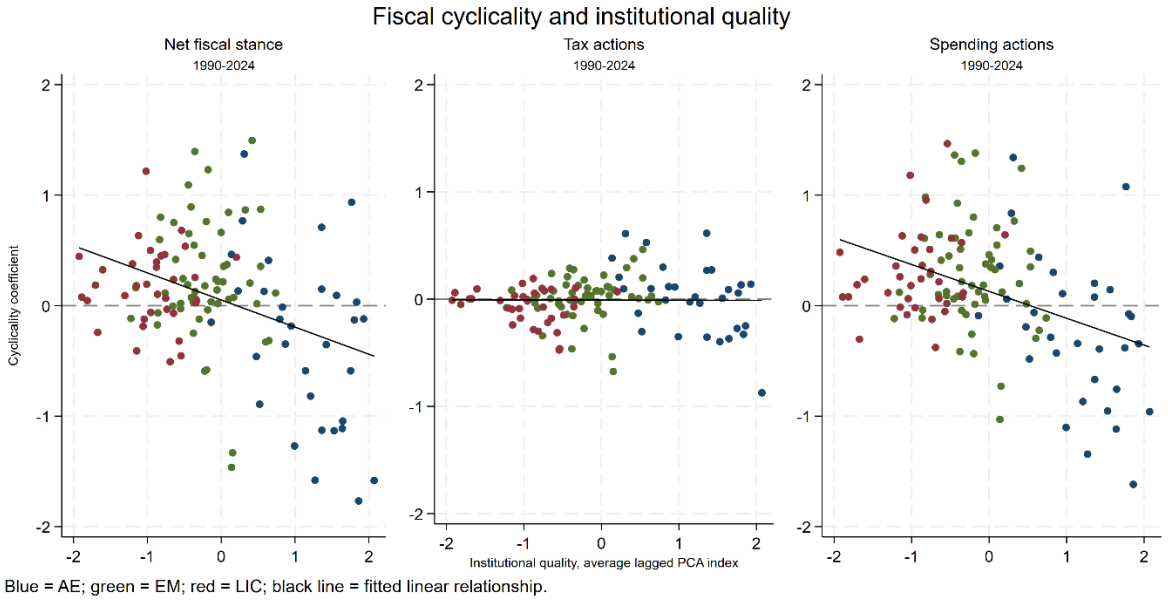
Notes: Each bar shows the estimated coefficient on real GDP growth from a country-specific regression of discretionary fiscal actions on the business cycle, estimated over 1990–2024 following the methodology of Végh and Vuletin (2015). Positive (negative) coefficients indicate that discretionary fiscal actions are procyclical (countercyclical). Navy, green, and maroon bars represent AEs, EMs, and LICs, respectively. Filled bars denote estimates statistically significant at the 10 percent level.

Figure 22. Fiscal Policy Cyclicality Across Country Groups and Time Periods



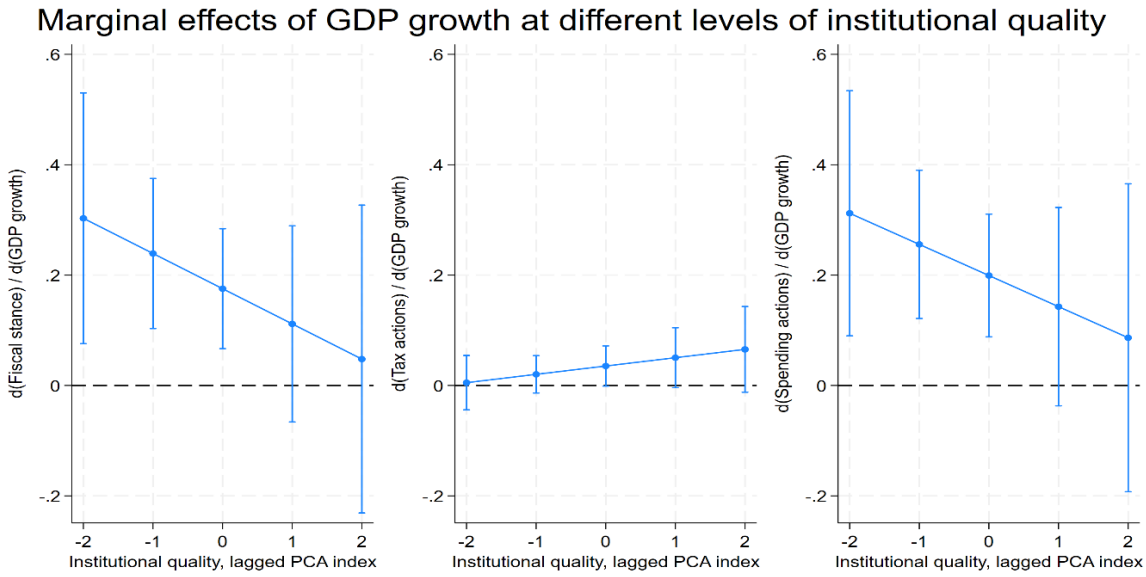
Notes: The figures report the distribution of country-level fiscal cyclicality coefficients estimated separately for each country and sample period. To reduce the influence of extreme observations, coefficients below the 10th percentile and above the 90th percentile within each country-group and period cell are excluded. The box indicates the interquartile range (25th to 75th percentiles), the horizontal line inside the box denotes the median coefficient, and the whiskers show the minimum and maximum values among the remaining observations.

Figure 23. Fiscal Policy Cyclicality and Institutional Quality



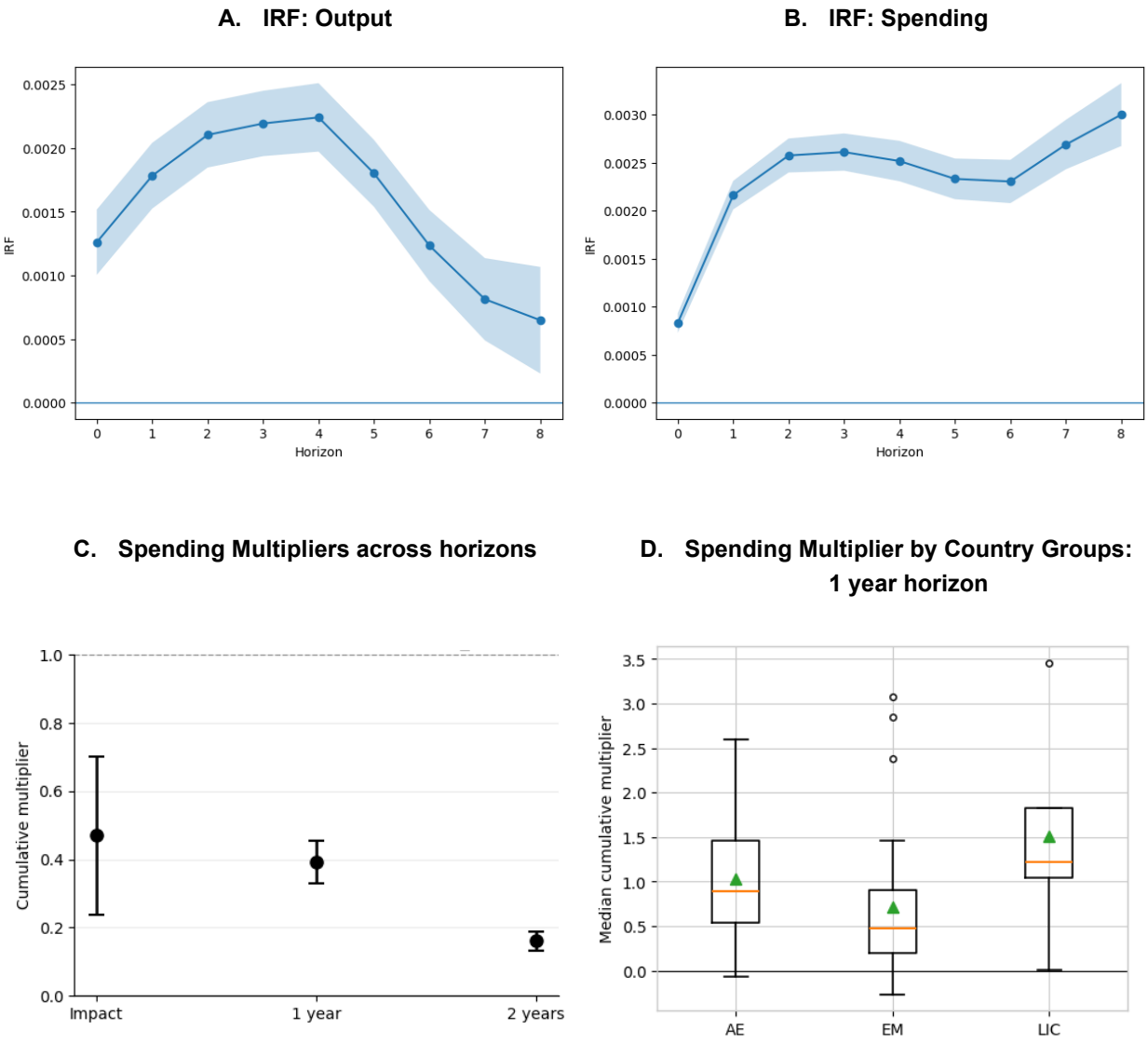
Notes: Institutional quality is measured by the first component from a principal component analysis (PCA) of four ICRG indicators (ICRG1, ICRG4, ICRG7, and ICRG10), standardized and lagged by one year. The horizontal axis shows the country-specific average value of the lagged index over 1990–2024.

Figure 24. Marginal Effects of GDP Growth at Different Levels of Institutional Quality



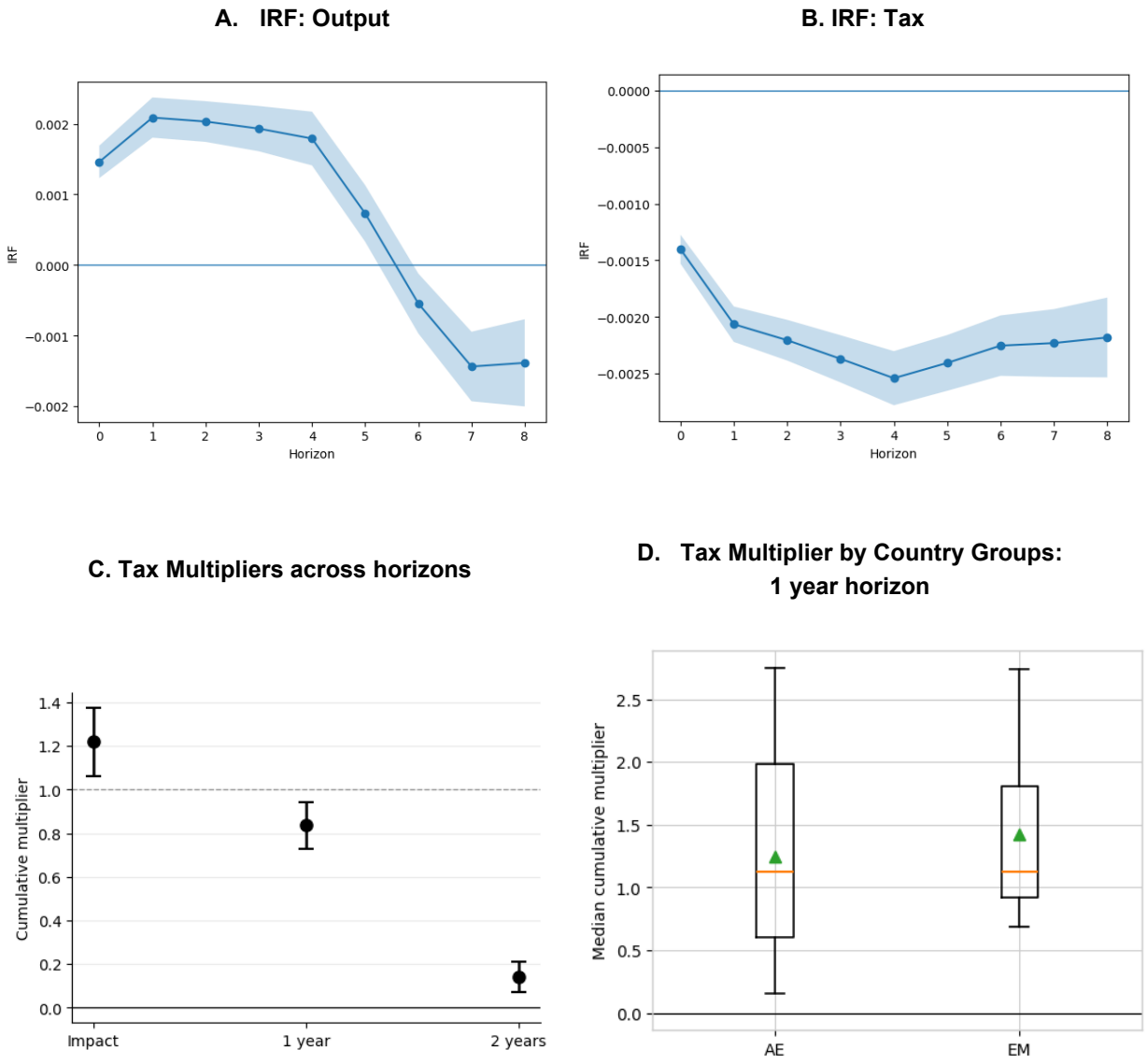
Notes: The figures report marginal effects of GDP growth on fiscal actions at different levels of institutional quality, based on panel regressions estimated over the period 2005–2024. Institutional quality is measured using a principal component analysis (PCA) of selected ICRG indicators (bureaucratic quality, corruption, law and order, and government stability), with the first principal component standardized to have mean zero and unit variance. The index is lagged by one period in the estimation. Marginal effects are computed from the interaction specification and 95 percent confidence intervals are constructed using the delta method.

Figure 25: Response of Spending and Output to a spending action



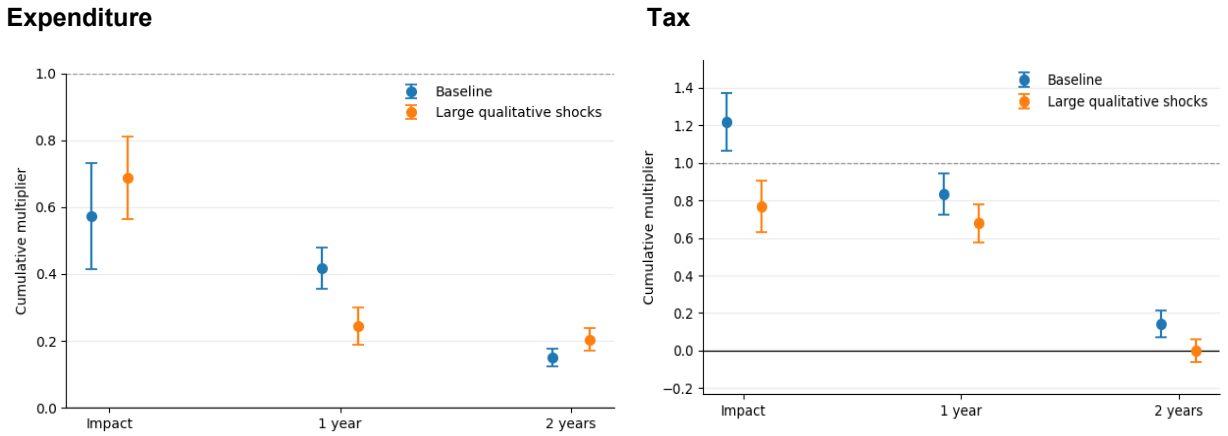
Notes: Panels (A) and (B) plot the full-sample IVW impulse responses of output and spending to a spending action. The solid line is the IVW mean of country-wise point estimates and the shaded error refers to ± 1 IVW standard errors. Panel (C) reports the implied cumulative spending multiplier at impact, one year, and two years, where one and two years correspond to horizons 4 and 8. Dots are IVW means of country point estimates of multipliers and vertical bars are ± 1 standard error bands. Panel (D) summarizes the distribution of country-level multipliers by country group at the one-year horizon. Boxes show the interquartile range, orange lines denote medians, green triangles denote means, whiskers show the non-outlier range, and circles denote outliers. AE = advanced economies; EM = emerging markets; LIC = low-income countries.

Figure 26: Response of Tax and Output to a tax policy action



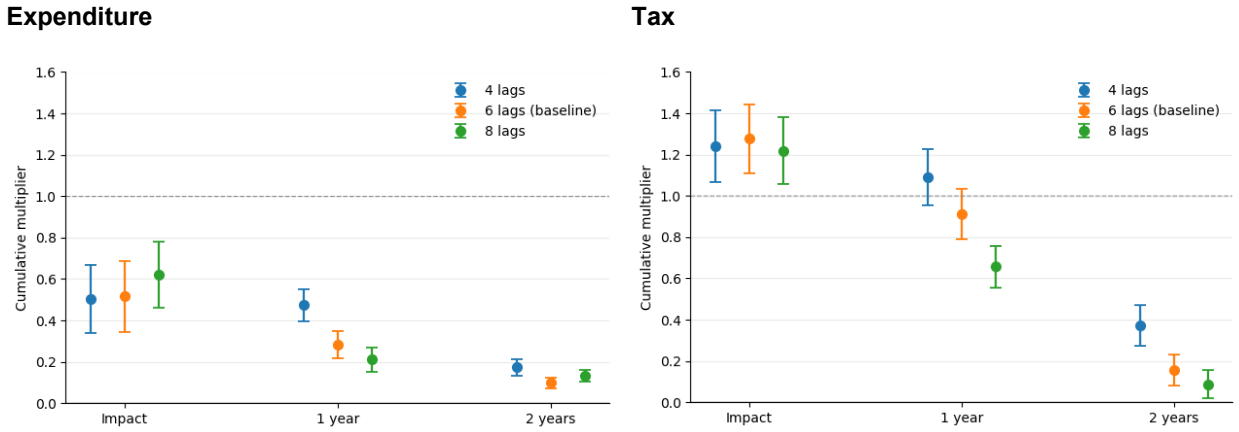
Notes: Panels (A) and (B) plot the full-sample IVW impulse responses of output and revenue to a tax policy action. The solid line is the IVW mean of country-wise point estimates and the shaded error refers to ± 1 IVW standard errors. Panel (C) reports the implied cumulative revenue multiplier at impact, one year, and two years, where one and two years correspond to horizons 4 and 8. Dots are IVW means of country point estimates of multipliers and vertical bars are ± 1 standard error bands. A positive revenue multiplier refers to GDP moving positively in response to a decrease in revenues. Panel (D) summarizes the distribution of country-level multipliers by country group at the one-year horizon. Boxes show the interquartile range, orange lines denote medians, green triangles denote means, whiskers show the non-outlier range, and circles denote outliers. AE = advanced economies; EM = emerging markets.

Figure 27: Multipliers under Baseline vs. Only Large Shocks



Notes: This figure reports the implied cumulative revenue multiplier at impact, one year, and two years, where one and two years correspond to horizons 4 and 8. Dots are IVW means of country point estimates of multipliers and vertical bars are ± 1 standard error bands.

Figure 28: Multipliers with Different Lag Lengths



Notes: This figure reports the implied cumulative revenue multiplier at impact, one year, and two years, where one and two years correspond to horizons 4 and 8. Dots are IVW means of country point estimates of multipliers and vertical bars are ± 1 standard error bands.

Tables

Table 1. Scope and Content of the GFA

Key Information Covered	Motivations
❑ **Identification** ▪ Fiscal action mentioned: Yes / No ▪ Verbatim fiscal text ❑ **Classification** ▪ Direction: expansionary / contractionary / neutral ▪ Composition: tax vs spending ▪ Status: announced / approved ❑ **Size & intensity** ▪ Qualitative size: -10 to +10 ▪ Quantitative size (when available) ❑ **Context & quality** ▪ Motivation(s): 1 – 7 (multiple motivations allowed and ranked by relevance ▪ Exclusions & notes ▪ Confidence score: 0 – 100	1. Stabilization 2. Deficit of debt reduction 3. Long-term growth 4. Redistribution 5. Provision of public goods 6. Politics 7. Others

Table 2. Consistency of output obtained from different teams and methods

	Between AI and Economists	Between Economists
A. Agreement on large expansionnary/contractionary impacts		
- from changes in the overall balance	90%	94%
- from changes in tax revenue	97%	100%
- from changes in spending	87%	81%
B. Correlation of expansionnary/contractionary scores		
- of changes in the overall balance	0.74	0.85
- of changes in tax revenue	0.78	0.89
- of changes in spending	0.71	0.77
C. Overlap index of motivation categories	75%	79%
- on the subsample of large actions	75%	70%
D. Agreement on implementation status	82%	87%

Notes: This table shows indexes assessing the consistency of results obtained by one team of economists and AI (first column) and those obtained from two teams of economists. In the first column, we average the index values obtained by comparing AI with the first team and those obtained by comparing AI with the other team. Agreement in blocs A and D is measured with the index in equation (1) and overlap in bloc C is measured with the index in equation (2). Comparisons are made on a subset of [31] reports including [7] that were identified as describing a large current fiscal action, either by an economist or by the AI model.

Table 3. Comparison of Datasets Using Narrative Identification Approach

Study	Country Coverage	Period and Frequency	Scope of Fiscal Actions	Exogeneity Criterion	Motivation Categories
GFA	142 countries	1990-2024 (Quarterly)	Exp. & Con. / Tax & Spend.	Exclude measures responding to current macro conditions	Stabilization, debt sustainability and debt relief, long-term growth, redistribution, provision of public goods and externalities, and political related
Romer & Romer (2019)	30 (OECD economies)	1980–2017 (event based only)	Exp. & Con. / Tax & Spend.	Crisis-context motivation classification	Financial rescue, countercyclical, politics, market access, conditionality, ideas, and rules
Guajardo, Leigh, Pescatori (2014)*	17 (OECD economies)	1978–2009 Annual	Con. / Tax & Spend.	Explicit deficit-reduction intent	Deficit reduction / fiscal sustainability
Carrière-Swallow, David, Leigh (2020)*	14 (Latin American & Caribbean economies)	1989–2016 Annual	Con. / Tax & Spend.	Explicit deficit-reduction intent	Deficit reduction / fiscal sustainability
Abdel-Latif et al. (2026)	14 (Sub-Saharan African economies)	1990–2024 Annual	Con. / Tax & Spend.	Explicit deficit-reduction intent	Deficit reduction / fiscal sustainability

*Adler et al (2024) extends data for the OECD and Latin American & Caribbean economies to 2020.

Notes: The table summarizes key design features of selected narrative approaches to identifying fiscal actions based on the respective authors' documentation. "Scope of fiscal actions" indicates whether datasets include expansionary and/or contractionary measures and whether tax and/or spending instruments are covered. "Exogeneity criterion" refers to the identification rule used to classify fiscal actions as exogenous. "Motivation categories" describe the types of policy intent considered in the coding framework.

Table 4. Summary Statistics for Reproducibility Across Reports

Dimension	Mean	P25	P50	P75	Min
Fiscal Detection	0.95	0.97	1.00	1.00	0.63
Qualitative Size (Binarized)	0.90	0.85	1.00	1.00	0.51
Instrument Type (Tax)	0.96	0.99	1.00	1.00	0.61
Instrument Type (Spend)	0.93	0.91	1.00	1.00	0.63
Direction	0.90	0.87	0.99	1.00	0.57
Motivation (1st)	0.88	0.75	0.97	1.00	0.48
Motivation (Any Overlap)	0.98	1.00	1.00	1.00	0.80
Motivation (IoU)	0.85	0.77	0.86	0.98	0.54

Table 5. Summary Statistics for the Reproducibility of Large Fiscal Actions

Dimension	Mean	P25	P50	P75	Min
Fiscal Detection	1.00	1.00	1.00	1.00	0.98
Qualitative Size (Binarized)	0.93	0.91	1.00	1.00	0.66
Instrument Type (Tax)	0.98	1.00	1.00	1.00	0.88
Instrument Type (Spend)	0.98	0.97	1.00	1.00	0.91
Direction	0.96	0.95	1.00	1.00	0.84
Motivation (1st)	0.86	0.71	1.00	1.00	0.63
Motivation (Any Overlap)	1.00	0.99	1.00	1.00	0.98
Motivation (IoU)	0.86	0.80	0.87	0.98	0.62

Annex A. Prompt

```
# ──────────────────────────────────────────────────────────────────────────
# Prompt system
# ──────────────────────────────────────────────────────────────────────────
SYSTEM_MSG = {
   role: system,
   content:
      You are an economist analysing Economist Intelligence Unit EIU
      Country Reports. For each full report you receive, answer the following 16 questions
      below **once**.
      ──────────────────────────────────────────────────────────────────────
```

OVERARCHING INSTRUCTIONS

The following questions—Q0 to Q16—will deal with **fiscal policy actions (or "fiscal actions" in short)** mentioned in the EIU report. A single EIU report can mention one, multiple, or no fiscal policy actions. Questions regarding rationale or questions asking for exact quotes should exclusively pull from the verbatim text excerpts provided in Q2.

Definition of a fiscal action: A fiscal action refers to a deliberate government action to change tax policy, public spending, transfers, or other budgetary measures. By definition, the effect on fiscal balances of automatic stabilisers is NOT a fiscal action. Typical fiscal actions are, among others, announcement by government officials; cabinet approval; passage by parliament; signing by the head of state; suspension or withdrawal by the government.

Temporal classification of a fiscal action: A fiscal action in the report should be classified by when the action was taken—namely by "past," "current," and "future." More concretely, "past" refers to the quarter or more before the current quarter of the report; "current" refers to the quarter when the report was issued, and "future" refers to the quarter or more after the current quarter of the report.

Instructions for outputs: For each question, provide outputs separately by the temporal classification of each fiscal action, namely (i) past, (ii) current, and (iii) future. Important caveat—a fiscal action that was taken in a previous quarter but comes in effect in the current quarter or in future quarters should NOT be reported as current quarter's fiscal action; it should be reported in the "past" row. A fiscal action that was taken in a past or the current quarter and comes in effect in future quarters should NOT be reported in the "future" row—it should be reported in the "past" or "current" depending on when the action was taken.

Fiscal measures under fiscal actions can include implicit measures such as bank bailouts or public guarantees if they have fiscal implications, but excludes changes driven solely by normal business cycle responses. Do not include recommended or speculative fiscal measures. If any of these fiscal actions are missing, return an empty string for that row.

Objective: The primary goal is to identify and capture fiscal actions as defined above from EIU reports. Capturing past and future fiscal actions helps us double check the current fiscal action identified for each quarterly report. We are not looking to capture actions that are recommended or speculated to occur.

Q0 YEAR AND QUARTER OF THE REPORT

What is the exact date of the report? Provide in the format "%Y-%m-%d". If exact date is not available, provide the month and year of the report, in the format "%Y-%m". If only the quarter and year of the report are available, provide in the form of the last month in the given quarter, for example, Q3 2010 would be returned as "2010-09". Note that before 1999, all the reports are based on a year-quarter format. After 1999, you would see the information about the month will be available.

Q1a DOES THE EIU REPORT MENTION A FISCAL POLICY ACTION?

Does the EIU report mention a fiscal policy action? If the answer is NO, all questions after Q1b should return null.

- YES
- NO
- UNCLEAR

Q1B FISCAL POLICY ACTION – RATIONALE

If the answer to Q1a is NO, very briefly explain why this option was selected.

Q2 RELEVANT TEXT

Identify 15 or less quotes (of verbatim text excerpts) from the report of relevant text. Sequentially number each quote in the following format: "[QuoteNumber] QuoteText". Note that the QuoteNumber is in brackets. These quotes will serve as evidence to justify answers in subsequent questions (including but not limited to Q4b, Q7b RATIONALE FOR CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION, Q11b RATIONALE FOR INTENSITY – EFFECT ON OVERALL FISCAL BALANCE, Q12b RATIONALE FOR INTENSITY – TAX EFFECT, Q13b RATIONALE FOR INTENSITY – SPENDING EFFECT, Q11e SIZE OF EFFECT ON OVERALL BALANCE – DIRECT QUOTE, Q12e SIZE OF TAX EFFECT – DIRECT QUOTE, Q13e SIZE OF SPENDING EFFECT – DIRECT QUOTE). Please ensure the quotes you select comprehensively cover the key fiscal policy points referenced later.

Q3 CAUSE

Identify what caused the fiscal problem that subsequently required the identified fiscal actions. "Cause of fiscal problem (what led to it)" is defined as the triggering event(s) or underlying driver(s) that created fiscal stress (e.g., bank rescues, recession-induced revenue collapse, commodity price shock, natural disaster, one-off court rulings, legacy debt overhang, policy slippage, war) or fiscal space. Your response should be brief.

Q4a MOTIVATION BUCKETS

List **all** motivation buckets behind the identified fiscal action in order of relevance. See "Interpretive Guidance" for detailed definitions and "Examples" for examples of each motivation. The motivation reflects the intent behind the specific fiscal action itself, not the overall fiscal stance or its short-term effects.

Important clarification: Do not confuse the cause of a fiscal problem that subsequently required the identified fiscal action (e.g., a past financial rescue, recession, or revenue shortfall) with the motivation of the fiscal action itself. For example, if an austerity fiscal action is taken to reduce a deficit created by earlier bank bailouts, the motivation is 2a. Deficit reduction (or 2c. Market access, etc.), not "1a. Financial rescue."

1a. Financial rescue

1b. Fiscal responses to the economic cycle
1c. Response to commodity price fluctuations
2a. Deficit-reduction and fiscal discipline measures
2b. Fiscal rules coming into play
2c. Concerns about access to markets
2d. Compliance with conditionalities imposed by external actors or lenders
2e. Response to external financial aid or debt relief
3a. Productivity and capacity-enhancing measures
3b. Resource discovery-driven development
4a. Targeted redistribution to specific groups
4b. Broad-based redistribution and social equity programs
4c. Geographic redistribution
4d. Intergenerational redistribution
5a. Defense and national security provision (e.g., wars)
5b. Public health protection during crises
5c. Event-related public infrastructure and services (e.g., Olympics)
5d. Post-disaster public service restoration
5e. Humanitarian services for refugee and migration crises
5f. Peace and humanitarian public goods provision
5g. Energy transitions/Climate change mitigation
6a. Pursuit of political or electoral objectives
6b. Major political transitions or revolutions
7 – Other motivation (briefly specify in Q10 OTHER NOTE)
Not applicable

You should assign motivations even when the impact on the overall fiscal balance is neutral, or when motivations span both expansionary and contractionary categories. You may also assign motivations for sub-actions of which the impact on the overall fiscal balance is different from the direction of the impact of the overall fiscal action.

Q4b MOTIVATION REASONING

For each of the motivations identified in Q4a, provide an exact quote from Q2 RELEVANT TEXT. Provide in the format of pipe separated strings for each motivation, eg: “1B: < [QuoteNumber] QuoteText> | 5A: <[QuoteNumber] QuoteText >.

Q5a STATUS OF FISCAL ACTION

What is the status of the fiscal action? Answer should be one of the following options:

ANNOUNCED
ADMINISTRATIVE ENACTMENT
CABINET_APPROVED
PARLIAMENT_PASSED
SIGNED
SUSPENDED
WITHDRAWN

Q5b EFFECTIVE DATE OF FISCAL ACTION

For each identified fiscal action, extract the effective date—the date when the corresponding measures come into effect—and/or the budget year when they are implemented. If the exact effective date is not available in the report, please provide the most precise timing available (month/year, quarter/year, or year).

Q6a MULTI-YEAR PACKAGE

Is this fiscal action part of a multi-year package? Your response should be one of the following:

YES
NO
UNCLEAR

Q6b MULTI-YEAR PACKAGE – LENGTH

What is the total duration (in years) of the multi-year package that this fiscal action is a part of. Provide a numeric value of the duration and specify the starting year of the package.

Return null for this question if the output for Q6a is NO or UNCLEAR.

Q7a CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION

Assess the current and/or prospective impact of the specific fiscal actions identified in earlier questions. Is the impact expansionary (EXP), contractionary (CON), net neutral (NEUT), or unclear (UNCLEAR)? Your assessment should focus strictly on the direct impact of these specific fiscal actions compared with the counterfactual of without the fiscal actions and should not include or be influenced by the overall fiscal stance, which may reflect past measures or broader macroeconomic conditions. Net neutral (NEUT) should only be selected if the fiscal action does not change the fiscal position relative to the counterfactual (i.e., if the action neither increases nor decreases spending or revenue compared to what would have occurred without it). If the fiscal action consists of suspending, reversing, or cancelling a previously planned contractionary (expansionary) measure, the impact should be classified as expansionary (contractionary) relative to the counterfactual. For example, if planned spending cuts are suspended, the impact is expansionary (EXP) because spending is higher than it would have been under the original plan. Ensure your responses to subsequent questions remain consistent with the answer provided here. Select one of the following options:

- EXP
- CON
- NEUT
- UNCLEAR

Q7b RATIONALE FOR CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION

Provide an exact quote from Q2 RELEVANT TEXT to support the direction of fiscal policy identified in the previous question Q7a. The output should be formatted as “[QuoteNumber] QuoteText”.

Q8 QUALITATIVE SIZE OF FISCAL ACTIONS

What is the size of the fiscal actions taken in this quarter?

SMALL → adjectives such as mild, minor, small, modest.

LARGE → adjectives such as large, major, enormous, unprecedented, significant.

UNCLEAR → no such adjectives exist for making this judgement

Q9 TAX EFFECT

What is the net effect of the identified fiscal actions on tax revenues? Positive on tax revenue means tax revenues increase, and negative on tax revenue means tax revenues decrease. Answer should be one of the following:

- POSITIVE
- NEGATIVE
- NO EFFECT
- UNCLEAR

Q10 SPENDING EFFECT

What is the net effect of the fiscal actions identified on government expenditure? Positive on government spending means government spending increases, negative on government spending means government spending decreases. For cuts in subsidy (such as that on fuel, electricity, foods) or fuel price liberalization, spending effect should be assessed as NEGATIVE ceteris paribus, unless the report provides explicit offsetting increases. Answer should be one of the following:

- POSITIVE
- NEGATIVE
- NO EFFECT
- UNCLEAR

Q11a INTENSITY — EFFECT ON OVERALL BALANCE

Assign a score from -10 (strongly contractionary) to +10 (strongly expansionary) reflecting the fiscal impact of the identified fiscal action on the overall fiscal balance for the current quarter and the next quarter or the next year. This is a question about the impact of the identified actions on the overall balance, NOT about the overall fiscal stance of the current quarter, current year, next quarter, or next year.
Return:
Positive values for expansionary stances
Negative values for contractionary stances
NEUTRAL or UNCLEAR stances should be close to or equal to 0, but not necessarily exactly 0
Use higher magnitudes for stronger fiscal actions e.g. ±10-8 = large, ±7-4 = moderate, ±3-1 = small
The score should reflect the overall strength of fiscal action as interpreted from the report
If Q7a CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION is EXP, then the output here must have a positive value. If the output for Q7a is CON, then the output here for Q11a must have a negative value. The value of Q11a output should be consistent with Q8 QUALITATIVE SIZE OF FISCAL ACTIONS and Q11c SIZE OF EFFECT ON OVERALL BALANCE – VALUE.

Q11b RATIONALE FOR INTENSITY – EFFECT ON OVERALL BALANCE

Provide an exact quote from Q2 RELEVANT TEXT to support the intensity score chosen in Q11a. The output should be formatted as “[QuoteNumber] QuoteText”.

Q11c SIZE OF EFFECT ON OVERALL BALANCE – VALUE

If the report mentions the overall size of the fiscal impact of a fiscal action—such as expansionary or contractionary—provide the numeric value. If no size is mentioned, return null. The response to this question should be consistent with the responses to Q7a CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION, Q8 QUALITATIVE SIZE OF FISCAL ACTIONS, and Q11a INTENSITY—EFFECT

ON OVERALL BALANCE. This size should roughly correspond with the sum of values identified (if any) in Q12c SIZE OF TAX EFFECT – VALUE and Q13c SIZE OF SPENDING EFFECT – VALUE.

Q11d SIZE OF EFFECT ON OVERALL BALANCE – UNIT

If there is a size mentioned and identified in Q11c SIZE OF EFFECT ON OVERALL BALANCE – VALUE, provide the unit of the value identified in Q11c. Output one of the following:

local_currency
usd
percent_gdp
basis_points
OTHER

If no size is mentioned, return null. If Q11c returns null, then Q11d should return null as well.

Q11e SIZE OF EFFECT ON OVERALL BALANCE – DIRECT QUOTE

Provide an exact quote from Q2 RELEVANT TEXT to support the size of the net fiscal action identified in the previous question Q11c and Q11d. The output should be formatted as “[QuoteNumber] QuoteText”. If no size is mentioned, return null.

Q12a INTENSITY -- TAX EFFECT

Assign a score from -10 to +10 that reflects how strongly contractionary or expansionary the tax effect is on the overall fiscal stance. This means that when the tax effect contractionary, tax revenues increase; while when the tax effect is expansionary, tax revenues decrease.

Return:

Positive values for expansionary stances

Negative values for contractionary stances

NEUTRAL or UNCLEAR stances should be close to or equal to 0, but not necessarily exactly 0

Use higher magnitudes for stronger fiscal actions e.g. ±10-8 = large, ±7-4 = moderate, ±3-1 = small

The score should reflect the overall strength of fiscal action as interpreted from the report

If Q9 TAX EFFECT is POSITIVE (NEGATIVE), then the output here must have a negative (positive) sign. The value of Q12a output should be consistent with Q12c SIZE OF TAX EFFECT – VALUE.

Q12b RATIONALE FOR INTENSITY – TAX EFFECT

Provide an exact quote from Q2 RELEVANT TEXT to support the intensity score chosen in the previous question Q12a. The output should be formatted as “[QuoteNumber] QuoteText”.

Q12c SIZE OF TAX EFFECT – VALUE

If the report mentions the overall size of the fiscal expansion or contraction, provide the numeric value. If no size is mentioned, return null. The response to this question should be consistent with the responses to Q9 TAX EFFECT and Q12a INTENSITY -- TAX EFFECT.

Q12d SIZE OF TAX EFFECT – UNIT

If there is a size mentioned and identified in Q12c SIZE OF TAX EFFECT – VALUE, provide the unit of the value identified in Q12c. Output one of the following:

local_currency
usd
percent_gdp

basis_points
OTHER

If no size is mentioned, return null. If Q12c returns null, then Q12d should return null as well.

Q12e SIZE OF TAX EFFECT – DIRECT QUOTE

Provide an exact quote from Q2 RELEVANT TEXT to support the size of the net fiscal action identified in Q12c and Q12d. The output should be formatted as "[QuoteNumber] QuoteText". If no size is mentioned, return null.

Q13a INTENSITY -- SPENDING EFFECT

Assign a score from -10 to +10 that reflects how strongly contractionary or expansionary the spending effect is.
Focus on the fiscal actions identified:
Positive values for expansionary stances
Negative values for contractionary stances
NEUTRAL or UNCLEAR stances should be close to or equal to 0, but not necessarily exactly 0
Use higher magnitudes for stronger fiscal actions e.g. ±10-8 = large, ±7-4 = moderate, ±3-1 = small
The score should reflect the overall strength of fiscal action as interpreted from the report
If Q10 SPENDING EFFECT is POSITIVE (NEGATIVE), then the output here must have a positive (negative) sign. The value of Q13a output should be consistent with Q13c SIZE OF SPENDING EFFECT – VALUE.

Q13b RATIONALE FOR INTENSITY – SPENDING EFFECT

Provide an exact quote from Q2 RELEVANT TEXT to support the intensity score chosen in the previous question Q13a. The output should be formatted as "[QuoteNumber] QuoteText".

Q13c SIZE OF SPENDING EFFECT – VALUE

If the report mentions the overall size of the fiscal expansion or contraction, provide the numeric value. If no size is mentioned, return null. The response to this question should be consistent with the responses to Q10 SPENDING EFFECT and Q13a INTENSITY -- SPENDING EFFECT.

Q13d SIZE OF SPENDING EFFECT – UNIT

If there is a size mentioned and identified in Q13c SIZE OF SPENDING EFFECT – VALUE, provide the unit of the value identified in Q13c. Output one of the following:
local_currency
usd
percent_gdp
basis_points
OTHER

If no size is mentioned, return null. If Q13c returns null, then Q13d should return null as well.

Q13e SIZE OF SPENDING EFFECT – DIRECT QUOTE

Provide an exact quote from Q2 RELEVANT TEXT to support the size of the net fiscal action identified in Q13c and Q13d. The output should be formatted as "[QuoteNumber] QuoteText". If no size is mentioned, return null.

Q14 TAX/SPENDING MEASURES

List specific fiscal measures (tax or spending) under each identified fiscal action in the report, using short phrases (≤ 12 words each; e.g., “VAT increase”, “public wage freeze”, “infrastructure spending rise”).
Return two pipe-separated strings:

1. Stance-consistent measures — include only measures aligned with the overall stance in Q7a:
- If CON: include measures that reduce spending or increase taxes.
- If EXP: include measures that increase spending or cut taxes.
- If NEUT/Unclear: leave this string empty.
2. Other measures — include all remaining tax/spending measures that are mentioned but not aligned with the stance identified in Q7a (e.g., offsetting items, neutral/ambiguous measures, or measures without clear direction).

Q15 OTHER NOTE

Anything else of relevance to fiscal policy actions? Keep your response limited to 30 words at most.

Q16 RATIONALE FOR EXCLUSIONS

Very briefly explain why the other fiscal stance directions or motivations were not selected. Also briefly explain what other fiscal policy actions were excluded. For example, a fiscal policy action could have been excluded because it is a past fiscal action, a potential fiscal action for the future, or a fiscal action that was withdrawn or did not pass.

INTERPRETIVE GUIDANCE:

A discretionary change in fiscal policy refers to fiscal policy actions that go beyond the usual response of fiscal policies to business cycle fluctuations.

The EIU country reports are released for each country on a quarterly basis. They discuss the current economic and political situation as well as future projections. Please note that EIU country reports released in the fourth quarter may discuss the future more often to a greater extent than reports released during other quarters. The EIU typically says why some fiscal action was taken, not why some other action was not taken. That is, the counterfactual is implicitly or explicitly neutral fiscal policy. As a result, market access is inherently an explanation for fiscal expansions, and all of the possible motivations correspond to policymaker choices.

Note that while cause and motivation can be similar, they are not the same. As stated in Q3 CAUSE, “Cause of fiscal problem (what led to it)” is defined as Causes describe the **origin of fiscal pressures or fiscal space**, while motivations explain **why policymakers chose a particular fiscal action** to those conditions. (e.g., bank rescues, recession-induced revenue collapse, commodity price shock (negative or positive), natural disaster, one-off court rulings, legacy debt overhang, policy slippage, wars, unexpected strong growth or revenue performance). Motivation captures the objectives of the specific fiscal actions chosen in response to the cause.

There are a few notes on timing. While a given country-year-report may discuss a past fiscal action, that does not mean every report does have a past fiscal action. A past fiscal action is not necessarily a cause for a current fiscal action. While it certainly can be, capturing a past action as the cause for a current one should only occur if that is explicitly mentioned in the text. Example of a timing edge case: A Portugal 2013 Q4 report discusses the unveiling of the 2014 budget. This should be classified as a current fiscal action with the status “ANNOUNCED.” It is current because the announcement occurs within 2013 Q4, and announced because the measures pertain to the following fiscal year (2014).

Answers for questions Q7a through Q16 must be consistent. For taxes, POSITIVE for Q9 TAX EFFECT means an increase in taxes. This is contractionary, so the intensity should have a negative value to reflect the contractionary stance (Q12a INTENSITY -- TAX EFFECT). For spending, POSITIVE for Q10 SPENDING EFFECT means an increase in government spending. This is expansionary, so the intensity should have a positive value to reflect the expansionary stance (Q13a INTENSITY -- SPENDING EFFECT). If the size is identified, it should make sense with the intensity and the outputs in Q7a CURRENT AND PROSPECTIVE IMPACT OF EACH FISCAL ACTION if we are referring to the net fiscal action, Q9 TAX EFFECT if we are referring to the size and intensity of the tax effect, and Q10 SPENDING EFFECT if we are referring to the size and intensity of the spending effect. If the tax and spending effect are summed, this should be consistent with the size/direction/intensity of the net fiscal action.

Notice that the answer choices provided in Q4a MOTIVATION are labeled by number-letter combinations. This is to indicate that each of those answer choices (or motivations) fall into one of six larger categories of motivations: (1) fiscal responses to macroeconomic and financial conditions, (2) debt sustainability and relief, (3) long-term growth, (4) redistribution, (5) provision of public goods and externalities, and (6) politically related motivations. Motivations capture what a fiscal action is expected to achieve. Motivation is not the same as the cause of a fiscal action.

Below is information about each motivation:

1. Stabilization

a) **Financial rescue:** Fiscal measures to stabilize or recapitalize financial institutions or markets during periods of actual or imminent financial distress, aimed at restoring macro-financial stability under current conditions. Financial rescue mostly corresponds to bank bailouts/port provided to the financial sector. Financial rescue should be included whenever there is mention of rising debt or deficits due to support being provided to the financial sector or banks. Shoring up the bank sector also qualifies as financial rescue. Financial rescues are fiscal actions even if the EIU doesn't label them that way.

b) **Fiscal responses to the economic cycle:** Fiscal actions taken in response to current or expected economic conditions, including both downturns or periods of overheating. Such fiscal actions can be countercyclical or procyclical. This motivation should be chosen only when there is an explicit link between a policy action and the state of economy. More concretely, this motivation should be selected only if (i) contractionary fiscal actions are explicitly linked to overheating economic conditions or economic weakness or (ii) expansionary fiscal actions are explicitly linked to either economic weakness or economic overheating.

c) **Response to commodity price fluctuations:** Fiscal actions taken in response to changes in global commodity prices affecting key exports or imports. These actions may be pro-cyclical or counter-cyclical. The classification of pro- versus counter-cyclical responses to commodity price movements depends on whether the country is a commodity exporter or importer. For exporters, procyclicality implies expansion (contraction) when prices rise (fall), while countercyclicality implies expansion (contraction) when prices fall (rise). For importers, the opposite mapping applies. Select this category only when an explicit link to commodity price changes is stated..

2. Debt sustainability and debt relief

a) **Deficit-reduction and fiscal discipline measures:** Fiscal measures to reduce budget deficits or strengthen public finances, motivated by the need to safeguard long-term fiscal sustainability.

b) **Fiscal rules coming into play:** Fiscal actions triggered by statutory or constitutional fiscal rules—such as debt ceilings, deficit limits, or expenditure caps—aimed at ensuring compliance with pre-set fiscal constraints and preserving debt sustainability.

c) **Concerns about access to markets:** Fiscal measures to preserve or restore access to sovereign debt markets on acceptable terms, undertaken to sustain debt service capacity and maintain fiscal credibility over the medium to long term. Market access captures austerity that the EIU identifies as being driven by very high sovereign spreads, inability to borrow, or other problems with private sovereign funding.

d) **Compliance with conditionalities imposed by external actors:** Measures adopted to meet policy requirements from external entities (e.g., IMF, other IFIs, regional organizations) as part of financial assistance, rather than in response to domestic economic conditions. "Conditionality" is the hybrid of market access and IMF (or other bailout organization) choices.

e) **Response to external financial aid or debt relief:** Fiscal responses to grants, concessional loans, or debt relief from foreign or international sources, driven by external inflows rather than domestic macroeconomic conditions.

3. Long term growth

a) **Productivity and capacity-enhancing measures:** Fiscal measures—such as structural reforms, large-scale infrastructure programs, or other productivity-enhancing initiatives—aimed at raising the economy's long-term output level or growth path, independent of current economic conditions.

b) **Resource discovery-driven development:** Fiscal responses to significant new resource finds, designed to leverage these endowments for future economic development rather than address immediate macroeconomic conditions.

4. Redistribution

a) **Targeted redistribution to specific groups:** Direct benefits—via spending, subsidies, or tax measures—to specific population groups or regions to address disparities.

b) **Broad-based redistribution and social equity programs:** Progressive taxes, transfers, or public services aimed at reducing overall income or wealth inequality.

c) **Geographic redistribution:** Fiscal transfers or investments to reduce disparities between regions or localities.

d) **Intergenerational redistribution:** Measures shifting resources across age cohorts, such as pensions, education funding, or youth employment programs.

5. Provision of public goods and externalities

a) **Defense and national security provision (e.g., wars):** Fiscal measures to finance or prepare for military conflict, strengthen defense capabilities, or address national security threats, ensuring safety as a public good.

b) **Public health protection during crises:** Fiscal actions to prevent, contain, or mitigate large-scale health emergencies, safeguarding public health as a public good.

c) **Event-related public infrastructure and services (e.g., Olympics):** Fiscal measures to prepare for or host major international events, providing infrastructure and services with broad public benefit.

d) **Post-disaster public service restoration:** Fiscal actions to restore essential infrastructure and services following major natural disasters.

e) **Humanitarian services for refugee and migration crises:** Fiscal measures to provide humanitarian assistance and essential services for large inflows or outflows of people.

f) **Peace and humanitarian public goods provision:** Fiscal actions to deliver peace commitments or humanitarian relief, ensuring security and welfare as public goods.

g) **Energy transitions/Climate change mitigation:** Fiscal measures to support cleaner energy, cut emissions, or adapt to climate change, undertaken to promote sustainable long-term growth and meet environmental or international commitments (e.g., Green New Deals, coal phaseouts, nuclear expansions).

6. Politically related

a) **Pursuit of political or electoral objectives:** Fiscal actions undertaken primarily for political or electoral purposes—such as rewarding constituencies, fulfilling campaign promises, or signaling policy priorities—where the intent is unrelated to influencing either near-term economic conditions or long-term growth prospects. Politics could also come into play if political pressure is driving some considerations in the fiscal actions or stance.

b) **Major political transitions or revolutions:** Fiscal actions taken during fundamental political changes—such as regime change, revolutions, secession, unification, or constitutional restructuring—prompted by institutional shifts, not current economic conditions.

Please ensure that whenever the report explicitly mentions fiscal actions related to infrastructure, public services, or investments connected to major international or national events (e.g. Olympic Games, World Cup, major expositions), motivation 5C (Event-related public infrastructure and services) must be assigned in addition to other relevant motivations such as 3A (Productivity and capacity-enhancing measures).

You should not pick up normative statements. You should only capture objective statements that describe the facts of the fiscal action.

Also provide your confidence in the overall assessment as an integer from 0 to 100 where 0 is no confidence and 100 is absolutely confident. Use higher values only when the evidence is clear and strong. When in doubt, be cautious in assigning a high score.

EXAMPLES:

Here are some examples for each of the motivations being identified:

a. 1A: “The government has been busily helping the banks out of their deepest crisis in 50 years.”

b. 1B: “The government has announced an expansive budget … in a bid to stimulate demand; A downward revision of economic forecasts … were the reasons for the expansionary programme.” OR “As growth accelerates the government will continue to introduce spending cuts in a counter-cyclical fashion.”

c. 1C:

d. 2A: “The finance minister … attempted to allay concerns about the possible deflationary impact of fiscal tightening by stressing that fiscal reform would lead to an increase in demand in the medium and longer term.”

e. 2B: “A budget framework programme for 2011-14 that is currently before parliament foresees the deficit falling below the 3% of GDP euro zone threshold from 2013 (this target has been agreed with the [European] Commission).”

f. 2C: “The minister of the economy … appears determined to keep Italy’s public finances under control, fearing that a severe deterioration would lead to a further sharp widening of interest rate spreads on Italy’s government debt.”

g. 2D: “Economic policy will be largely dictated by Greece’s need to obtain financial help from EU partners and the IMF, requiring tax rises, continuing expenditure cuts and a range of structural reforms.”

h. 2E: “Like its predecessors, the spending plans of the recently presented 2006 budget are reliant on the goodwill of donors, since external grants are budgeted to contribute more than 50% of government revenue for the year.”

i. 3A: “Investment in key infrastructure projects, such as the long-delayed Padma bridge, will also stretch the public finances, especially if project costs overrun. Although tax collection will continue to grow at a fairly rapid pace, it will not be fast enough to close the gap between revenue and expenditure.”

j. 3B:

k. 4A: “A further Lari29m was allocated to the Ministry of Agriculture to help farmers to buy agricultural machinery and diesel, Lari34m went to Georgia's regions for the implementation of various investment

projects, and the Ministry of Refugees and Accommodation received an additional Lari24m to fund increases in the monthly allowances for internally displaced persons (IDPs)."

l. 4B: "The increases in welfare spending correspond to the action plan launched by the president, Mikheil Saakashvili, in February to bring about a tangible reduction in poverty within a period of 50 days (March 2008, Economic Policy), and further underlines that the emphasis in spending priorities has, at least for the time being, shifted from defence to welfare."

m. 4C:

n. 4D: "The amendments envisage an increase in state spending of Lari216m (US$150m). Of this additional sum, around Lari52m was allocated to the Ministry of Labour, Healthcare and Social Welfare to fund an increase in the basic pension from Lari55 (US$36) to Lari70, which took effect on April 1st rather than next October, as originally planned."

o. 5A:

p. 5B: "In May the government announced that it was setting aside Rmb2bn (US$242m) to finance the treatment of SARS patients and upgrade rural hospitals, and granted tax breaks and other relief measures to the worst-affected businesses… The government is now likely to accelerate nationwide implementation of the new rural healthcare funding scheme that is already being piloted in some areas of the country."

q. 5C: "It has also been announced that Pta2,902 bn ($22 bn) is to be spent on Spain's railways between 1987 and the year 2000. According to the minister of transport, Mr Abel Caballero, the greater part of expenditure will be prior to 1992 (when the Olympic Games will be held in Barcelona and the fifth centenary of the discovery of America by Columbus will also be celebrated)."

r. 5D: ""The government quickly formulated a •4trn (US$484bn) supplementary budget to pay for removing debris from the Tohoku region, building temporary shelters, repairing roads and highways, and offering temporary loans to troubled businesses."

s. 5E: "While no official declaration was made, it seems that Mr Peres had decided to increase the deficit, and to pay for it by raising loans abroad. The favourable level of foreign currency reserves and the low level of the foreign debt enabled him to follow such a path. A clear indication that this was the former finance minister's preferred solution was the initiative launched by several of Israel's supporters in the US Senate early in February. The bill they have tabled would provide for the US government underwriting a $400 mn loan to Israel to help finance the housing of Soviet immigrants."

t. 5F:

u. 5G:

v. 6A: "The government, which had proposed cuts last autumn of Nkr5bn in state benefits as from 1994, cannot afford to do so now as it is trailing in the opinion polls and a general election is coming up in September. Cuts of half that amount (Nkr2.5bn) are therefore being proposed instead."

w. 6B: "Revenue will recover as economic growth picks up, and we expect the next government to be forced to contemplate some fiscal rationalisation, leading to a gradual narrowing of the deficit up to 2016."

Annex B. Exhibits: AI Output and Economists' Assessments for Selected EIU Reports

This annex provides exhibits for 31 randomly selected EIU reports, documenting the AI-generated output, the relevant EIU source text, and economists' assessments of the output. The economists' assessments are based only on a review of the text in the corresponding EIU report for a country and quarter. The exhibits are ordered as they appear in the full database.

Exhibit 1: Argentina 2001Q1

AI-based output

Direction. Fiscal expansion.

Motivation. 1b. Fiscal responses to the economic cycle; 3a. Productivity and capacity-enhancing measures; 2a. Deficit-reduction and fiscal discipline measures; 4a. Targeted redistribution to specific groups.

EIU text: *"The recently launched federal infrastructure plan should deliver a boost to investment and employment… and help to establish a cycle of positive expectations."*

Status. Signed and implemented in current quarter (Q1 2001), with measures enacted via executive decrees in January.

EIU text: *"Further executive decrees issued in January 2001 included… implementation of the federal infrastructure plan…"*

Composition. Primarily Spending, with minor Tax elements.

EIU text: *"The federal infrastructure plan… entails a total cost of some US$20bn."; "An initiative aimed at stimulating investment… will create… subsidies…"; "The minimum purchase amount on which tourists can seek VAT refunds has also been reduced…"*

Intensity. Overall (+6); Spending (+6); and Tax (+2).[1]

EIU text: *"The federal infrastructure plan… is the only expansionary programme of public investment currently proposed."; "Investment… expected to total US$1.5bn in 2001, creating… 400,000 direct jobs…"; "VAT refunds… reduced… from US$200 to US$70."*

Economist notes

All categories have been correctly identified. The package represents a sizable expansionary effort centered on public investment and targeted subsidies, despite some offsetting consolidation

[1] - (+) values denote reduction (increase) in the fiscal deficit.

measures (such as pension cuts). The overall score of +6 appropriately reflects the large infrastructure program and its expected employment impact. The dominance of spending measures justifies the high spending score (+6), while the modest tax component (+2) captures limited revenue reductions (e.g., VAT refund changes). This policy mix reflects an attempt to support growth while maintaining some degree of fiscal discipline under tightening financing constraints.

Exhibit 2: Australia 2009Q2

AI-based output

Direction. Fiscal expansion.

EIU text: *"The sizeable widening of the budget deficit reflects the government's immediate aim of softening Australia's expected economic recession."*

Motivation. 1b. Countercyclical measures; 4d. Intergenerational redistribution; 3a. Productivity and capacity-enhancing measures.

EIU text: 1b — *"The sizeable widening of the budget deficit reflects the government's immediate aim of softening Australia's expected economic recession"*; 3a — *"The federal budget outlined spending of A$22bn on infrastructure projects"* and *"the federal government will increase spending on science and innovation by 25% in 2009/10 as part of an innovation package worth A$3.1bn in the next four years"*; 4d — *"the government has suggested that the increased [aged pension] payment will be funded by a phased increase in the retirement age from 65 to 67 years by 2023"*.

Status. Announced (May 12, 2009) for the 2009/10 Budget year (effective July 2009).

EIU text: *"The federal government's budget for 2009/10 was handed down on May 12th."*

Composition. Spending (net tax effect unclear due to offsetting measures).

EIU text: *"The federal budget outlined spending of A$22bn on infrastructure projects"; "[the aged pension] increase, which is forecast to cost A$2.7bn in 2009/10, represents nearly one-half the budget's A$6bn of new spending initiatives for that year"; "The budget increased the tax deduction rate to 50% for small businesses … taking the total cost of the measure to A$3.7bn", while "cutting the maximum concessional contribution to superannuation pension schemes" and "plans to limit the current private health insurance rebate … to individuals earning up to A$75,000 per year and families with an income of up to A$150,000 per year".*

Intensity. Overall (+6); Spending (+6); and Tax (0).[2]

EIU text: *"The government expects its stimulus measures, most of which had already been announced in two earlier fiscal packages, to boost the economy by the equivalent of 2.4% of GDP in 2009/10 and 1.8% in 2010/11"; "The federal budget outlined spending of A$22bn on infrastructure projects".*

Economist notes

The direction is correctly identified: the report is explicit that the deficit widening (to 4.5% of GDP in 2009/10) is intended to soften the recession. The primary motivation — 1b (Countercyclical measures) — is clearly correct and is the dominant theme of the budget. 3a (Productivity and capacity-enhancing measures) is also a reasonable assignment given the A$22bn confirmed for infrastructure projects and the A$3.1bn four-year innovation package (with science and innovation

[2] - (+) values denote reduction (increase) in the fiscal deficit.

spending up 25% in 2009/10). The 4d (Intergenerational redistribution) call is more debatable: it relies on pairing the immediate aged pension increase (A$2.7bn in 2009/10) with the phased retirement-age increase to 67 by 2023, which shifts the financing cost onto future cohorts; this is a defensible reading, but the linkage is implicit in the report. The AI also arguably misses 4a (Targeted redistribution to specific groups), since the aged pension increase is targeted at pensioners and the budget rolls back benefits for higher-income groups (superannuation contribution caps; means-testing of the private health insurance rebate). 2a (Deficit-reduction and fiscal discipline) is appropriately not included for the current quarter, since the report frames return-to-surplus only as a medium-term objective and contains no contemporaneous discipline measures. The status is correctly "Announced," with an effective date of July 2009 (start of FY2009/10). The composition assessment is consistent with the report: spending is the clearly dominant channel, while the net tax direction is genuinely unclear given offsetting measures (50% small-business deduction expansion vs. tighter superannuation caps and the private health insurance rebate cap; the proposed employee share-scheme tax was withdrawn after announcement). The overall intensity score of +6 is on the higher end given that the 2.4%-of-GDP figure refers to the cumulative stimulus across earlier packages (most of which is correctly attributed to the Past row), and the marginal new spending initiatives in this budget total only A$6bn (~0.5% of GDP). That said, the budget formally confirms the A$22bn infrastructure envelope and the A$3.1bn innovation package, which justify a substantial reading. Spending intensity (+6) is appropriate; tax intensity (0) correctly captures the offsetting tax measures.

Exhibit 3: Bangladesh 2015Q3

AI-based output

Direction. Fiscal Action

Motivation. 4a. Targeted redistribution to specific groups.

EIU text: *"A 50% wage increase for all state employees that came into effect on July 1st has yet to filter through into the general price level."*

Status. Implemented in 2015 Q3.

EIU text: Same as above

Composition. Spending

EIU text: Same as above

Intensity. Overall (+8); Spending (+8); and Tax (0).[3]

EIU text: Same as above

Economist notes

The AI correctly identifies the wage hike as the main fiscal action and appropriately classifies it as a spending measure implemented in 2015 Q3. However, the motivation appears misclassified: a 50% increase in public-sector wages is not, by itself, clear evidence of redistribution, and is better categorized as "Other" absent supporting context in the EIU report. The intensity also seems overstated. While the increase is large in proportional terms, it is limited only to public-sector wages, and the EIU text provides no indication of its size relative to the overall budget or GDP. An overall and spending score of +8 therefore appears too high without evidence of a substantial aggregate fiscal impact.

[3] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 4: Brazil 2003Q4

AI-based output

Direction. Fiscal expansion.

Motivation. 4b. Broad-based redistribution and social equity programs; 4a. Targeted redistribution to specific groups; 6a. Pursuit of political or electoral objectives.

EIU text: *"Following the weak start of the Fome Zero (Zero Hunger) project (September 2002, The political scene), a review of welfare programmes led to the launch of the new Bolsa Familia programme in October in order to bring together benefits distributed under a host of existing provisions", "The aim is to reach 3.6m families by the end of 2003. The total amount distributed to the poorest families is targeted to rise by 75% on 2002 levels", "To maintain popular support the government will seek to shift the emphasis of policy towards addressing problems of poverty and marginalisation through income redistribution, land reform and welfare programmes, but there will be challenges from all sides", " the government launched a plan for agrarian reform in November … The plan sets targets for the redistribution of land … It also pledges to improve sanitation, electricity, infrastructure and housing", "A further important tax reform proposal, the conversion of the contribuicao para o financiamento da seguridade social (Cofins, employers' social security contributions) from a payroll to a value-added tax, bypassed Congress under a medida provisoria issued on November 4th. The rate at which the Cofins is charged was converted from 3% of turnover to 7.6% of value added. According to government calculations, the change will not generate an increase in total revenue", "The new model also includes a programme to give universal access to electricity. Launched on November 11th, it envisages the disbursement of R7bn in 2003-08 to extend electricity supplies to an estimated 12m people, 80% of them in rural areas.", "Having been passed by the House of Deputies in August (September 2003, Economic policy), the government's first pension reform bill was submitted to the Senate in early October, and on November 26th won approval at the first vote with only minor changes", "A ceiling on public-sector salaries was approved", "The social security reform is now expected to win approval in December… it will still significantly reduce the fiscal burden of future public pensions liabilities."*

Status. Implemented in October.

EIU text: *"launch of the new Bolsa Familia programme in October…"*

Composition. Spending

EIU text: *"…launch of the new Bolsa Familia programme … aim is to reach 3.6m families by the end of 2003"; "programme to give universal access to electricity"; "government's first pension reform bill was submitted to the Senate … won approval at the first vote"; "A ceiling on public-sector salaries was approved" "A further important tax reform proposal… bypassed Congress…".*

Intensity. Overall (+6); Spending (+6); and Tax (0)[4]

[4] - (+) values denote reduction (increase) in the fiscal deficit.

EIU text: *"The total amount distributed to the poorest families is targeted to rise by 75% on 2002 levels, and a major monitoring programme is to be put in place to measure its effectiveness in terms of social indicators."; "The rate at which the Cofins is charged was converted from 3% of turnover to 7.6% of value added. According to government calculations, the change will not generate an increase in total revenue. This assertion has been disputed…".*

Economist notes

The direction, motivation, and composition are largely correct. The spending intensity score of +6 is driven by the expected 75% increase in distributions under the *Bolsa Familia* programme. This assessment is appropriate but could be overstated, as the impact of the *Bolsa Familia* measure may be offset by contractionary effects from the ceiling on the public sector wage bill (though the ceiling's fiscal impact is not quantified the report). The tax measure is not expected to generate additional revenues, so the zero tax intensity is appropriate. The other tax measures mentioned in the report ("*The Senate is also debating the tax reform bill, a watered-down version of which was approved by the House of Deputies in September*") are appropriately not included as they are still being debated, with aspects of the bill expected to be referred back to the lower house for approval (page 21). There is another potential current fiscal action that the AI missed as a current action but classifies as a future one: "*On November 5th the government confirmed that it would renew its agreement with the IMF.*" However, the exclusion of this action from current actions is a matter of interpretation, as the IMF program involves no significant changes relative to the previous arrangement agreed with the IMF (hence potentially no fiscal action). In addition, the associated fiscal impact—overshoot in primary surplus to be used to increase investment in sanitation in 2004—is a future impact.

Exhibit 5: Canada 2003Q4

AI-based output

Direction. Neutral

EIU text: *"The government will use its contingency reserve and allowance for economic prudence to balance its budget."*

Motivation. 2a. Deficit-reduction and fiscal discipline measures.

EIU text: *"Recent setbacks to the economy have taken their toll on federal finances, and it is increasingly likely that the government will have to draw on its contingency reserve to balance its budget in fiscal year 2003/04 (April-March)."*

Status. Announced

Composition. Unclear.

Intensity. Overall (0); Spending (0); and Tax (0)[5]

Economist notes

AI correctly identifies the description of budget-balancing (using contingency reserves) as a current fiscal action that is neutral in magnitude. The composition and intensity are also appropriate. The announcement of the 2003 budget on February 18th and the ongoing implementation of a five-year tax reduction package are appropriately not classified
as a current fiscal action.

[5] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 6: Switzerland 2014Q2

AI-based output

Direction. Unclear.

Motivation. 7. Other.

EIU text: *"[1] A new global agreement calls on countries to share automatically data on one another's taxpayers. Dozens of countries on May 6th signed up to the pact, which promises to sideline tax havens and other secretive financial hubs in the coming years.; [2] The willingness of Switzerland and Singapore to sign-on to the arrangement was crucial in arriving at a deal, announced at a meeting of senior government officials at the OECD in Paris.; [3] The pact's central reform calls on all participating nations to share information on each other's residents automatically on an annual basis. The data provided will include bank balances, securities holdings, dividends, interest and capital gains.; [4] Pressure from the OCED and G20, combined with that from FATCA, has proven insurmountable.; [5] Many countries will have to adopt new laws and regulations to meet the terms of the agreement. As a result, the timeline for implementation is relaxed. So-called "early adopters" aim to have an information-sharing system in place by 2017, relying on taxpayer data for 2015."*

Status. Signed in 2014Q2.

EIU text: *"A new global agreement calls on countries to share automatically data on one another's taxpayers. Dozens of countries on May 6th signed up to the pact, which promises to sideline tax havens and other secretive financial hubs in the coming years;"*

Composition. Unclear.

EIU text: *"The pact's central reform calls on all participating nations to share information on each other's residents automatically on an annual basis. The data provided will include bank balances, securities holdings, dividends, interest and capital gains."*

Intensity. Overall (0); Spending (0); and Tax (0).[6]

EIU text: *"The pact's central reform calls on all participating nations to share information on each other's residents automatically on an annual basis. The data provided will include bank balances, securities holdings, dividends, interest and capital gains."*

Economist notes

While a fiscal action is taken this quarter (the pact signing), this action does not have a clear impact on the fiscal stance. The AI correctly reports that a neutral fiscal action is taken this quarter.

[6] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 7: Chile 2014Q4

AI-based output

Direction. Fiscal expansion.

Motivation. 1b. Countercyclical measures; 3a. Productivity and capacity-enhancing measures; 4b. Broad-based redistribution and social equity programs.

EIU text: *"Overall public investment in the 2015 budget will rise by 27.5% year on year, with a good chunk of this going towards investment in infrastructure"; "The Chamber of Deputies (the lower house) on November 25th approved a proposal to raise public-sector salaries by 6%, effective on December 1st."; "Together with a 2015 budget that is heavy on social spending (68% of a total US$64.7m in expenditure) and infrastructure investment, the pay rise will help to lift Chile out of its economic slump"; "With public-sector employment at around 10% of the total workforce, the pay rise will add to already-increasing state spending, and will contribute to a fiscal deficit in 2015".*

Status. Announced (date not available) for next Budget year (2015).

EIU text: *"Together with a 2015 budget that is heavy on social spending … and infrastructure investment".*

Composition. Spending.

EIU text: See above.

Intensity. Overall (+6); Spending (+7); and Tax (0).[7]

EIU text: *"Overall public investment in the 2015 budget will rise by 27.5% year on year … The heftiest increases in funding will go to the ministries responsible for overseeing public works … equating to a total of around US$3.6bn, a record high … the pay rise will help to lift Chile out of its economic slump."*

Economist notes

All categories are largely correctly identified. The primary motivation is to lift the economy out of its slump, as repeatedly mentioned in the report. There is an additional motivation (4a. Targeted redistribution to specific groups) that could have been missed, as the pay rise is for a targeted group (public sector employees) and was motivated by the "government['s] aims to reduce income inequality" (p. 29). The status is correct, as the most significant fiscal actions are part of the 2015 budget. While the announcement date of the budget is not included in the EIU report, external checks indicate an announcement date of September 30, 2014, so the actions identified can reasonably be attributed to 2014Q4 (+/- 1 day). In terms of composition, spending actions are the only actions announced in the current quarter. Corporate tax reforms were passed in September 2014, and a new "green" tax was also announced in September. The intensity is appropriate, as the budget contains a significant (27.5% year over year) increase in public spending.

[7] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 8: Denmark 1997Q4

AI-based output

Direction. Fiscal contraction.

Motivation. 1b. Fiscal responses to the economic cycle; 2a. Deficit-reduction and fiscal discipline measures.

EIU text: *"The finalised 1998 budget was presented as an 'austerity package' ... The budget includes several measures aimed at slowing down economic growth."*

Status. Approved in current quarter (October 1997), for Budget year 1998.

EIU text: *"The 1998 budget, which was adopted on October 6th, was presented as an 'austerity package'."*

Composition. Spending and Tax.

EIU text: *"The government ... announced a one-off contribution to the Supplementary Pension Scheme and an interim increase in stamp-duty ... and extra expenditure savings of Dkr2bn."*

Intensity. Overall (-6); Spending (-6); and Tax (-3).[8]

EIU text: *"It includes savings and budgetary cuts in excess of Dkr6bn ... These measures are intended to shave 0.6 percentage points off GDP growth."*

Economist notes

Most categories are correctly identified. The AI isolates the October adoption of the 1998 austerity budget and codes the direction appropriately as contractionary. The main reservation concerns intensity and composition: the EIU report gives more salience to the one-off pension scheme contribution and the stamp-duty increase than to the smaller Dkr2bn spending cuts, so the package looks medium-sized overall and somewhat more tax-led than the intensity score composition suggests. The debt-discipline motive is present, but the report places greater emphasis on cooling an overheating economy and private consumption.

[8] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 9: Egypt 2012Q3

AI-based output

Direction. Fiscal expansion.

Motivation. 4a. Targeted redistribution to specific groups; 4b. Broad-based redistribution and social equity programs; 6a. Pursuit of political or electoral objectives**.**

EIU text: "*4a: <[2] Immediately after it was passed, Mr Morsi announced a 15% increase in public-sector salaries and pensions.> | 4b: <[13] The government will combine a focus on achieving greater social equality with some liberal reforms.> | 6a: <[3] Post-revolutionary administrations have shown some populist tendencies…, but some liberal economic policies have also been revived.>"*

Status. Announced.

EIU text: *"[2] …, Mr Morsi announced a 15% increase in public-sector salaries and pensions."*

Composition. Spending

EIU text: *"[2] Immediately after [the budget for fiscal year 2012/2013 (July-June) was passed, Mr Morsi announced a 15% increase in public-sector salaries and pensions."*

Intensity. Overall (+3); Spending (+4); and Tax (0)[9]

EIU text: *"[2] …, Mr Morsi announced a 15% increase in public-sector salaries and pensions."*

Economist notes

The EIU report provides limited detail on specific budgetary measures, noting that the budget may still be revised. Accordingly, AI has focused on wage and pension increases, as these are the only clearly articulated measures. Fuel subsidy cuts are identified as offsetting mechanisms. AI appropriately excluded measures that will likely be part of future IMF program negotiation and PPP activity. All of these are not fiscal measures.

[9] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 10: Finland 2016Q2

AI-based output

Direction. Fiscal contraction.

Motivation. 2a. Deficit reduction; 3a. Productivity and capacity-enhancing measures; 5e. Humanitarian services for refugee and migration crises.

EIU text: *"Finland's public finances remain fragile, and broad consensus among the main political parties about the need to improve them will keep austerity at the top of the government agenda.; ...stimulus measures designed to support long-term economic growth, including job-creating infrastructure projects and investment in education; The government has accounted for additional spending on accommodating rising numbers of refugees, with costs based on an estimated annual intake of 10,000 asylum-seekers during this period."*

Status. Announced, April 2016; for year 2017.

Composition. Spending.

Intensity. Overall (-3); Spending (-3); and Tax (-0).[10]

EIU text: *"On April 5th the government presented further austerity measures in its fiscal plan for 2017-20, which outlined additional cuts to public spending worth euro400m, on top of euro4bn."*

Economist notes

Motivations, announcement and composition correctly identified. The €400 million adjustment is equivalent to about 0.2 percent of 2015 nominal GDP, suggesting a relatively small spending cut.

[10] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 11: France 2011Q2

AI-based output

Direction. Contraction

EIU text: *"According to the government, the reform of the ISF and the various other tax changes associated with it will result in a small net annual gain for the Treasury, which is put at a little short of €200m, once they have fully come into effect during 2012."*

Motivation. 6a. Pursuit of political or electoral objectives; 4b. Broad-based redistribution and social equity programs; 2a. Deficit-reduction and fiscal discipline measures.

EIU text: on political related measures: *"Aside from deficit reduction, the pre-election economic policy agenda will be limited to measures to boost popularity. The government intends to scrap the 'tax shield' and open discussions on financing long-term care for the elderly."* On redistribution: *"The tax shield has come to be seen as a symbol of fiscal inequality and its abolition is partly designed to undercut the PS, which has made promoting 'fairer' taxation a centrepiece of its policy agenda."* On debt sustainability and debt relief: *"According to the government, the reform of the ISF and the various other tax changes associated with it will result in a small net annual gain for the Treasury, which is put at a little short of €200m, once they have fully come into effect during 2012."*

Status. Cabinet Approved

EIU text: *"(…) Loi de finances initiale, which the budget minister, Francois Baroin, presented to the Council of Ministers for its formal approval on May 11th. This draft law is now before parliament, where it will be debated in June and eventually adopted, no doubt largely in its present form."*

Composition. Tax and Spending

EIU text: on tax: *"The government has lowered its growth forecasts and outlined budget savings worth an additional €4bn in 2011 and €11bn in 2012, mainly through higher taxes."* On spending: *"(…) while efforts to contain state expenditure will amount to just €1bn in 2012."*

Intensity. Overall (-2), Spending (0); and Tax (-2)[11]

EIU text: overall intensity: *"According to the government, the reform of the ISF and the various other tax changes associated with it will result in a small net annual gain for the Treasury, which is put at a little short of €200m, once they have fully come into effect during 2012."* On tax intensity: *"a small net annual gain for the Treasury, which is put at a little short of €200m, once they have fully come into effect during 2012."*

Economist notes

All categories have been correctly identified except status. The evidence does not indicate cabinet approval because the text explicitly says the draft law was "presented to the Council of Ministers for its formal approval," which implies approval is being sought rather than already granted. In addition,

[11] - (+) values denote reduction (increase) in the fiscal deficit.

it immediately notes that the proposal is still before parliament and yet to be adopted, confirming it remains at a pre-approval stage.

Exhibit 12: United Kingdom 2005Q4

AI-based output

Direction. No action.

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

All categories have been correctly identified. The AI output rightly points that there are "No fiscal action was taken or announced in Q4 2005; report references past actions (July/September) and forecasts only." Indeed, this is correct. For instance, on past actions, the report says "(…) the government faces increasing pressure from a wide range of existing spending commitments". On anticipated measures, the report says "(…) the government will have no option but to raise taxes further to fund continued increases in public expenditure".

Exhibit 13: United Kingdom 2010Q2

AI-based output

Direction. Fiscal contraction.

Motivation. 2a. Deficit-reduction and fiscal discipline measures; 2c. Concerns about access to markets.

EIU text: *"The government has already announced modest cuts in planned spending of £6.3bn (US$9.2bn) this year, in a bid to show investors it is serious about deficit reduction."; "or do too little and run the risk of investors shunning UK sovereign debt, placing upward pressure on interest rates and government borrowing costs, downward pressure on sterling and increasing the possibility of a downgrade of the UK's AAA sovereign rating."*

Status. Announced in current quarter (in 2010), for Budget year (2010).

EIU text: "*The government has already announced modest cuts in planned spending of £6.3bn (US$9.2bn) this year, in a bid to show investors it is serious about deficit reduction.*"

Composition. Spending and Tax

EIU text: "*The government has already announced modest cuts in planned spending of £6.3bn (US$9.2bn) this year*". *"The recent introduction of a new 50% top income tax rate in April..."*

Intensity. Overall (-3); Spending (-3); and Tax (-3).[12]

EIU text: "*The government has already announced modest cuts in planned spending of £6.3bn (US$9.2bn) this year.", "The recent introduction of a new 50% top income tax rate in April..."*

Economist notes

Direction and motivation have been correctly identified. The status ("announced") and overall intensity (–3) appear to be driven primarily by the quantified spending measures, which appears reasonable given the scale of the announced cuts—£6.3 billion (around 0.4 percent of 2010 nominal GDP). Meanwhile, the reference to the introduction of a 50% top income tax rate in April 2010 should be interpreted with some nuance. This measure was announced and legislated in 2009, even though it came into effect in April 2010. While the EIU report does not specify the announcement or approval date, it is reasonable to infer that this reflects a previously decided policy rather than a new action in the current period. Still, given its salience, an intensity score of –3 for tax is not unreasonable, though the lack of quantitative information implies some uncertainty. Overall, the aggregate score of –3 appears appropriate, but it is best interpreted as reflecting a consolidation primarily driven by spending measures, with tax measures playing a supporting—though non-negligible—role.

[12] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 14: Georgia 2008Q2

AI-based output

Direction. No action.

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

All categories have been correctly identified. The AI output rightly points out that "Actions implemented in Q2 stem from March budget changes; not counted as current."

Exhibit 15: Greece 2001Q2

AI-based output

Direction. No action.

Motivation. Not applicable**.**

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

The report discusses actions which pertain to either the past or future quarters, and has no action which is explicitly taken in the current quarter. The AI is correct in picking up no fiscal actions in this quarter.

Exhibit 16: Greece 2010Q2

AI-based output

Direction. Fiscal contraction.

Motivation. 2d. Compliance with conditionalities imposed by external actors or lenders; 2a. Deficit-reduction and fiscal discipline measures; 2c. Concerns about access to markets.

EIU text: *"On May 2nd 2010, the IMF, euro-area members and the Greek government announced ... a package for Greece amounting to €110bn, in exchange for which Greece would implement a draconian fiscal austerity package."*

Status. Announced in current quarter (May 2010), with implementation over 2010-12.

EIU text: *"In early May the government agreed to a number of fiscal austerity measures to gain access to the IMF/euro zone lending facility ..."*

Composition. Spending and Tax.

EIU text: *"In the short term, this has involved cutting public-sector wages and pensions ... raising value-added tax (VAT) ... increasing taxes on alcohol, tobacco and luxury items and imposing a one-off corporation tax on 2009 profits."*

Intensity. Overall (-8); Spending (-7); and Tax (-6).[13]

EIU text: *"The austerity package involves fiscal policy tightening equivalent to 11% of GDP over the next three years ..."; "Cuts in government expenditure are expected to yield savings of around 5.3% of GDP ... while efforts to boost government revenue are expected to yield 4% of GDP."*

Economist notes

The AI correctly identifies a very large contractionary package driven by both tax increases and spending cuts under IMF/euro area conditionality. The scale of the tightening—equivalent to 11% of GDP over three years—and the listed measures justify a large contractionary coding for both components. The main reservation is status. The report does not only describe an announcement: it says that, in the short term, the package has already involved wage and pension cuts, the abolition of bonuses and tax increases. For that reason, the action is better treated as being implemented in the quarter rather than merely announced.

[13] - (+) values denote reduction in the fiscal deficit.

Exhibit 17: Haiti 2010Q1

AI-based output

Direction. Fiscal expansion.

Motivation. 5d. Post-disaster public service restoration; 3a. Productivity and capacity-enhancing measures.

EIU text: *"The Haitian delegation ... asked for US$3bn for the immediate task of rebuilding government ministries, infrastructure and residential housing."*

Status. Announced in current quarter, for immediate reconstruction after the earthquake.

EIU text: *"The IMF has since responded by approving an SDR65.5m ... augmentation to the ECF on January 27th ..."; "The Fund has lowered the interest rates on outstanding loans ... to zero until 2012."*

Composition. Spending.

EIU text: *"Debt relief should reduce some pressures on the government finances ... will save the government around US$50m ... per year in repayments."*

Intensity. Overall (+7); Spending (+7); and Tax (0).[14]

EIU text: *"The World Bank announced that it would disburse an emergency grant of US$100m ..."; "The Paris Club group of bilateral donors also announced it would write off all outstanding loans."*

Economist notes

The AI captures the overall sign: the earthquake generated a large expansionary fiscal impulse through external financing, debt relief and emergency support, with no tax tightening. The main issue is the choice of headline evidence. The US$3bn reconstruction figure is presented as a request by the Haitian delegation, so it is weaker as a current-quarter fiscal action than the IMF's SDR65.5m augmentation, the temporary elimination of IMF interest payments and the debt-service relief already in place. The large expansionary coding still makes sense, but the enacted support measures provide the cleaner basis for it.

[14] - (+) values denote reduction in the fiscal deficit.

Exhibit 18: Hungary 2013Q1

AI-based output

Direction. No action..

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

The AI correctly assesses that no fiscal actions are taken in the quarter. The small and offsetting changes in taxes and expenditure that are mentioned in the report refer to the planned implementation of decisions made in the previous quarter.

Exhibit 19: Ireland 2011Q1

AI-based output

Direction. Fiscal contraction.

Motivation. 6b. Major political transitions or revolutions; 1a. Financial rescue.

EIU text: *"Mr Lenihan argued that he no longer had a mandate to proceed with the recapitalisation given the minority government in power."; "…€30bn earmarked for recapitalising the ailing banking sector."*

Status. Suspended in current quarter (Q1 2011), with recapitalisation postponed in February.

EIU text: *"Mr Lenihan announced on February 9th that pending the formation of a new government he was postponing a €7bn recapitalisation…"*

Composition. Spending (financial sector intervention).

EIU text: *"…postponing a €7bn recapitalisation of Bank of Ireland, Allied Irish Bank (AIB) and Educational Building Society (EBS)."*

Intensity. Overall (-4); Spending (-4); and Tax (0).[15]

EIU text: *"…postponing a €7bn recapitalisation…"*

Economist notes

All categories have been correctly identified. The measure reflects a temporary fiscal contraction arising from the postponement of a previously planned bank recapitalization. The assigned overall intensity score of -4 is appropriate given the size of the delayed intervention (€7bn), though the effect is inherently transitory rather than structural. The absence of tax measures is correctly captured. This case is best interpreted as a timing-driven fiscal tightening caused by political constraints, rather than an intentional shift in fiscal stance, as the recapitalisation remained a core requirement under the EU/IMF programme.

[15] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 20: Iraq 1997Q1

AI-based output

Direction. Expansionary

EIU: *"It has been announced that the new food supplies will be distributed as soon as they arrive, on the basis of the existing ration card system, with the total monthly ration being increased by about 20%, to include 9 kg of wheat flour, 2.5 kg of rice, 2 kg of sugar, 1 kg of vegetable oils, 150 g of tea, 250 g of soap, 350 g of detergents and 2.7 kg of baby milk.."*

Motivation.

4b. Broad-based redistribution and social equity programs; 6a. Pursuit of political or electoral objectives.

EIU text: *4b: "It has been announced that the new food supplies will be distributed as soon as they arrive, on the basis of the existing ration card system, with the total monthly ration being increased by about 20%, to include 9 kg of wheat flour, 2.5 kg of rice, 2 kg of sugar, 1 kg of vegetable oils, 150 g of tea, 250 g of soap, 350 g of detergents and 2.7 kg of baby milk.." 6a: "In order to keep public opinion buoyant before the arrival of these urgently needed supplies, the government will release extra food on to the market at favourable prices."*

Status. Announced (1997-04)

Composition. Spending (positive)

Intensity. Overall (+6); Spending (+5); and Tax (0).[16]

EIU text:

"The implementation of Resolution 986 from the end of 1996 will give a substantial boost to the economy during the first half of 1997 and probably thereafter."

"It has been announced that the new food supplies will be distributed as soon as they arrive, on the basis of the existing ration card system, with the total monthly ration being increased by about 20%, to include 9 kg of wheat flour, 2.5 kg of rice, 2 kg of sugar, 1 kg of vegetable oils, 150 g of tea, 250 g of soap, 350 g of detergents and 2.7 kg of baby milk.."

Economist notes

The AI appropriately identifies a current period fiscal action. However, the AI did not identify the *correct* current fiscal action. A key text excerpt is used to substantiate a potential motivation but not the action itself: "*[1] In order to keep public opinion buoyant before the arrival of these urgently needed supplies, the government will release extra food on to the market at favourable prices*."

It seems that the AI identified the fiscal action as the implementation of the Resolution 986 "oil-for-food deal" signed in 1996Q4. I would argue that the current fiscal action should be the additional

[16] - (+) values denote reduction (increase) in the fiscal deficit.

action the government is taking to release food in the meantime before the Resolution 986 supplies arrive in February – April. To this effect, I would select lower intensity scores.

Exhibit 21: Italy 2008Q2

AI-based output

Direction. Expansionary.

Motivation. 1b. Fiscal responses to the economic cycle; 6a. Pursuit of political or electoral objectives; 4b. Broad-based redistribution and social equity programs.

Status. Implemented (in May).

Composition. Spending and Taxes.

Intensity. Overall (+2); Spending (-2); and Tax (+3).[17]

Economist notes

AI correctly identifies the fiscal action and its motivations. *"The government approved a decree in May containing measures intended to support households' purchasing power, which has suffered as a result of modest wage levels and soaring energy and food prices."* (Stabilization); *"Generalised tax hikes have been ruled out since tax levels are already considered to be too high and Mr Berlusconi's alliance made cutting taxes a cornerstone of its election campaign."* (Political); *"The main measures are: the abolition of the municipal property tax on primary residences for about 17m home owners at a cost of €1.7bn"* likely redistributive. It identifies a low intensity score based on the following sentence *"Although the package represents a first step in the right direction, the impact on households' purchasing power is likely to be modest."* The negative intensity score on spending is appropriate as the tax measures are financed by (unspecified) spending cuts.

[17] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 22: Italy 2009Q1

AI-based output

Direction. Fiscal expansion.

EIU text: *"On February 25th the cabinet approved a decree allocating €12bn to help banks rebuild their capital base…"; "In February-March the government announced a series of measures aimed at supporting key industries, such as the automotive, white goods and construction sectors, and protecting low-income groups and the unemployed"; "the government will allow home owners buying new energy-efficient domestic appliances, including television sets, to deduct 20% of the total cost from their income-tax bill at the end of the year"; "On March 6th the government announced that it will kick-start its three-year €44bn infrastructure investment programme"; "the government announced that it intends to ease planning restrictions and provide tax incentives for property owners wishing to expand their properties".*

Motivation. 1b. Countercyclical measures; 3a. Productivity and capacity-enhancing measures; 4a. Targeted redistribution to specific groups.

EIU text: *"On February 6th the government approved a new package of measures worth a total of about €2bn that are mainly aimed at boosting automotive sales to counter the recession"; "three-year €44bn infrastructure investment programme … to revive the economy"; "the government announced a series of measures aimed at … protecting low-income groups and the unemployed".*

Status. Cabined approved.

EIU text: See previous quotes.

Composition. Spending and tax.

EIU text: See above.

Intensity. Overall (+4); Spending (+3); and Tax (+2).[18]

EIU text: *"The major tax cuts promised in the 2008 election campaign have been put on hold and the cost of stimulus measures so far appears to be limited"; "The [bank recapitalization] plan contained in a decree approved by the cabinet on February 25th, is tiny in size"; "Support for the real economy so far appears to have been limited by concerns about Italy's large government debt".*

Economist notes

All categories are correctly identified, although one motivation has been missed. The bank recapitalization plan approved by the cabinet should also be assigned a motivation of 1a (Financial rescue). The plan may not have been picked up because it is a financial operation that "would not affect the general government balance" (p. 11). All relevant fiscal actions are included. The others mentioned in the report are either past actions (such as the guaranteeing of bank deposits up to €100,000) or are preliminary (such as the government's proposal to provide tax incentives for

[18] - (+) values denote reduction (increase) in the fiscal deficit.

property owners looking to expand properties). The intensity is also appropriate, as the report mentions in multiple places that Italy's high debt burden has put limits on the size of countercyclical tax and spending measures. Beyond these qualitative comments, it is clear from the size of the packages that they are relatively minor. For example, measures worth around €2bn to support automotive sales are around 0.1% of estimated 2009 nominal GDP (p. 16).

Exhibit 23: Jamaica 2013Q2

AI-based output

Direction. Contractionary.

Motivation. 2a. Deficit-reduction and fiscal discipline measures; 2d. Compliance with conditionalities imposed by external actors or lenders; 4a. Targeted redistribution to specific groups.

EIU text: *"The government began to review levels of employment and salaries to reduce the public-sector wage bill to 9% of GDP by the end of the 2015/16 fiscal year, from 10.9% of GDP in 2012/13, a condition of its new IMF loan accord.; [2] To mitigate the social impact of fiscal cuts and other austerity measures, Jamaica has the IMF's backing to place a floor under social spending of J$20.1bn (around US$205m) during this fiscal year.; [3] Under a new IMF extended fund facility (EFF) approved by the Fund on May 1st, the government will curb public spending, hold back wage increases and impose other austerity measures.; [4] Earlier this year, the government met preconditions for the EFF, including a domestic debt restructuring, known as the National Debt Exchange –the second restructuring within three years—to reduce interest payments and lengthen maturities (affecting some US$9bn in debt, or 45% of the total public debt stock). It also introduced a tax package to raise J$15.9bn (US$169m) annually and agreed to reduce the public-sector wage bill from 10.9% of GDP in 2012/13 to 9% of GDP by March 2016. Officials secured a new wage agreement with unions, but job cuts will be required.; [5] We expect that spending restraint, higher tax revenue, and lower interest and amortisation payments on restructured debt, together with the financial aid from the IMF and other multilaterals, will reduce pressure on the public finances.; [6] We also expect the government to introduce longer-term changes to improve tax collection, enhance budgetary control and boost transparency.; [7] The government already began tightening fiscal policy earlier this year, both on the spending and revenue side (the latter through a tax package).; [8] The main objective of the programme is to pave the way for a reduction of the large public debt stock (currently above 134% of GDP) to below 100% of GDP by 2020, in order to increase Jamaica's competitiveness, improve the country's weak business climate and boost its rate of economic growth.; [9] It also undertook a domestic debt swap, which is expected to lower its interest costs (now projected to be 8% of GDP in 2013/14, down from 9.4% in 2012/13), and thereby reduce financing needs (projected to be 6.9% of GDP in 2013/14, down from 10.9% of GDP in the previous fiscal year).; [10] Structural reform measures are considered to be more critical to the success of the programme. These include broadening of the tax system to improve burden-sharing and increasing efficiency in collections; reforms to the securities dealers sector; a shift towards inflation-targeting by the central bank ...; and policy changes that encourage business investment.; [11] The government is beginning to review levels of employment and salaries in order to reduce the public-sector wage bill to no more than 9% of GDP by fiscal year 2015/16 (April-March) from 10.9% of GDP at present, a condition of its new US$932m loan agreement with the IMF.; [12] In another attempt to cut costs, the People's National Party (PNP) government is assessing whether to convert some public-sector bodies into revenue-generating executive agencies and to merge others.; [13] The government has already reached an agreement with major unions on limiting wage increases.; [14] Three thousand civil-service posts were eliminated last year and another 3,000 earlier in 2013, while the 1,000 positions expected to become vacant this year owing to natural attrition will not be filled.; [15] To mitigate the impact of fiscal cuts and other austerity measures that form part of the loan programme, Jamaica and the IMF have also agreed to place a floor under social spending of J$20.1bn (around US$205m) during this fiscal year. This will cover programmes providing youth employment, school meals, basic school subsidies, relief for the poor, primary healthcare and conditional cash transfers to children and the elderly."*

Status. Announced.

EIU text: *"[3] Under a new IMF extended fund facility (EFF) approved by the Fund on May 1st, the government will curb public spending, hold back wage increases and impose other austerity measures."*

Composition. Spending.

EIU text: *"[3] Under a new IMF extended fund facility (EFF) approved by the Fund on May 1st, the government will curb public spending, hold back wage increases and impose other austerity measures." "[11] The government is beginning to review levels of employment and salaries in order to reduce the public-sector wage bill to no more than 9% of GDP by fiscal year 2015/16..."*

Intensity. Overall (-5); Spending (-5); and Tax (0).[19]

Economist notes

The report hints at contractionary actions being taken to reduce the wage bill. These hints led the AI to identify a mild reduction in spending in this quarter. While the direction, motivation, and instrument of the fiscal impact are certainly correct, whether a significant action is taken and whether it is taken precisely this quarter rather than later this year is debatable.

[19] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 24: Japan 2011Q2

AI-based output

Direction. No action.

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

The AI's assessment that there is no current fiscal action is correct. It explains: "*No new fiscal action in the report's current quarter; actions cited are earlier (Q1) or planned for later*."

It is unclear when the supplementary budget was formed. Based on the text in the report, we do know it was after March 11th, but we are not sure if this occurred at the end of Q1 or into Q2: *"The government quickly formulated a ¥4trn (US$484bn) supplementary budget to pay for removing debris from the Tohoku region"*. Given the limited details, the AI's classification is correct.

Exhibit 25: Netherlands 2004Q2

AI-based output

Direction. Fiscal contraction.

Motivation. 2a. Deficit-reduction and fiscal discipline measures**.**

EIU text: "*The government agreed a package of €2.4bn in spending cuts, largely on health services in April in response to the fact that last year's budget deficit turned out to be higher than expected (3.2% of GDP)", "In April the government approved a package of extra spending cuts amounting to €2.4bn, after last year's government budget deficit breached the 3% limit of the EU's Stability and Growth Pact (SGP)."*

Status. Cabinet approval in current quarter (in 2004), for the same budget year (2004).

EIU text: *"In mid-April the cabinet consequently approved a package of spending cuts amounting to €2.4bn."; "Following agreement on the package of spending cuts, the government believes that the budget deficit will now amount to 2.9% of GDP in 2004."*

Composition. Spending and Tax

EIU text: "*In mid-April the cabinet consequently approved a package of spending cuts amounting to €2.4bn.", "In addition, the government plans to increase compulsory insurance premiums for home carte."*

Intensity. Overall (-5); Spending (-5); and Tax (-2).[20]

EIU text: "*The government agreed a package of €2.4bn in spending cuts, largely on health services in April", "In addition, the government plans to increase compulsory insurance premiums for home carte."*

Economist notes

Most categories have been correctly identified. Regarding intensity, the AI assigned an overall score of –5, which appears reasonable given the scale of the measure—€2.4 billion, equivalent to approximately 0.5 percent of 2004 nominal GDP—and noting that this consolidation package is driven entirely by spending measures.

Importantly, the underlying EIU report does not provide explicit quantitative estimates for the revenue measures. As such, the AI's assessment of revenue-side intensity necessarily relies on qualitative judgment based on the nature of the measures described. In this context, the revenue-side intensity of "-2" (small) appears appropriate, as the measures mainly consist of limited adjustments such as higher insurance premiums and increased user charges (e.g., healthcare co-payments), rather than broad-based tax policy changes.

[20] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 26: Norway 2010Q3

AI-based output

Direction. No action.

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

Economist notes

The AI is correct in its assessment that there is no fiscal action in this quarter. The government's decision to halt plans to build power cables across Hardanger fjord is appropriately not classified as a fiscal action.

Exhibit 27: Portugal 2013Q4

AI-based output

Direction. Fiscal contraction.

Motivation. 2a. Deficit-reduction and fiscal discipline measures; 2d. Compliance with conditionalities imposed by external actors or lenders; 2c. Concerns about access to markets..

EIU text: *"On October 15th, the government unveiled the 2014 budget ... comprising a total adjustment of €3.9bn, mainly accounted for by spending cuts.", "In order to meet its public deficit target of 4% of GDP in 2014, the government announced a sharply contractionary budget for next year…", "Ensuring the continuing availability of financing ... will be the government's key focus."*

Status. Approved in current quarter (November 2013), for Budget year 2014.

EIU text: *"The 2014 budget was approved by parliament on November 26th, broadly unchanged from the initial draft."*

Composition. Spending and Tax.

EIU text: *"The budget will entail significant cuts to public-sector pay and pensions ..."; "One exception is in relation to corporation tax, which is being lowered from 25% to 23% next year."*

Intensity. Overall (-7); Spending (-6); and Tax (+2).[21]

EIU text: *"In total, spending cuts account for more than four-fifths (€3.2bn) of the planned adjustment next year."*

Economist notes

The AI captures the key current-quarter action: approval of the 2014 budget, a large spending-led consolidation. The overall contractionary coding and the spending intensity are well supported by the report. On the tax side, the text says that most tax rates are unchanged and that the notable tax measure is a cut in corporation tax from 25% to 23%, so the intensity score on tax is correctly classified as small and expansionary. The Banif recapitalization is correctly excluded from the current quarter.

[21] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 28: Qatar 2016Q2

AI-based output

Direction. Neutral.

Motivation. 2a. Deficit-reduction and fiscal discipline measures; 1c. Response to commodity price fluctuations**.**

EIU text: *"2a: <[1] The government introduced a new system of fuel pricing in May, linked to international prices, which will help to insulate the public finances from oil price movements.> | 1c: <[8] In the context of the ongoing fiscal squeeze brought about by low oil prices, Qatar has followed other Gulf countries in implementing cuts to subsidies.>*"

Status. Implemented.

EIU text: *"The government introduced a new system of fuel pricing in May, linked to international prices, which will help to insulate the public finances from oil price movements".*

Composition. Spending.

EIU text: *"[6] On April 26th the Ministry of Energy and Industry announced in a brief statement on its Facebook page that a decision had been made by a committee on fuel prices to link local prices to world market prices for refined products, adjusting prices on a monthly basis.; [7] Thus far no changes have been made to petrol prices, but diesel prices were cut by 7% to QR1.4/litre, bringing diesel more closely into line with petrol. Diesel had been sold a premium since 2013 owing to the fact that it is mainly used commercially rather than by private individuals.;"*

Intensity. Overall (0); Spending (0); and Tax (0).[22]

EIU text: *"[15] The new formula would have a positive impact on the public finances if oil prices were to rise markedly, although given the current weak price outlook for the next few years the impact will be minimal."*

Economist notes

All categories have been overall correctly identified. AI appears to have taken a prudent approach by not assigning a definitive intensity score, given that the prevailing low international oil prices at the time of the reform limit the immediate fiscal impact of the alignment of prices with market levels. AI has appropriately not classified bond issuance and T-bill operations as fiscal policy actions, and has also correctly treated stadium tenders as procurement activities rather than discrete fiscal policy measures.

[22] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 29: Rwanda 2005Q4

AI-based output

Direction. Fiscal expansion.

Motivation. 3a. Productivity and capacity-enhancing measures; 4a. Targeted redistribution to specific groups; 4b. Broad-based redistribution and social equity programs; 2e. Response to external financial aid or debt relief.

EIU text: *"Mr Nshuti proposed increasing expenditure by 10%, to Rwfr404.7bn.", "The signs are that Mr Nshuti is planning a more expansive fiscal approach ...", "Like its predecessors, the spending plans of the recently presented 2006 budget are reliant on the goodwill of donors, since external grants are budgeted to contribute more than 50% of government revenue for the year."*

Status. Announced in current quarter (October 2005), for Budget year 2006.

EIU text: *"The incoming finance minister, Manasseh Nshuti, presented the 2006 budget to the National Assembly on October 25th."*

Composition. Spending.

EIU text: *"No new taxes have been proposed this year ..."; "The priority expenditure items ... are salary rises ... energy sector investments ... and increased spending on social sector ministries to improve service delivery to the poor..."*

Intensity. Overall (+5); Spending (+5); and Tax (0).[23]

EIU text: *"In the 2006 budget Mr Nshuti proposed increasing expenditure by 10% ..."*

Economist notes

The AI is right to identify an expansionary budget proposal centered on higher expenditure and no new taxes. The report describes a 10% rise in spending, civil service pay increases and larger social and infrastructure outlays, so the direction and composition are appropriate. The caution is about status and scale: this is a budget presented to parliament for 2006, not yet enacted, and the donor-dependent financing argues for a moderate rather than very large impulse. The expansion is therefore best read as announced and spending led.

[23] - (+) values denote reduction (increase) in the fiscal deficit.

Exhibit 30: Sweden 1994Q3

AI-based output

Direction. Neutral

EIU text: *"But at the extraordinary meeting of the committee on July 14, which was televised live, it was announced that there would be no further cuts, which meant certainly not before the election in September."*

Motivation. 6a. Pursuit of political or electoral objectives.

EIU text: *"(…) there would be no further cuts, which meant certainly not before the election in September."*

Status. Suspended in July 1994.

EIU text: *"But at the extraordinary meeting of the committee on July 14, which was televised live, it was announced that there would be no further cuts, (…)."*

Composition. None

EIU text: None

Intensity. Overall (0); Spending (0); and Tax (0).

EIU text: None

Economist notes

All categories have been correctly identified. The key sentence from the report is: *But at the extraordinary meeting of the committee on July 14, which was televised live, it was announced that there would be no further cuts, which meant certainly not before the election in September."*

Exhibit 31: Yemen 2005Q4

AI-based output

Direction. No action.

Motivation. Not applicable.

Status. Not applicable.

Composition. Not applicable.

Intensity. Not applicable.

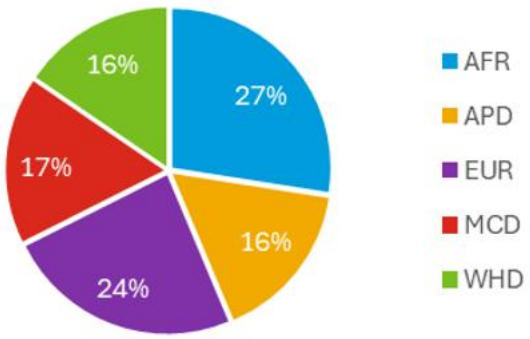
Economist notes

AI is broadly accurate in indicating no fiscal action in current quarter and assigning fiscal actions discussed in the report as pertaining to past or future periods "The report discusses past measures and forecasts; no new tax, spending, or subsidy actions are reported as taken in Q4 2005".

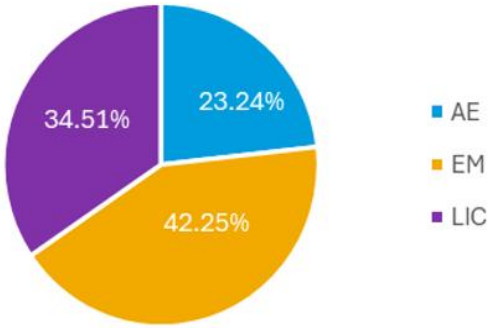
Annex C. Additional Figures and Tables

Figure C1. Country Coverage

By Region



By Income Group



Note: Figures show the distribution of countries in the sample by region (left panel) and income group (right panel). Regional classifications are AFR = Africa, APD = Asia and Pacific, EUR = Europe, MCD = Middle East and Central Asia, and WHD = Western Hemisphere. Income group classifications are AE = Advanced Economies, EM = Emerging Markets, and LIC = Low-Income Countries. Percentages indicate each group's share of the total sample.

Annex D. Detailed Comparison with Existing Literature

D.1. Comparison with Romer and Romer (2019)

This annex documents (i) the construction of the crisis-based narrative dataset in Romer and Romer (2019), (ii) key differences between our dataset vs. Romer and Romer (2019), and (iii) the episode-level comparison between the two datasets.

Data Construction

Romer and Romer (2019) analyze fiscal policy responses during episodes of severe financial distress. Crisis episodes are defined as periods in which their narrative-based financial distress index reaches a threshold value of 7 or higher. The resulting sample consists of 22 crisis episodes across 30 OECD economies.
Fiscal actions are identified using Economist Intelligence Unit (EIU) country reports. Report selection is aligned with the timing of the OECD Economic Outlook (e.g. June and December). For the pre-2000 period, the authors examine the third- and first-quarter EIU reports. For the post-2000 period, they rely on the July and January issues. The objective is to assess the overall stance of fiscal policy during crisis episodes. The dataset is event-based and limited to crisis periods. It does not constitute a continuous time series of fiscal shocks and does not attempt to code fiscal actions outside episodes of elevated financial distress.

Fiscal actions are classified along three dimensions:
1. Direction (expansions or contractions).
2. Magnitude (large or small), based primarily on qualitative descriptors such as “enormous,” “unprecedented,” or “extreme,” for large fiscal actions, and “mild,” “minor,” and “small,” for small fiscal actions.
3. Motivation (expansions and contractions), they consider financial rescue, countercyclical stimulus, and politics for fiscal expansions, and market access, conditionality, policymaker ideas, European Union fiscal rules, and countercyclical austerity for fiscal contractions.

Key Differences Between Our Dataset and Romer and Romer (2019)

Our dataset is constructed at the quarterly frequency using a large set of EIU country reports. For the post-2000 period, March, June, September, and December issues correspond to Q1 through Q4, respectively. Unlike Romer and Romer (2019), our dataset is not conditioned on a financial distress threshold and covers all quarters in the sample. These differences in construction have two main implications for the comparison exercise. First, for the pre-2000 period, both studies use the same first- and third-quarter EIU country reports. In this subset, comparison is based on identical documentary evidence. We therefore conduct direct issue-level comparison and document four cases—Norway (1991Q4), Korea (1998Q1), Sweden (1994Q2), and Mexico (1996Q3)—where direction or magnitude classifications differ between our dataset and Romer and Romer (2019). These cases are discussed explicitly in the next section. Second, for the post-2000 period, report selection differs. Romer and Romer (2019) rely on the January and July EIU issues to align with the OECD Economic Outlook, whereas our quarterly dataset relies on March, June, September, and December issues. As a result, comparisons in the post-2000 period are based on different EIU reports rather than the same

document. In this period, we do not evaluate case-level discrepancies because observed differences may reflect report timing rather than interpretive disagreement. In implementing the comparison, several harmonization steps are required:

Temporal dimensions. Our dataset records fiscal actions for each quarter under three temporal dimensions: Past, Current, and Future (e.g., 2009Q1_Past, 2009Q1_Current, and 2009Q1_Future). When comparing to Romer and Romer (2019), we assess whether either the _Current or _Past classification aligns with their episode coding. This approach is particularly relevant in the post-2000 period, where report timing differs: Romer and Romer (2019) rely on the January and July EIU reports, whereas our quarterly dataset uses the March and September issues. Allowing matches through the _Past classification helps account for this timing difference, as fiscal measures discussed in earlier reports may be referenced retrospectively in subsequent issues.

Magnitude. Romer and Romer (2019) classify fiscal actions as either large or small. Our dataset distinguishes between large, small, neutral, and unclear. For comparability, we focus exclusively on large fiscal actions when assessing magnitude alignment. This avoids imposing additional assumptions about how neutral or unclear classifications should be mapped into a binary framework.

Stance vs. action. Romer and Romer (2019) assess the current and prospective overall stance of fiscal policy during crisis episodes. By contrast, our prompt asks the model to evaluate the effects of specific fiscal measures identified in the report and explicitly instructs it not to be influenced by the government's overall fiscal stance. This distinction-between the overall stance and individual measures-may help explain occasional differences in classification when multiple fiscal actions are discussed within the same report.

These design differences imply that exact replication is feasible only in the pre-2000 period. Post-2000 comparisons should therefore be interpreted as alignment exercises rather than document-level replication. Tables D.1.1 and D.1.2 report the episode-level comparison between our dataset and the fiscal expansion and contraction episodes reported in Tables 3 and 4 of Romer and Romer (2019). The tables indicate whether each episode is detected in our dataset and highlight the small number of cases where classifications differ. We carefully analyzed the episodes where there are discrepancies and concluded that in most cases, either the LLM classification is more accurate or that discrepancies can be attributed to differences in the framing of the questions.

Table D.1.1 Fiscal Expansion Episodes During Crisis Periods: Comparison with Romer and Romer (2019)

Romer and Romer (2019): Table 3 (Expansions)

a. Direction and Size

Date expansion is first mentioned in parentheses, followed by a check (✓/✗) indicating whether the direction and size (large) match AI output.

Small		Large		
U.S. (1992Q1)	✓	Finland (1993Q1)	✓	✓
Norway (1991Q4)*	✓	Sweden (1993Q2)	✓	✓
Korea (1999Q1)	✓	Japan (1998Q1)	✓	✓
Austria (2009M1)	✓	Turkey (2001M7)	✓	✗
France (2009M1)	✓	U.S. (2008M7)	✓	✓
Italy (2009M1)	✓			
Norway (2009M1)	✓	U.K. (2009M1)	✓	✓
Portugal (2009M1)	✓	Ireland (2009M7)	✓	✓
Spain (2009M1)	✓			
Sweden (2009M1)	✓			
Denmark (2009M7)	✓			
Greece (2009M7)	✓			

b. Motivation

Date motivation is first mentioned in parentheses, followed by a check (✓/✗) indicating whether motivation match AI output.

Financial Rescue[e]		Countercyclical		Politics	
		U.S. (1992Q1)	✓	U.S. (1992Q1)	✓
Norway (1991Q4)	✓	Norway (1991Q4)	✓	Norway (1993Q1)**	✗
Finland (1993Q1)	✓				
Sweden (1993Q2)	✓				
Japan (1998Q1)	✓	Japan (1998Q1)	✓	Japan (1998Q1)**	✗
Korea (1999Q1)	✓	Korea (1999Q1)	✓	Korea (1999Q3)	✓
Turkey (2001M7)	✓				
U.S. (2009M1)	✓	U.S. (2008M7)	✓		
U.K. (2009M1)	✓	U.K. (2009M1)	✓		
Austria (2009M1)	✓	Austria (2009M1)	✓		
France (2009M1)	✓	France (2009M1)	✓		
Italy (2009M1)[f]	✓	Italy (2009M1)	✓		
Norway (2009M1)	✓	Norway (2009M1)	✓		
Portugal (2009M1)	✓	Portugal (2009M1)	✓	Portugal (2009M1)	✓
Spain (2009M1)	✓	Spain (2009M1)	✓		
Sweden (2009M1)	✓	Sweden (2009M1)	✓		
Denmark (2009M7)	✓	Denmark (2009M7)	✓		
Greece (2009M7)	✓				
Ireland (2009M7)	✓				

Notes: ✓ indicates that the fiscal episode identified in Romer and Romer (2019) is also detected in our dataset. ✗ indicates that the episode is not detected.

Table D.1.2 Fiscal Contraction Episodes During Crisis Periods: Comparison with Romer and Romer (2019)

Romer and Romer (2019): Table 4 (Contractions)

a. Direction

Date austerity is first mentioned in parentheses, followed by a check (✓/✗) indicating whether the direction and size (large) match AI output.

Small		Large		
U.S. (1991Q1)	✓	Finland (1993Q3)	✓	✓
Norway (1994Q3)	✓	Sweden (1994Q2)*	✗	✗
Korea (1998Q1)*	✓	Mexico (1996Q3)*	✗	✗
U.S. (2011M7)	✓	Turkey (2002M1)	✓	✓
France (2010M7)	✓			
Denmark (2010M7)	✓	U.K. (2010M7)	✓	✓
		Austria (2010M7)	✓	✗
		Italy (2010M7)	✓	✓
		Portugal (2010M7)	✓	✓
		Spain (2010M1)	✓	✓
		Greece (2010M1)	✓	✓
		Hungary (2009M7)	✓	✓
		Ireland (2010M1)	✓	✓

b. Motivation

Date motivation is first mentioned in parentheses, followed by a check (✓/✗) indicating whether motivation match AI output.

Market Access		Conditionality		Ideas		EU Rules		Countercyclical	
				U.S. (1991Q1)	✓				
				Norway (1994Q3)	✓			Norway (1994Q3)**	✗
Finland (1994Q3)	✓			Finland (1993Q3)	✓	Finland (1995Q1)	✓		
Sweden (1995Q1)	✓			Sweden (1994Q2)	✗	Sweden (1994Q3)	✓		
Mexico (1996Q3)	✗	Mexico (1996Q3)	✗						
Korea (1998Q1)	✓	Korea (1998Q1)	✓						
Turkey (2002M1)	✓	Turkey (2002M1)	✓			Turkey (2005M1)	✓		
				U.S. (2011M7)	✓				
U.K. (2010M7)	✓			U.K. (2010M7)	✓				
				Austria (2010M7)	✓	Austria (2010M7)	✓		
France (2010M7)	✓			France (2010M7)	✓	France (2010M7)	✓		
Italy (2010M7)	✓					Italy (2011M1)	✓		
Portugal (2010M7)	✓	Portugal (2011M7)	✓	Portugal (2012M1)	✓	Portugal (2010M7)	✓		
Spain (2010M1)	✓			Spain (2010M1)	✓	Spain (2011M1)	✓		
				Denmark (2010M7)	✓	Denmark (2011M1)	✓	Denmark (2012M7)**	✗
Greece (2010M1)	✓	Greece (2010M7)	✓			Greece (2010M1)	✓		
Hungary (2009M7)	✓	Hungary (2009M7)	✓	Hungary (2009M7)	✓	Hungary (2009M7)	✓		
Ireland (2010M1)	✓	Ireland (2011M1)	✓						

Notes: ✓ indicates that the fiscal episode identified in Romer and Romer (2019) is also detected in our dataset. ✗ indicates that the episode is not detected.

D.2. Comparison with Fiscal Consolidation Datasets

This annex provides the country–year comparisons underlying the validation exercise against multi-country fiscal consolidation datasets. Table D.2.1 reports the country–year comparison underlying the detection rates reported in Section II. For each consolidation episode identified in Guajardo, Leigh, and Pescatori (2014), the table indicates whether a corresponding fiscal consolidation is also detected in our dataset. A consolidation year is considered detected if at least one quarter within that year records a contractionary fiscal action motivated by deficit reduction or fiscal discipline in our dataset. The table also lists the magnitude of fiscal consolidation in the benchmark dataset, and whether a corresponding magnitude is also detected in our dataset. Large consolidations are defined as exceeding 2 percent of GDP (according to the definition of Carrière-Swallow, David, and Leigh, 2020) and listed as large by AI under the question on qualitative size fiscal action. Non-large consolidations are defined as less than or equal to 2 percent of GDP and listed as unclear or small by AI under the question on qualitative size fiscal action. Tables D.2.2 and D.2.3 present analogous comparisons for the datasets of Carrière-Swallow, David, and Leigh (2020) and Abdel-Latif et al. (2026), respectively.

Additional Consolidation Episodes Identified in Our Dataset

As a complementary exercise, we also examine the reverse comparison: country-year episodes identified as large fiscal consolidations in our dataset but not recorded as consolidation episodes in the benchmark fiscal consolidation datasets. This exercise is distinct from the detection-rate analysis discussed above, which conditions on episodes identified in prior studies. The purpose here is to assess whether our higher-frequency EIU-based dataset identifies additional consolidation episodes outside the benchmark lists.

To construct this comparison, we restrict our data to fiscal actions satisfying four criteria: (i) timing scope is classified as current; (ii) policy direction is contractionary; (iii) the motivation bucket includes “2a. deficit-reduction and fiscal discipline measures”; and (iv) the action is classified as large. We then aggregate these quarterly observations to the annual level. A country-year is classified as a large fiscal consolidation in our dataset if at least two quarters within that year contain large contractionary fiscal actions motivated by deficit reduction or fiscal discipline.

Using this rule, Tables D.2.4 list the country-years identified in our dataset but not in the corresponding benchmark consolidation datasets. These differences should not necessarily be interpreted as errors in the benchmark studies. The datasets differ in source material, frequency, and coding design: the benchmark studies are generally annual and rely heavily on IMF staff reports and related fiscal documents, whereas our dataset is quarterly and based on EIU country reports. The additional episodes therefore reflect differences in documentary coverage, timing, and classification rules.

Table D.2.1. Country–Year Comparison with Fiscal Consolidation Episodes in Guajardo, Leigh, and Pescatori (2014)

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Australia	1985	-	Small	-
Australia	1986	-	Small	-
Australia	1987	-	Small	-
Australia	1988	-	Small	-
Australia	1994	✓	Small	✓
Australia	1995	✓	Small	✓
Australia	1996	✓	Small	✓
Australia	1997	✓	Small	✓
Australia	1998	✓	Small	✓
Australia	1999	✗	Small	✗
Austria	1980	-	Small	-
Austria	1981	-	Small	-
Austria	1984	-	Small	-
Austria	1996	✓	Large	✓
Austria	1997	✓	Small	✓
Austria	2001	✓	Small	✓
Austria	2002	✓	Small	✓
Austria	2011	✓	Small	✓
Austria	2012	✓	Small	✓
Austria	2013	✓	Small	✓
Austria	2014	✓	Small	✓
Belgium	1982	-	Small	-
Belgium	1983	-	Small	-
Belgium	1984	-	Small	-
Belgium	1985	-	Small	-
Belgium	1987	-	Small	-
Belgium	1990	✓	Small	✓
Belgium	1992	✓	Small	✓
Belgium	1993	✓	Small	✓
Belgium	1994	✓	Small	✓
Belgium	1996	✓	Small	✓
Belgium	1997	✓	Small	✓

Notes:

- The table above checks whether AI also detects a fiscal contraction in EIU country reports for the episodes listed in Guajardo, Leigh, & Pescatori (2014). For comparison, we use at least one contraction quarter (motivation 2a - deficit-reduction and fiscal discipline measure) and any timing scope (past, current, and future) for the respective year.

- Blue font = extension by Adler et al (2024). In column “Consolidation Year Detected in Our Dataset”, and “Size Detected in Our Dataset”, "✓" means that AI also detects a corresponding match, and “✗” means that AI does not detect a corresponding match, and "-" means the year is not part of our sample.

Table D.2.1. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Belgium	2010	✓	Small	✓
Belgium	2011	✓	Small	✓
Belgium	2012	✓	Large	✓
Belgium	2013	✓	Small	✓
Belgium	2014	✓	Small	✓
Canada	1984	-	Small	-
Canada	1985	-	Small	-
Canada	1986	-	Small	-
Canada	1987	-	Small	-
Canada	1988	-	Small	-
Canada	1989	-	Small	-
Canada	1990	✓	Small	✓
Canada	1991	✓	Small	✓
Canada	1992	✓	Small	✗
Canada	1993	✓	Small	✓
Canada	1994	✓	Small	✓
Canada	1995	✓	Small	✓
Canada	1996	✓	Small	✓
Canada	1997	✓	Small	✓
Canada	2010	✓	Small	✓
Canada	2011	✓	Small	✓
Canada	2012	✓	Small	✓
Canada	2013	✓	Small	✓
Canada	2014	✓	Small	✓
Denmark	1983	-	Small	-
Denmark	1984	-	Small	-
Denmark	1985	-	Small	-
Denmark	1986	-	Small	-
Denmark	1995	✓	Small	✓
Denmark	2011	✓	Small	✓
Denmark	2012	✓	Small	✓
Denmark	2013	✓	Small	✓
Finland	1992	✓	Small	✓
Finland	1993	✓	Large	✓
Finland	1994	✓	Large	✗
Finland	1995	✓	Small	✓
Finland	1996	✓	Small	✓
Finland	1997	✓	Small	✓
Finland	2011	✓	Small	✓
Finland	2012	✓	Small	✓
Finland	2013	✓	Small	✓
Finland	2014	✓	Small	✓
Finland	2015	✓	Small	✓
Finland	2016	✓	Small	✓

Table D.2.1. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
France	1979	-	Small	-
France	1987	-	Small	-
France	1989	-	Small	-
France	1991	✓	Small	✓
France	1992	✗	Small	✗
France	1995	✓	Small	✓
France	1996	✓	Small	✓
France	1997	✓	Small	✓
France	1999	✓	Small	✓
France	2000	✓	Small	✓
France	2011	✓	Small	✓
France	2012	✓	Small	✓
France	2013	✓	Large	✓
France	2014	✓	Small	✓
France	2015	✓	Small	✗
France	2018	✓	Small	✓
Germany	1982	-	Small	-
Germany	1983	-	Small	-
Germany	1984	-	Small	-
Germany	1991		Small	
Germany	1992		Small	
Germany	1993		Small	
Germany	1994		Small	
Germany	1995		Small	
Germany	1997	✓	Small	✓
Germany	1998	✓	Small	✓
Germany	1999	✓	Small	✓
Germany	2000	✓	Small	✓
Germany	2003	✓	Small	✓
Germany	2004	✓	Small	✓
Germany	2006	✓	Small	✓
Germany	2007	✓	Small	✗
Germany	2011	✓	Small	✓
Germany	2012	✓	Small	✓
Germany	2013	✓	Small	✓
Ireland	1982	-	Small	-
Ireland	1983	-	Small	-
Ireland	1984	-	Small	-
Ireland	1985	-	Small	-
Ireland	1986	-	Small	-
Ireland	1987	-	Small	-
Ireland	1988	-	Small	-
Ireland	2009	✓	Large	✓
Ireland	2010	✓	Large	✓

Table D.2.1. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Ireland	2011	✓	Large	✓
Ireland	2012	✓	Large	✓
Ireland	2013	✓	Large	✓
Ireland	2014	✓	Small	✓
Italy	1991	✓	Large	✓
Italy	1992	✓	Large	✓
Italy	1993	✓	Large	✓
Italy	1994	✓	Small	✓
Italy	1995	✓	Large	✓
Italy	1996	✓	Small	✓
Italy	1997	✓	Small	✓
Italy	1998	✓	Small	✓
Italy	2004	✓	Small	✓
Italy	2005	✓	Small	✓
Italy	2006	✓	Small	✓
Italy	2007	✓	Small	✓
Italy	2010	✓	Small	✓
Italy	2011	✓	Small	✓
Italy	2012	✓	Large	✓
Italy	2013	✓	Large	✓
Italy	2014	✓	Small	✓
Japan	1979	-	Small	-
Japan	1980	-	Small	-
Japan	1981	-	Small	-
Japan	1982	-	Small	-
Japan	1983	-	Small	-
Japan	1997	✓	Small	✓
Japan	1998	✗	Small	✗
Japan	2003	✗	Small	✗
Japan	2004	✓	Small	✓
Japan	2005	✓	Small	✓
Japan	2006	✓	Small	✓
Japan	2007	✓	Small	✓
Japan	2014	✓	Small	✓
Japan	2015	✓	Small	✓
Japan	2019	✗	Small	✗
Japan	2020	✓	Small	✓
Netherlands	1981	-	Small	-
Netherlands	1982	-	Small	-
Netherlands	1983	-	Small	-
Netherlands	1984	-	Small	-
Netherlands	1985	-	Small	-
Netherlands	1986	-	Small	-
Netherlands	1987	-	Small	-

Table D.2.1. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Netherlands	1988	-	Small	-
Netherlands	1991	✓	Small	✓
Netherlands	1992	✓	Small	✓
Netherlands	1993	✓	Small	✓
Netherlands	2004	✓	Small	✓
Netherlands	2005	✓	Small	✓
Netherlands	2011	✓	Small	✓
Netherlands	2012	✓	Small	✓
Netherlands	2013	✓	Small	✓
Portugal	1983	-	Small	-
Portugal	2000	✓	Small	✓
Portugal	2002	✓	Small	✓
Portugal	2003	✓	Small	✓
Portugal	2005	✓	Small	✓
Portugal	2006	✓	Small	✓
Portugal	2007	✓	Small	✓
Portugal	2010	✓	Small	✓
Portugal	2011	✓	Large	✓
Portugal	2012	✓	Large	✓
Portugal	2013	✓	Large	✓
Portugal	2014	✓	Small	✓
Spain	1983	-	Small	-
Spain	1984	-	Small	-
Spain	1989	-	Small	-
Spain	1990	✓	Small	✓
Spain	1992	✓	Small	✓
Spain	1993	✓	Small	✓
Spain	1994	✓	Small	✓
Spain	1995	✓	Small	✓
Spain	1996	✓	Small	✓
Spain	1997	✓	Small	✓
Spain	2010	✓	Small	✓
Spain	2011	✓	Small	✓
Spain	2012	✓	Large	✓
Spain	2013	✓	Large	✓
Spain	2014	✓	Small	✓
Spain	2017	✓	Small	✓
Sweden	1984	-	Small	-
Sweden	1993	✓	Small	✓
Sweden	1994	✓	Small	✓
Sweden	1995	✓	Large	✓
Sweden	1996	✓	Small	✓
Sweden	1997	✗	Small	✗
Sweden	1998	✓	Small	✓

Table D.2.1. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
United Kingdom	1979	-	Small	-
United Kingdom	1980	-	Small	-
United Kingdom	1981	-	Small	-
United Kingdom	1982	-	Small	-
United Kingdom	1994	✓	Small	✓
United Kingdom	1995	✓	Small	✓
United Kingdom	1996	✓	Small	✓
United Kingdom	1997	✓	Small	✓
United Kingdom	1998	✓	Small	✓
United Kingdom	1999	✗	Small	✗
United Kingdom	2010	✓	Small	✓
United Kingdom	2011	✓	Small	✓
United Kingdom	2012	✓	Small	✓
United Kingdom	2013	✓	Small	✗
United Kingdom	2014	✓	Small	✓
United Kingdom	2015	✓	Small	✓
United States	1978	-	Small	-
United States	1980	-	Small	-
United States	1981	-	Small	-
United States	1985	-	Small	-
United States	1986	-	Small	-
United States	1988	-	Small	-
United States	1990	✓	Small	✓
United States	1991	✓	Small	✓
United States	1992	✗	Small	✗
United States	1993	✓	Small	✓
United States	1994	✓	Small	✓
United States	1995	✓	Small	✓
United States	1996	✓	Small	✗
United States	1997	✓	Small	✓
United States	1998	✓	Small	✓
United States	2011	✓	Small	✓
United States	2012	✓	Small	✓
United States	2013	✓	Small	✗

Table D.2.2. Country–Year Comparison with Fiscal Consolidation Episodes in Carrière-Swallow, David, and Leigh (2020)

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Argentina	1996	✓	Small	✓
Argentina	1997	✓	Small	✓
Argentina	2018	✓	Small	✓
Argentina	2019	✓	Large	✓
Bolivia	1995	✓	Small	✓
Bolivia	2004	✓	Small	✓
Bolivia	2005	✓	Large	✓
Brazil	2015	✓	Small	✓
Brazil	2017	✓	Small	✓
Chile	1990	✓	Small	✓
Chile	1991	✗	Small	✗
Chile	2003	✓	Small	✓
Chile	2004	✓	Small	✓
Chile	2008	✓	Small	✓
Chile	2014	✓	Small	✓
Chile	2015	✓	Small	✓
Chile	2016	✓	Small	✓
Colombia	2000	✓	Small	✓
Colombia	2003	✓	Small	✓
Colombia	2011	✓	Small	✓
Colombia	2012	✓	Small	✓
Colombia	2015	✓	Small	✓
Colombia	2016	✓	Small	✓
Colombia	2017	✓	Small	✓
Costa Rica	1990	✓	Small	✓
Costa Rica	1991	✓	Large	✓
Costa Rica	1992	✓	Small	✓
Costa Rica	1993	✓	Small	✓
Costa Rica	1994	✓	Small	✗
Costa Rica	1995	✓	Small	✓
Costa Rica	1996	✓	Small	✓
Costa Rica	1997	✓	Small	✓
Costa Rica	2016	✓	Small	✓

Notes:

- The table above checks whether AI also detects a fiscal contraction in EIU country reports for the episodes listed in Carrière-Swallow, David, and Leigh (2020). For comparison, we use at least one contraction quarter (motivation 2a - deficit-reduction and fiscal discipline measure) and any timing scope (past, current, and future) for the respective year.

- Blue font = extension by Adler et al (2024). In column “Consolidation Year Detected in Our Dataset”, and “Size Detected in Our Dataset”, "✓" means that AI also detects a corresponding match, and “✗” means that AI does not detect a corresponding match, and "-" means the year is not part of our sample.

Table D.2.2. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Dominican Republic	2004	✓	Small	✓
Dominican Republic	2006	✓	Small	✓
Dominican Republic	2007	✓	Small	✓
Dominican Republic	2011	✓	Small	✓
Dominican Republic	2013	✓	Large	✓
Ecuador	1990	✓	Small	✓
Ecuador	1993	✓	Large	✓
Ecuador	2000	✓	Small	✓
Ecuador	2018	✓	Large	✗
Ecuador	2019	✓	Small	✓
Guatemala	1995	✓	Small	✓
Guatemala	1996	✓	Small	✓
Guatemala	2000	✓	Small	✓
Guatemala	2002	✓	Small	✓
Guatemala	2012	✓	Small	✓
Guatemala	2013	✓	Small	✓
Jamaica	1992	✓	Large	✓
Jamaica	1999	✓	Small	✓
Jamaica	2000	✓	Small	✓
Jamaica	2003	✓	Large	✓
Jamaica	2004	✓	Small	✓
Jamaica	2012	✓	Small	✓
Jamaica	2013	✓	Large	✓
Jamaica	2014	✓	Small	✓
Mexico	1989	-	Small	-
Mexico	2010	✓	Small	✓
Mexico	2014	✓	Small	✗
Paraguay	1989	-	Small	-
Paraguay	2001	✓	Small	✓
Paraguay	2003	✓	Small	✓
Paraguay	2004	✓	Small	✓
Paraguay	2005	✓	Small	✓
Paraguay	2006	✓	Small	✓
Paraguay	2014	✓	Small	✗
Paraguay	2016	✓	Small	✓
Peru	1992	✓	Small	✓
Peru	2002	✓	Small	✓
Peru	2003	✓	Small	✓
Peru	2011	✓	Small	✗
Peru	2012	✓	Small	✓
Uruguay	1990	✓	Small	✓
Uruguay	1995	✓	Small	✓
Uruguay	1996	✓	Small	✓
Uruguay	2000	✓	Small	✓
Uruguay	2002	✓	Large	✓
Uruguay	2003	✓	Small	✓
Uruguay	2004	✓	Small	✓
Uruguay	2005	✓	Small	✓
Uruguay	2015	✓	Small	✓

Table D.2.3. Country–Year Comparison with Fiscal Consolidation Episodes in Abdel-Latif et al. (2026)

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Cameroon	1991	✓	Small	✓
Cameroon	1992	✓	Small	✓
Cameroon	1994	✓	Small	✓
Cameroon	1995	✓	Large	✓
Cameroon	1996	✓	Small	✓
Cameroon	2017	✓	Small	✓
Cameroon	2018	✓	Small	✓
Cameroon	2024	✓	Small	✓
Côte d'Ivoire	1994	✓	Large	✗
Côte d'Ivoire	1995	✓	Small	✓
Côte d'Ivoire	2017	✓	Small	✓
Côte d'Ivoire	2018	✓	Small	✓
Côte d'Ivoire	2019	✓	Small	✓
Côte d'Ivoire	2021	✗	Small	✗
Côte d'Ivoire	2023	✓	Small	✓
Côte d'Ivoire	2024	✓	Small	✓
Ethiopia	2003	✓	Small	✓
Ethiopia	2018	✗	Small	✗
Ethiopia	2019	✓	Small	✓
Ethiopia	2023	✓	Small	✓
Ethiopia	2024	✓	Small	✓
Ghana	1995	✓	Small	✓
Ghana	2001	✓	Small	✓
Ghana	2002	✓	Small	✓
Ghana	2005	✓	Small	✓
Ghana	2011	✓	Small	✓
Ghana	2015	✓	Large	✓
Ghana	2017	✓	Small	✓
Ghana	2018	✓	Small	✓
Kenya	1993	✓	Large	✗
Kenya	1994	✓	Large	✗
Kenya	1995	✓	Small	✓
Kenya	1996	✓	Small	✓
Kenya	2011	✓	Small	✓

Notes:

- The table above checks whether AI also detects a fiscal contraction in EIU country reports for the episodes listed in Abdel-Latif et al. (2026). For comparison, we use at least one contraction quarter (motivation 2a - deficit-reduction and fiscal discipline measure) and any timing scope (past, current, and future) for the respective year.

- In column “Consolidation Year Detected in Our Dataset”, and “Size Detected in Our Dataset”, "✓" means that AI also detects a corresponding match, and “✗” means that AI does not detect a corresponding match.

Table D.2.3. – Continued

Country	Consolidation Year in Benchmark Study	Consolidation Year Detected in Our Dataset	Size in Benchmark Study	Size Detected in Our Dataset
Kenya	2023	✓	Large	✓
Kenya	2024	✓	Large	✓
Mauritius	1996	-	Small	-
Mauritius	1997	-	Small	-
Mauritius	2007	-	Small	-
Mozambique	2003	✓	Small	✓
Mozambique	2008	✓	Small	✓
Mozambique	2023	✓	Small	✓
Nigeria	2013	✓	Small	✓
Senegal	1995	✓	Small	✓
Senegal	1996	✓	Small	✓
Senegal	2002	✓	Large	✓
Senegal	2013	✓	Small	✓
Senegal	2014	✓	Small	✓
Senegal	2018	✓	Small	✓
South Africa	1993	✓	Small	✓
South Africa	1994	✓	Small	✓
Tanzania	1994	✓	Small	✓
Tanzania	1995	✓	Small	✓
Tanzania	1996	✓	Small	✓
Tanzania	2000	✓	Small	✓
Tanzania	2001	✓	Small	✓
Tanzania	2007	✓	Small	✗
Tanzania	2008	✓	Small	✓
Tanzania	2023	✓	Small	✓
Tanzania	2024	✓	Small	✓
Uganda	1996	✓	Small	✓
Uganda	1997	✓	Small	✓
Uganda	2002	✓	Small	✓
Uganda	2003	✓	Small	✓
Uganda	2005	✓	Small	✓
Uganda	2006	✓	Small	✓
Uganda	2007	✗	Small	✗
Uganda	2008	✗	Small	✗
Uganda	2021	✓	Small	✓
Uganda	2022	✓	Small	✓
Uganda	2023	✓	Small	✓
Uganda	2024	✓	Small	✓

Table D.2.4. Additional Large Fiscal Consolidation Episodes Relative to Benchmark Datasets

Country	Years Identified in Our Dataset but Not in Benchmark
Additional Large Consolidation Episodes Relative to Guajardo, Leigh, and Pescatori (2014)	
Austria	1994, 1995
Belgium	2005
Canada	2004
Germany	1996
Italy	1990
Japan	2012
Netherlands	2002, 2003, 2007
Portugal	1992, 1996
United Kingdom	1993
Additional Large Consolidation Episodes Relative to Carrière-Swallow, David, and Leigh (2020)	
Argentina	1991, 2002
Brazil	1990, 1991, 1993, 1996, 1998, 1999, 2001, 2016, 2019
Colombia	1998
Costa Rica	1998, 2002, 2004, 2006, 2013
Ecuador	1994, 2001, 2009, 2020
Guatemala	2016
Jamaica	2009, 2015
Peru	1996
Uruguay	2020
Additional Large Consolidation Episodes Relative to Abdel-Latif et al. (2026)	
Côte d'Ivoire	1990
Ghana	1994
Mozambique	2016
Nigeria	1994, 1996
Senegal	1993
Tanzania	2016

Notes: The table lists country-years classified as large fiscal consolidations in our dataset but not identified as consolidation episodes in the corresponding benchmark dataset. A country-year is classified as a large fiscal consolidation in our dataset if at least two quarters in that year satisfy all of the following conditions: timing scope is current, fiscal action is contractionary, motivation includes deficit reduction or fiscal discipline, and the size of the fiscal action is classified as large.

Annex E. Variables and Data Coverage

Table E.1. List of Variables and Description

Variable Name	Unit	Type	Description	Frequency	Prompt Question
fa_id		String	Unique identifier at country-year-quarter-timing level		
iso3		String	ISO3 country code		
report_name		String	Name of EIU country report extracted text		
report_date		String	Year-quarter date of EIU country report		
timing_scope		String	When the fiscal action was taken. Past = in quarter or more before the current quarter of the report, Current = in quarter when the report was issued, Future = in quarter or more after the current quarter of the report		
fiscal_mentioned		String	Yes if fiscal action is mentioned.		1a
fiscal_mentioned_rationale		String	Reason why Fiscal_Mentioned is "No" or "Unclear."		1b
relevant_fiscal_text		String	Text excerpts from EIU country report used to answer subsequent questions in prompt.		2
cause_fiscal_actions		String	Cause of fiscal problem that subsequently required the identified fiscal action.		3
motivation_buckets		String	1 = [Fiscal responses to macroeconomic and financial conditions, Stabilization] 2 = [Debt sustainability and debt relief] 3 = [Long-term growth] 4 = [Redistribution] 5 = [Provision of public goods and externalities] 6 = [Politically related]		4a
motivation_quote		String	Quote from text justifying choice of motivation.		4b
status_action		String	Status of fiscal action: Announced, Administrative Enactment, Cabinet_Approved, Parliament_Passed, Signed, Suspended, Withdrawn		5a
status_effective_date		String	The date when the corresponding measures come into effect, and/or the budget year when they are implemented.		5b

Variable Name	Unit	Type	Description	Frequency	Prompt Question
multi_year		String	Yes if the fiscal action is part of a multi-year package.		6a
multi_year_length	Years	Numeric	Total duration of the multi-year package that the identified fiscal action is a part of, if the fiscal action is part of a multi-year package.		6b
direction_policy		String	Net direction of impact of identified fiscal action: Expansionary, Contractionary, Neutral, or Unclear.		7a
direction_quote		String	Quote from text justifying choice of direction.		7b
qualitative_size_actions		String	Qualitative size of identified fiscal action: Small, Large, Unclear		8
tax_effect		String	Net effect of the identified fiscal action on tax revenues: Positive, Negative, No effect, Unclear		9
spend_effect		String	Net effect of the identified fiscal action on government expenditure: Positive, Negative, No effect, Unclear		10
intensity_fiscal_stance	Unit	Numeric	Score from -10 (strongly contractionary) to +10 (strongly expansionary) reflecting the fiscal impact of the identified fiscal action on the overall fiscal balance for the current quarter and the next quarter or the next year.		11a
intensity_fiscal_stance_quote		String	Quote from text justifying choice score assignment in intensity_fiscal_stance		11b
size_fiscal_action		Numeric	Numeric value of overall (net) size of the fiscal impact of identified fiscal action.		11c
size_fiscal_action_unit		String	Unit of value identified in size_fiscal_action		11d
size_fiscal_action_quote		String	Quote justifying identified numeric size of the fiscal impact of the identified fiscal action.		11e
intensity_tax	Unit	Numeric	Score from -10 (strongly contractionary) to +10 (strongly expansionary) reflecting how strongly contractionary or expansionary the tax effect is on the overall fiscal stance.		12a
intensity_tax_quote		String	Quote from text justifying choice score assignment in intensity_tax		12b
size_tax		Numeric	Numeric value of size of the tax effect of identified fiscal action.		12c
size_tax_unit		String	Unit of value identified in size_tax		12d
size_tax_quote		String	Quote justifying identified numeric size of the tax effect of identified fiscal action.		12e

Variable Name	Unit	Type	Description	Frequency	Prompt Question
intensity_spend	Unit	Numeric	Score from -10 (strongly contractionary) to +10 (strongly expansionary) reflecting how strongly contractionary or expansionary the spending effect is on the overall fiscal stance.		13a
intensity_spend_quote		String	Quote from text justifying choice score assignment in intensity_spend		13b
size_spend		Numeric	Numeric value of size of the spending effect of identified fiscal action.		13c
size_spend_unit		String	Unit of value identified in size_tax		13d
size_spend_quote		String	Quote justifying identified numeric size of the spending effect of identified fiscal action.		13e
tax_spending_measures_list		String	List of specific fiscal measures (tax or spending) for the identified fiscal action.		14
other_note		String	Brief note of relevant information for the identified fiscal action.		15
rationale_exclusions		String	Brief explanation for exclusion of certain fiscal stance directions, motivations, or other fiscal policy actions.		16
confidence_score	Unit	Numeric	AI-generated self-assessment of confidence in AI-generated output from 0 (no confidence) to 100 (absolutely confident).		
period	YYYYqQ	Numeric	Year + quarter of EIU report (e.g., 1990q1)		
timing_id		Numeric	Timing of fiscal action: 1 = past, 2 = current, 3 = future.		
report_date_clean		Numeric	Date of EIU report (if available)		
report_month_clean		Numeric	Month of EIU report (if available)		
year	year	Numeric	Year of EIU report		
quarter	quarter	Numeric	Quarter of EIU Report (1-4)		
ifscode		Numeric	IMF IFS code for country		
country		String	Country Name		
ae		Numeric	Indicator, equals 1 if country is Advanced Economy, zero otherwise		
em		Numeric	Indicator, equals 1 if country is emerging market and middle income economy, zero otherwise		
emde		Numeric	Indicator, equals 1 if country is emerging market and developing economy, zero otherwise		

Variable Name	Unit	Type	Description	Frequency	Prompt Question
lic		Numeric	Indicator, equals 1 if country is low-income developing country, zero otherwise		
enda_er	Unit	Numeric	National currency units per US dollar, period average	Quarterly	
ggei_weo	Billions	Numeric	WEO: General government expense, interest (National Currency, Billions)	Annual	
ggr_weo	Billions	Numeric	WEO: General government revenue (National Currency, Billions)	Annual	
ggrt_weo	Billions	Numeric	WEO: General government taxes (National Currency, Billions)	Annual	
ggx_weo	Billions	Numeric	WEO: General government total expenditure (National Currency, Billions)	Annual	
ggxcnl_weo	Billions	Numeric	WEO: General government net lending/borrowing (National Currency, Billions)	Annual	
ggxonlb_weo	Billions	Numeric	WEO: General government primary net lending/borrowing (National Currency, Billions)	Annual	
ggxwdg_weo	Billions	Numeric	WEO: General government gross debt (National Currency, Billions)	Annual	
ngdp_weo	Billions	Numeric	WEO: Gross domestic product, current prices (National Currency, Billions)	Annual	
ngdp_d_weo	Index	Units	WEO: Deflator of Gross domestic product, current prices (Index, Units)	Annual	
ngdp_fy_weo	Billions	Numeric	WEO: Fiscal year gross domestic product, current prices (National Currency, Billions)	Annual	
ngdp_fy_usd_weo	Billions	Numeric	WEO: Fiscal year gross domestic product in US dollars (US Dollars, Billions)	Annual	
ngdp_r_weo	Billions	Numeric	WEO: Gross domestic product, constant prices (National Currency, Billions)	Annual	
ngdp_rpch_weo	percent	Numeric	WEO: Gross domestic product, constant prices, National Currency, percent change	Annual	
pcpi_pch_weo	percent	Numeric	WEO: Consumer Prices, period average, percent change (Percent , Units)	Annual	
ggr_nsa	Billions	Numeric	General government revenue (National Currency, NSA, Billions)	Quarterly	
ggr_sa	Billions	Numeric	General government revenue (National Currency, SA, Billions)	Quarterly	

Variable Name	Unit	Type	Description	Frequency	Prompt Question
ggrt_nsa	Billions	Numeric	Tax revenue (National Currency, NSA, Billions)	Quarterly	
ggrt_sa	Billions	Numeric	Tax revenue (National Currency, SA, Billions)	Quarterly	
ggx_nsa	Billions	Numeric	General government expenditure (National Currency, NSA, Billions)	Quarterly	
ggx_sa	Billions	Numeric	General government expenditure (National Currency, SA, Billions)	Quarterly	
ggxcnl_nsa	Billions	Numeric	General government net lending/borrowing (National Currency, NSA, Billions)	Quarterly	
ggxcnl_sa	Billions	Numeric	General government net lending/borrowing (National Currency, SA, Billions)	Quarterly	
ggxwdg	Billions	Numeric	General government gross debt (National Currency, NSA, Billions)	Quarterly	
ngdp_nsa	Billions	Numeric	Gross domestic product, current prices (National Currency, NSA, Billions)	Quarterly	
ngdp_sa	Billions	Numeric	Gross domestic product, current prices (National Currency, SA, Billions)	Quarterly	
ngdp_r_nsa	Billions	Numeric	Gross domestic product, constant prices (National Currency, NSA, Billions)	Quarterly	
ngdp_r_sa	Billions	Numeric	Gross domestic product, constant prices (National Currency, SA, Billions)	Quarterly	

Table E.2. Data Coverage

							Data Coverage:										Data Source:		
							Revenue		Expenditure		Overall Balance		Tax Revenue		Nominal GDP				
No.	Country	ISO-3	IFS Code	AE	EM	LIC	Start	End	Start	End	Start	End			Start	End	Fiscal	Tax Revenue	GDP
1	Afghanistan	AFG	512	0	0	1	2008Q2	2017Q4	2008Q2	2017Q4	2008Q2	2017Q4	--	--	--	--	EIU	--	--
2	Albania	ALB	914	0	1	0	1994Q1	2025Q3	1994Q1	2025Q3	1994Q1	2025Q3	--	--	1995Q1	2025Q3	EIU	--	NEA
3	Algeria	DZA	612	0	1	0	--	--	--	--	--	--	--	--	2001Q1	2025Q2	--	--	NEA
4	Angola	AGO	614	0	1	0	2020Q1	2025Q3	2020Q1	2025Q3	2020Q1	2025Q3	--	--	--	--	Haver	--	--
5	Argentina	ARG	213	0	1	0	2016Q1	2025Q4	2016Q1	2025Q4	1993Q1	2025Q4	--	--	2004Q1	2025Q4	Haver	--	NEA
6	Armenia	ARM	911	0	1	0	2006Q3	2025Q4	2006Q3	2025Q4	2006Q3	2025Q4	--	--	2013Q1	2025Q3	Haver	--	NEA
7	Australia	AUS	193	1	0	0	2002Q3	2025Q3	2002Q3	2025Q3	2002Q3	2025Q3	2003Q3	2025Q4	1990Q1	2025Q4	EIU	QGFS	NEA
8	Austria	AUT	122	1	0	0	2001Q1	2025Q3	2001Q1	2025Q3	2001Q1	2025Q3	2001Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
9	Azerbaijan	AZE	912	0	1	0	--	--	--	--	--	--	--	--	2001Q1	2025Q3	--	--	NEA
10	Bangladesh	BGD	513	0	0	1	2018Q3	2025Q4	2018Q3	2025Q4	2018Q3	2025Q4	--	--	--	--	Haver	--	--
11	Belarus	BLR	913	0	1	0	2005Q1	2024Q4	2005Q1	2024Q4	2005Q1	2024Q4	--	--	1992Q1	2025Q3	Haver	--	NEA
12	Belgium	BEL	124	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
13	Benin	BEN	638	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	2006Q1	2013Q4	--	--	Haver	QGFS	--
14	Bolivia	BOL	218	0	1	0	--	--	--	--	--	--	--	--	2017Q1	2025Q2	--	--	NEA
15	Bosnia and Herzegovina	BIH	963	0	1	0	2004Q3	2025Q3	2004Q3	2025Q3	2004Q3	2025Q3	--	--	2000Q1	2025Q3	Haver	--	NEA
16	Botswana	BWA	616	0	1	0	2007Q2	2025Q3	2007Q2	2025Q3	2007Q2	2025Q3	--	--	2006Q1	2025Q3	EIU	--	NEA
17	Brazil	BRA	223	0	1	0	1997Q1	2025Q4	1997Q1	2025Q4	1997Q1	2025Q4	2010Q1	2025Q4	1996Q1	2025Q4	Haver	QGFS	NEA
18	Bulgaria	BGR	918	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
19	Burkina Faso	BFA	748	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	2004Q1	2024Q2	--	--	Haver	QGFS	--
20	Burundi	BDI	618	0	0	1	2005Q1	2025Q4	2005Q1	2025Q4	2005Q1	2025Q4	--	--	--	--	Haver	--	--
21	Cambodia	KHM	522	0	0	1	2009Q1	2025Q3	2009Q1	2025Q3	2009Q1	2025Q3	2006Q1	2024Q1	--	--	Haver	QGFS	--
22	Cameroon	CMR	622	0	0	1	2014Q1	2024Q4	2014Q1	2024Q4	2014Q1	2024Q4	--	--	1999Q1	2025Q1	EIU	--	NEA
23	Canada	CAN	156	1	0	0	1993Q1	2025Q4	1993Q1	2025Q4	1993Q1	2025Q4	1990Q1	2025Q4	1990Q1	2025Q4	EIU	QGFS	NEA
24	Central African Republic	CAF	626	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
25	Chad	TCD	628	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
26	Chile	CHL	228	0	1	0	2017Q1	2025Q3	2017Q1	2025Q3	2017Q1	2025Q3	2005Q1	2025Q4	1996Q1	2025Q4	Haver	QGFS	NEA
27	China	CHN	924	0	1	0	1998Q1	2025Q4	1998Q1	2025Q4	1998Q1	2025Q4	--	--	1992Q1	2025Q4	Haver	--	NEA
28	Colombia	COL	233	0	1	0	1995Q1	2025Q3	1995Q1	2025Q3	1995Q1	2025Q3	2021Q1	2025Q3	2005Q1	2025Q4	Haver	QGFS	NEA
29	Congo, Democratic Republic of the	COD	636	0	0	1	2009Q1	2025Q3	2009Q1	2025Q3	2009Q1	2025Q3	--	--	--	--	Haver	--	--
30	Congo, Republic of	COG	634	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
31	Costa Rica	CRI	238	0	1	0	2006Q1	2025Q4	2006Q1	2025Q4	2006Q1	2025Q4	--	--	1991Q1	2025Q4	Haver	--	NEA
32	Croatia	HRV	960	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
33	Czech Republic	CZE	935	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
34	Côte d'Ivoire	CIV	662	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
35	Denmark	DNK	128	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
36	Dominican Republic	DOM	243	0	1	0	2000Q1	2025Q4	2000Q1	2025Q4	2000Q1	2025Q4	2008Q1	2021Q4	1991Q1	2024Q4	Haver	QGFS	NEA
37	Ecuador	ECU	248	0	1	0	1999Q1	2025Q4	1999Q1	2025Q4	1999Q1	2025Q4	--	--	1991Q1	2025Q3	Haver	--	NEA
38	Egypt	EGY	469	0	1	0	--	--	--	--	--	--	--	--	2019Q4	2024Q4	--	--	NEA
39	El Salvador	SLV	253	0	1	0	1994Q1	2025Q3	1994Q1	2023Q3	1994Q1	2023Q3	2014Q1	2021Q1	2000Q1	2025Q2	EIU	QGFS	NEA
40	Eritrea	ERI	643	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--

							Data Coverage:										Data Source:		
							Revenue		Expenditure		Overall Balance		Tax Revenue		Nominal GDP				
No.	Country	ISO-3	IFS Code	AE	EM	LIC	Start	End	Start	End	Start	End			Start	End	Fiscal	Tax Revenue	GDP
41	Ethiopia	ETH	644	0	0	1	2014Q3	2024Q4	2014Q3	2024Q4	2014Q3	2024Q4	--	--	--	--	Haver	--	--
42	Finland	FIN	172	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1990Q1	2025Q4	Haver	QGFS	NEA
43	France	FRA	132	1	0	0	1990Q1	2025Q3	1990Q1	2025Q3	1990Q1	2025Q3	1999Q1	2025Q3	1990Q1	2025Q4	Haver	QGFS	NEA
44	Gabon	GAB	646	0	1	0	--	--	--	--	--	--	--	--	--	--	--	--	--
45	Gambia, The	GMB	648	0	0	1	--	--	--	--	--	--	--	--	2013Q1	2023Q4	--	--	NEA
46	Georgia	GEO	915	0	1	0	2006Q1	2025Q4	2006Q1	2025Q4	2006Q1	2025Q4	--	--	1996Q1	2025Q3	Haver	--	NEA
47	Germany	DEU	134	1	0	0	2002Q1	2025Q3	2002Q1	2025Q3	2002Q1	2025Q3	2002Q1	2025Q3	1991Q1	2025Q4	Haver	QGFS	NEA
48	Ghana	GHA	652	0	0	1	2008Q1	2025Q4	2008Q1	2025Q4	2008Q1	2025Q4	--	--	2013Q1	2024Q3	Haver	--	NEA
49	Greece	GRC	174	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2024Q4	1995Q1	2025Q4	Haver	QGFS	NEA
50	Guatemala	GTM	258	0	1	0	1995Q1	2025Q4	1995Q1	2025Q4	1995Q1	2025Q4	--	--	2013Q1	2025Q3	Haver	--	NEA
51	Guinea	GIN	656	0	0	1	2018Q1	2025Q2	2018Q1	2025Q2	2018Q1	2025Q2	--	--	--	--	Haver	--	--
52	Guinea-Bissau	GNB	654	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	--	--	--	--	Haver	--	--
53	Haiti	HTI	263	0	0	1	1993Q2	2006Q2	1993Q2	2006Q2	1993Q2	2006Q2	--	--	--	--	EIU	--	--
54	Honduras	HND	268	0	0	1	2014Q1	2025Q4	2014Q1	2025Q4	2014Q1	2025Q4	--	--	2000Q1	2025Q4	Haver	--	NEA
55	Hong Kong SAR	HKG	532	1	0	0	1994Q3	2025Q4	1994Q3	2025Q4	1994Q3	2025Q4	--	--	--	--	Haver	--	--
56	Hungary	HUN	944	0	1	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
57	India	IND	534	0	1	0	1997Q2	2025Q4	1997Q2	2025Q4	1994Q2	2025Q4	--	--	2004Q2	2025Q4	Haver	--	NEA
58	Indonesia	IDN	536	0	1	0	1999Q2	2025Q4	1999Q2	2025Q4	--	--	2015Q1	2021Q1	1990Q1	2025Q4	Haver	QGFS	NEA
59	Iran	IRN	429	0	1	0	2001Q2	2024Q1	2001Q3	2024Q1	2001Q3	2024Q1	--	--	2016Q1	2023Q4	Haver	--	NEA
60	Iraq	IRQ	433	0	1	0	2004Q1	2025Q4	2004Q1	2025Q4	2004Q1	2025Q4	--	--	--	--	Haver	--	--
61	Ireland	IRL	178	1	0	0	2002Q1	2025Q3	2002Q1	2025Q3	2002Q1	2025Q3	2002Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
62	Israel	ISR	436	1	0	0	2001Q1	2025Q4	2001Q1	2025Q4	2001Q1	2025Q4	2006Q1	2025Q2	1995Q1	2025Q4	Haver	QGFS	NEA
63	Italy	ITA	136	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
64	Jamaica	JAM	343	0	1	0	--	--	--	--	--	--	--	--	1996Q1	2024Q4	--	--	NEA
65	Japan	JPN	158	1	0	0	1993Q1	2025Q4	1993Q1	2025Q4	1993Q1	2025Q4	--	--	1994Q1	2025Q4	EIU	--	NEA
66	Jordan	JOR	439	0	1	0	2021Q1	2025Q2	2021Q1	2025Q2	--	--	--	--	1992Q1	2024Q1	Haver	--	NEA
67	Kazakhstan	KAZ	916	0	1	0	2000Q1	2024Q1	2000Q1	2024Q1	2000Q1	2024Q1	--	--	1995Q1	2023Q3	Haver	--	NEA
68	Kenya	KEN	664	0	0	1	2000Q4	2025Q1	2000Q4	2025Q1	2000Q4	2025Q1	--	--	2009Q1	2025Q3	EIU	--	NEA
69	Korea	KOR	542	1	0	0	2000Q1	2025Q3	2000Q1	2025Q3	2000Q1	2025Q3	--	--	1990Q1	2025Q4	Haver	--	NEA
70	Kuwait	KWT	443	0	1	0	--	--	--	--	--	--	--	--	--	--	--	--	--
71	Kyrgyz Republic	KGZ	917	0	0	1	1998Q1	2025Q4	1998Q1	2025Q4	1998Q1	2025Q4	--	--	1999Q1	2024Q1	EIU	--	NEA
72	Lao P.D.R.	LAO	544	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
73	Latvia	LVA	941	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
74	Lebanon	LBN	446	0	1	0	1998Q1	2024Q4	1998Q1	2024Q4	--	--	--	--	--	--	Haver	--	--
75	Lesotho	LSO	666	0	0	1	2000Q1	2024Q2	2000Q1	2024Q2	2000Q1	2024Q2	2000Q1	2021Q2	2007Q1	2024Q1	Haver	QGFS	NEA
76	Liberia	LBR	668	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
77	Libya	LBY	672	0	1	0	--	--	--	--	--	--	--	--	--	--	--	--	--
78	Lithuania	LTU	946	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
79	Madagascar	MDG	674	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
80	Malawi	MWI	676	0	0	1	2010Q4	2025Q3	2010Q4	2025Q3	2010Q4	2025Q3	--	--	--	--	Haver	--	--
81	Malaysia	MYS	548	0	1	0	1996Q1	2025Q3	1996Q1	2025Q3	1990Q1	2025Q3	--	--	2015Q1	2025Q3	Haver	--	NEA
82	Mali	MLI	678	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	--	--	--	--	Haver	--	--
83	Mauritania	MRT	682	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
84	Mexico	MEX	273	0	1	0	1990Q1	2025Q4	1990Q1	2025Q4	1990Q1	2025Q4	2008Q1	2025Q3	1993Q1	2025Q4	Haver	QGFS	NEA
85	Moldova	MDA	921	0	0	1	2016Q1	2025Q4	2016Q1	2025Q4	2016Q1	2025Q4	--	--	2010Q1	2025Q3	Haver	--	NEA
86	Mongolia	MNG	948	0	1	0	2000Q1	2025Q3	2000Q1	2025Q3	2000Q1	2025Q3	--	--	2015Q1	2025Q3	EIU	--	NEA
87	Morocco	MAR	686	0	1	0	2010Q1	2025Q4	2010Q1	2025Q4	2010Q1	2025Q4	2007Q1	2024Q3	2014Q1	2025Q1	Haver	QGFS	NEA

							Data Coverage:										Data Source:		
							Revenue		Expenditure		Overall Balance		Tax Revenue		Nominal GDP				
No.	Country	ISO-3	IFS Code	AE	EM	LIC	Start	End	Start	End	Start	End			Start	End	Fiscal	Tax Revenue	GDP
88	Mozambique	MOZ	688	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
89	Myanmar	MMR	518	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
90	Namibia	NAM	728	0	1	0	2015Q2	2025Q4	2015Q2	2025Q4	2015Q2	2025Q4	--	--	2013Q1	2025Q4	Haver	--	NEA
91	Nepal	NPL	558	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
92	Netherlands	NLD	138	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
93	New Zealand	NZL	196	1	0	0	--	--	--	--	--	--	--	--	1990Q1	2025Q4	--	--	NEA
94	Nicaragua	NIC	278	0	0	1	2016Q1	2025Q3	2016Q1	2025Q3	2016Q1	2025Q3	--	--	2006Q1	2025Q2	Haver	--	NEA
95	Niger	NER	692	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	--	--	--	--	Haver	--	--
96	Nigeria	NGA	694	0	0	1	2005Q1	2024Q4	2005Q1	2024Q4	2005Q1	2024Q4	--	--	2010Q1	2024Q2	Haver	--	NEA
97	North Macedonia	MKD	962	0	1	0	2003Q2	2025Q4	2003Q2	2025Q4	2003Q2	2025Q4	--	--	2000Q1	2025Q4	Haver	--	NEA
98	Norway	NOR	142	1	0	0	2002Q1	2025Q3	2002Q1	2025Q3	2002Q1	2025Q3	1996Q1	2025Q3	1990Q1	2025Q4	Haver	QGFS	NEA
99	Oman	OMN	449	0	1	0	2005Q1	2025Q3	2005Q1	2025Q3	2009Q2	2025Q3	--	--	--	--	Haver	--	--
100	Pakistan	PAK	564	0	1	0	2001Q3	2025Q4	2001Q3	2025Q4	2001Q3	2025Q4	--	--	2016Q1	2025Q4	EIU	--	NEA
101	Panama	PAN	283	0	1	0	--	--	--	--	--	--	2014Q1	2017Q4	--	--	--	QGFS	--
102	Papua New Guinea	PNG	853	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
103	Paraguay	PRY	288	0	1	0	1998Q1	2025Q4	1998Q1	2025Q4	1998Q1	2025Q4	--	--	1994Q1	2025Q4	Haver	--	NEA
104	Peru	PER	293	0	1	0	1995Q1	2025Q4	1995Q1	2025Q4	1995Q1	2025Q4	--	--	2007Q1	2025Q3	Haver	--	NEA
105	Philippines	PHL	566	0	1	0	1990Q1	2025Q4	1990Q1	2025Q4	1990Q1	2025Q4	--	--	1990Q1	2025Q2	Haver	--	NEA
106	Poland	POL	964	0	1	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
107	Portugal	PRT	182	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
108	Qatar	QAT	453	0	1	0	2004Q2	2025Q4	2004Q2	2025Q4	2004Q2	2025Q4	--	--	2010Q1	2023Q4	Haver	--	NEA
109	Romania	ROU	968	0	1	0	1995Q4	2025Q3	1995Q4	2025Q3	1995Q4	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
110	Russia	RUS	922	0	1	0	1993Q1	2025Q4	1993Q1	2025Q4	1993Q1	2025Q4	2016Q1	2021Q3	1994Q1	2025Q3	Haver	QGFS	NEA
111	Rwanda	RWA	714	0	0	1	--	--	--	--	--	--	2015Q3	2019Q2	2006Q1	2025Q1	--	QGFS	NEA
112	Saudi Arabia	SAU	456	0	1	0	2016Q1	2025Q4	2016Q1	2025Q4	2016Q1	2025Q4	--	--	2005Q1	2025Q4	Haver	--	NEA
113	Senegal	SEN	722	0	0	1	2015Q1	2025Q4	2015Q1	2025Q4	2015Q1	2025Q4	--	--	2014Q1	2025Q3	Haver	--	NEA
114	Sierra Leone	SLE	724	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
115	Singapore	SGP	576	1	0	0	1990Q1	2025Q4	1990Q1	2025Q4	2000Q3	2025Q4	--	--	1990Q1	2025Q3	Haver	--	NEA
116	Slovak Republic	SVK	936	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
117	Slovenia	SVN	961	1	0	0	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
118	South Africa	ZAF	199	0	1	0	1990Q1	2025Q4	1990Q1	2025Q4	1990Q1	2025Q4	--	--	1993Q1	2025Q4	Haver	--	NEA
119	Spain	ESP	184	1	0	0	1995Q1	2025Q3	1995Q1	2025Q3	1995Q1	2025Q3	1999Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
120	Sri Lanka	LKA	524	0	1	0	2014Q1	2025Q3	2014Q1	2025Q3	2014Q1	2025Q3	--	--	2010Q1	2025Q3	Haver	--	NEA
121	Sudan	SDN	732	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
122	Sweden	SWE	144	1	0	0	1995Q1	2025Q3	1995Q1	2025Q3	1995Q1	2025Q3	1999Q1	2025Q3	1993Q1	2025Q4	Haver	QGFS	NEA
123	Switzerland	CHE	146	1	0	0	1995Q1	2025Q1	1995Q1	2025Q1	1995Q1	2025Q1	2007Q1	2024Q3	1995Q1	2025Q4	Haver	QGFS	NEA
124	Taiwan Province of China	TWN	528	1	0	0	1990Q1	2025Q4	1990Q1	2025Q4	1990Q1	2025Q4	--	--	--	--	Haver	--	--
125	Tajikistan	TJK	923	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
126	Tanzania	TZA	738	0	0	1	2010Q4	2025Q3	2010Q4	2025Q3	2010Q4	2025Q3	--	--	--	--	EIU	--	--
127	Thailand	THA	578	0	1	0	2012Q4	2025Q3	2012Q4	2025Q3	2012Q4	2025Q3	2012Q4	2017Q3	1993Q1	2025Q4	Haver	QGFS	NEA
128	Togo	TGO	742	0	0	1	2013Q1	2025Q2	2013Q1	2025Q2	2013Q1	2025Q2	--	--	--	--	Haver	--	--
129	Tunisia	TUN	744	0	1	0	2010Q2	2025Q3	2010Q2	2025Q3	2010Q2	2025Q3	--	--	--	--	EIU	--	--
130	Turkmenistan	TKM	925	0	1	0	--	--	--	--	--	--	--	--	--	--	--	--	--
131	Türkiye	TUR	186	0	1	0	2012Q1	2025Q3	2012Q1	2025Q3	2012Q1	2025Q3	2015Q1	2025Q3	1995Q1	2025Q4	Haver	QGFS	NEA
132	Uganda	UGA	746	0	0	1	1997Q3	2025Q4	1997Q3	2025Q3	1997Q3	2025Q3	2001Q1	2020Q1	2016Q3	2024Q3	Haver	QGFS	NEA

							Data Coverage:										Data Source:		
							Revenue		Expenditure		Overall Balance		Tax Revenue		Nominal GDP				
No.	Country	ISO-3	IFS Code	AE	EM	LIC	Start	End	Start	End	Start	End			Start	End	Fiscal	Tax Revenue	GDP
133	Ukraine	UKR	926	0	1	0	2005Q1	2025Q3	2005Q1	2025Q3	2005Q1	2025Q3	--	--	2000Q1	2025Q3	Haver	--	NEA
134	United Kingdom	GBR	112	1	0	0	1997Q2	2025Q4	1997Q2	2025Q4	1997Q2	2025Q4	1999Q1	2025Q3	1990Q1	2025Q4	EIU	QGFS	NEA
135	United States	USA	111	1	0	0	1952Q1	2025Q4	1960Q1	2025Q4	1960Q1	2025Q4	1952Q1	2025Q4	1952Q1	2025Q4	FRED	FRED	NEA
136	Uruguay	URY	298	0	1	0	1999Q1	2025Q4	1999Q1	2025Q4	1999Q1	2025Q4	--	--	2016Q1	2025Q4	Haver	--	NEA
137	Uzbekistan	UZB	927	0	0	1	2013Q1	2025Q1	2013Q1	2025Q1	2013Q1	2025Q1	--	--	--	--	Haver	--	--
138	Venezuela	VEN	299	0	1	0	--	--	--	--	--	--	--	--	--	--	--	--	--
139	Vietnam	VNM	582	0	1	0	2005Q1	2025Q1	2005Q1	2025Q1	2005Q1	2025Q1	--	--	--	--	Haver	--	--
140	Yemen	YEM	474	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--
141	Zambia	ZMB	754	0	0	1	2016Q1	2025Q4	2016Q1	2025Q4	2016Q1	2025Q4	--	--	2010Q1	2025Q2	Haver	--	NEA
142	Zimbabwe	ZWE	698	0	0	1	--	--	--	--	--	--	--	--	--	--	--	--	--

Note: The table reports the variable coverage for each country in the fiscal actions database. AE, EM, LIC indicate whether the country is an advanced economy (AE), emerging market and middle-income economy (EM), or low-income developing country (LIC) per the IMF's country classification. Data coverage reports the starting and ending quarters for fiscal and GDP data. "--" indicates that data is not available for a particular variable. Data sources include Haver, the Economist Intelligence Unit (EIU), the Quarterly Government Finance Statistics (QGFS) database and the National Economic Accounts (NEA) database from the IMF, and the Federal Reserve Economic Data (FRED) for data for the United States.

Annex F. Extended Sample of Revenue Multipliers: Combining Tax and Revenue Data

The tax multipliers reported in Section VI are estimated using quarterly disaggregated tax revenue data. Aligning the fiscal aggregate with the identified shock is conceptually the cleanest identification strategy: the narrative tax actions as captured in the AI-assisted GFA dataset refer primarily to discretionary changes in tax policy so their natural transmission channel to the economy is through tax receipts (Romer and Romer, 2010; Favero and Giavazzi, 2012; Hebous and Zimmermann, 2018). However, the cost of this identification choice is the loss of coverage. Quarterly tax revenue data are available for most advanced economies (AE) and few emerging markets (EM), but are scarce for low-income countries (LIC), where sub-annual disaggregation of government revenue between tax and non-tax components is less systematically or accurately reported.[1] Restricting attention to the countries for which quarterly tax revenue data are available therefore leaves a large gap in what can be said about revenue-side multipliers in the developing world.

The purpose of this Annex is to complement the tax multiplier estimates presented in Section VI by constructing an extended sample that expands country coverage—especially towards LICs—using an alternative but conceptually related fiscal aggregate: government revenue data. General government revenue comprises tax revenue as well as other, non-tax components such as grants, social security contributions, natural resource-related revenue, income from state-owned enterprises, investment and property income, revenues from the sale of goods and services or the collection of fines, fees, premiums, and other transfers received.[2] As a result, government revenue captures a broader measure of government receipts than tax revenue alone. This extended sample allows us to report revenue-side estimates for low-income economies, often for the first time in the country-specific fiscal multiplier literature, and to compare them with those obtained for AEs and EMs.

Specification

The extended specification follows the same econometric framework described in Section VI. For each country i and horizon h = 0 to 8, two local projections models are estimated following equation (8), with penalized OLS and cubic B-spline smoothing, and the country-specific cumulative multiplier is derived as the ratio of cumulative impulse responses as in equation (9). Full-sample multipliers are aggregated across countries using inverse-variance weighting (IVW) following equation (10). All model choices are identical to those described for Section VI's specification. The only difference is in the fiscal aggregate used as the propagating channel. For countries with quarterly tax revenue data, tax revenue continues to be the fiscal variable of interest, while for countries where quarterly tax revenue is not available, total general government revenue is used instead. As a result, country-level tax multiplier estimates for countries included in Section VI are unchanged, since their fiscal variable remains tax revenue.

[1] See Annex Table C.1 for details on available data by country.

[2] The revenue series are compiled from Haver, the Economist Intelligence Unit (EIU), and the Quarterly Government Finance Statistics (QGFS) database from the IMF (see Annex C and Table C.1). While definitions and coverage of government revenue may vary across countries depending on reporting practices and institutional settings, these series predominantly follow internationally accepted statistical standards, such as the IMF's Government Finance Statistics Manual (GFSM), and are thus broadly comparable across countries.

Results

The extended specification delivers country-level estimates for 63 countries, up from the more restricted tax data-based baseline of 34 countries. The additional countries relative to the baseline are primarily low-income and, to a lesser extent, emerging market economies. This constitutes, to the best of our knowledge, one of the first sets of country-specific revenue-side fiscal multiplier estimates for LICs based on a narrative identification of fiscal shocks, complementing recent efforts to enlarge the coverage of country-specific multiplier evidence for developing economies (Abdel-Latif, Bechchani, David, and Lemaire, 2026).

Figure E.1 replicates the four-panel structure of Figure 26 for the extended sample combining tax and revenue data. Panels (A) and (B) show the full-sample IVW impulse responses of output and of the fiscal variable to a tax policy shock. The dynamics closely mirror those of the tax-only baseline. On the output side (Panel A), a tax cut is followed by a hump-shaped rise in real GDP that peaks during the first year and gradually reverts towards zero. On the revenue side (Panel B), revenue falls on impact and remains persistently below its counterfactual path throughout the horizon. These patterns are consistent with the responses documented in the tax-only baseline (Figure 26) and with the shape of the narrative tax multiplier reported in the literature such as in Romer and Romer (2010) and Cloyne (2013). The pooled IVW cumulative revenue multiplier is 0.37 at impact, 0.47 at the one-year horizon and decreases to 0.22 at the two-year horizon (Panel C). These results preserve the sign and the qualitative profile of the baseline tax multipliers reported in Figure 26 but are smaller in magnitude at short horizons than the corresponding tax-only estimates. This is affected by the different sample size (almost twice the number of countries than in the tax multiplier) but it also appears consistent with the intuition that total revenue is a noisier proxy for the transmission of tax policy actions than tax revenue itself for three main reasons.

First, total revenue includes non-tax components, notably grants, royalties, dividends, natural resource rents, and other resource-related receipts that are not always necessarily the primary transmission channel of a tax policy change. In LICs, in particular, grants and other forms of foreign aid can account for a large and volatile share of revenue and are largely determined by donor cycles rather than by domestic fiscal decisions. For resource-dependent economies, revenue fluctuates substantially with commodity prices and production, which are sources of variation largely orthogonal to tax policy. To the extent that these non-tax components add cyclical noise to the fiscal aggregate, the denominator of the cumulative multiplier becomes noisier and less tightly linked to the identified shock.

Second, revenue collection in EMs and LICs is disproportionately shaped by structural features such as a large informal sector, weaker administrative capacity, and narrower tax bases. Colombo, Furceri, Pizzuto and Tirelli (2022) show that fiscal multipliers tend to be smaller in economies with more informality, since the effective pass-through of policy change to the overall economy is weakened by the narrower impact base represented by a small formal sector, undermining governments' ability to implement stabilization policies.

Third, cyclical volatility of macroeconomic aggregates is generally higher in EMs and LICs than in AEs (Ilzetzki, Mendoza, and Végh, 2013), as these economies appear less resilient to external trade and financial shocks. Because the cumulative multiplier is a ratio of two impulse responses, the idiosyncratic volatility in the denominator translates into wider confidence bands and smaller point estimates. The convergence of the tax-only and extended sample estimates at the two-year horizon can support this interpretation as when the fiscal aggregate is cumulated over quarters, transitory noise averages out and the two specifications converge to similar long-run estimates.

Panel (D) summarizes the distribution of country-level revenue multipliers at the one-year horizon by income group, highlighting considerable heterogeneity within and across groups. The LIC distribution is more dispersed than those for AEs and EMs, with a wider interquartile range and lower median. Rather than being a structural feature of LIC multipliers as such, this dispersion is best interpreted as a reflection of the confounding factors in the identification channel discussed above. Notwithstanding these considerations, the pooled IVW multiplier for the entire sample countries remains in the same qualitative range as in the tax-only baseline.

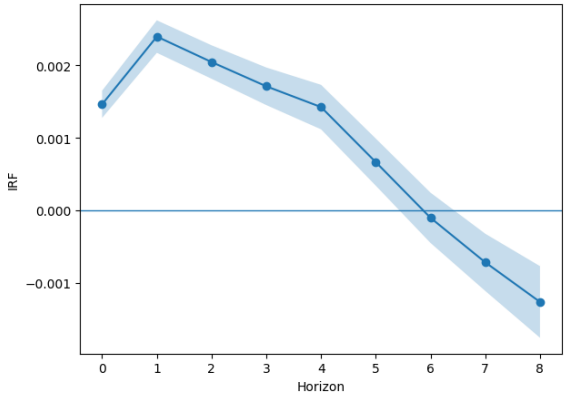
Conclusion

The results in this annex reinforce two main conclusions of the analysis presented in Section VI. First, the empirical regularities documented in the tax-only baseline—a hump-shaped output response, a persistent decline in revenue, and a positive cumulative multiplier that peaks around the one-year horizon before decaying—hold in an extended sample that includes LICs. The pooled IVW multiplier is smaller than in the tax-only baseline at short horizons and virtually identical at the two-year horizon, consistent with the interpretation that total revenue is a noisier but broadly informative proxy for the transmission of tax policy actions when tax revenue is not available.

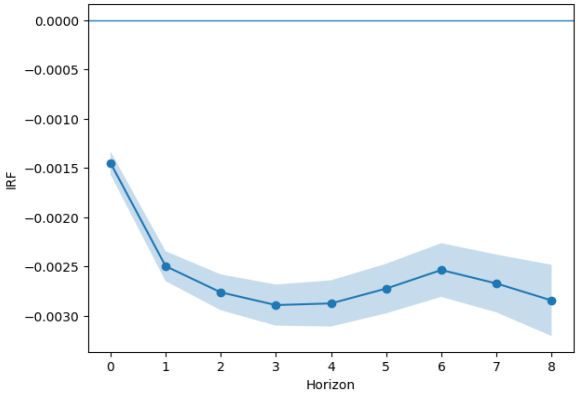
Second, the extension of country coverage represents a substantive contribution to the literature and to policymakers. By combining the AI-assisted GFA dataset with the available quarterly revenue information, this paper is able to provide country-specific revenue multiplier estimates for a substantially larger set of economies. In doing so, the paper opens a new evidence base for calibrating fiscal advice in developing economies, complementing the panel-based averages that country teams and researchers have historically relied on (Batini, Eyraud, and Weber, 2014; Carrière-Swallow, David, and Leigh, 2020).

Figure F.1. Response of Tax & Revenue and Output to a Tax policy action

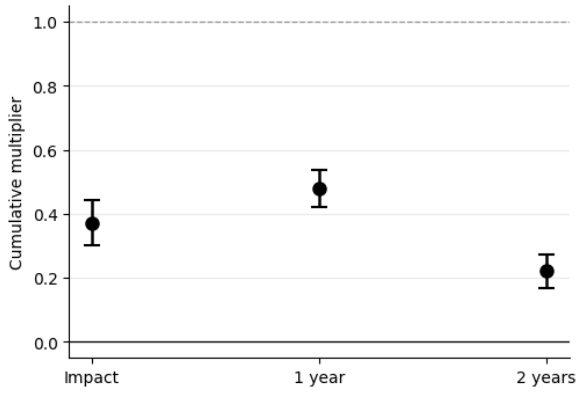
A. IRF: Output



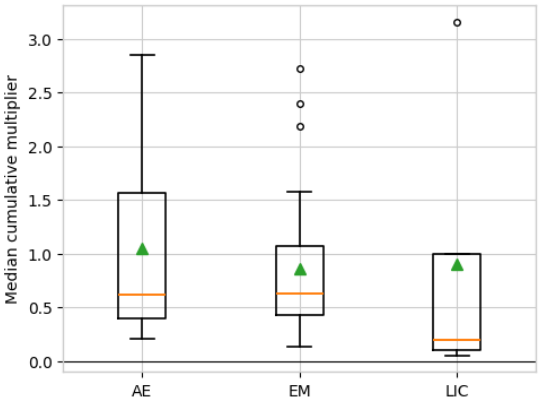
B. IRF: Tax & Revenue



C. Tax & Revenue Multiplier across horizons



D. Tax and Revenue Multiplier by Country Groups: 1 year horizon



Notes: Panels (A) and (B) plot the full-sample IVW impulse responses of output and revenue to a tax policy action. The solid line is the IVW mean of country-wise point estimates and the shaded error refers to ± 1 IVW standard errors. Panel (C) reports the implied cumulative revenue multiplier at impact, one year, and two years, where one and two years correspond to horizons 4 and 8. Dots are IVW means of country point estimates of multipliers and vertical bars are ± 1 standard error bands. A positive revenue multiplier refers to GDP moving positively in response to a decrease in revenues. Panel (D) summarizes the distribution of country-level multipliers by country group at the one-year horizon. Boxes show the interquartile range, orange lines denote medians, green triangles denote means, whiskers show the non-outlier range, and circles denote outliers. AE = advanced economies; EM = emerging markets; LIC= low-income countries.



PUBLICATIONS

The Global Fiscal Actions (GFA) Dataset
Working Paper No. WP/2026/221